CAPACITY DETERMINATION AND EXPANSION MODELS FOR RAIL NETWORKS

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Dedicated to:

My Brother, Kaveh Bevrani

Abstract

Efficient railways need to be designed, constructed and operated. Their performance must also be identifiable and their capacity must be transparent to users. Railway capacity analysis is an important approach to determine whether the network can handle an intended traffic flow and whether there is any free capacity left for additional train services. As capacity planning is highly combinatorial and technical, there are many alternatives to compare. Hence mathematical approaches are critical in finding the best solution and course of action. This thesis has considered that topic and tested that presumption. The shortcomings of a prominent model for capacity analysis have been addressed and a number of innovations have been made. In particular a number of mathematical optimization models are developed. To verify their applicability and validity they were applied and tested on a case study of the Iranian national railway.

As all railways need to be expanded to meet increasing demands, the presumption that analytical railway capacity models can be used to identify how best to improve an existing network at least cost, was first investigated. Track duplication was specifically considered as the primary means of expanding a network's capacity, and two variant capacity expansion models were formulated. One is static, and the other is over time (i.e. time varying). The static model was introduced to identify the best sections to duplicate immediately to obtain the maximum capacity, or else, a specified level of capacity. The time varying model determines a plan over time and can consider inflation and other economic phenomena.

A secondary outcome of this thesis is the development and validation of bi objective models for capacity analysis. These models are very beneficial as they are able to regulate the competition for track access and perform a trade-off analysis. An opportunity to develop more general multi-objective approaches with more than two objectives was identified. This thesis has also identified that capacity expansion and bi-objective capacity analyses are not dissimilar topics. For example, any expansion that is performed must be equitable and should satisfy as many clients as possible. This is essentially a multi-objective situation with a maximise-capacity and minimise-spending objective.

In summary the models presented are important and useful because they can reduce the time required to perform decision making activities that planners and managers need to perform. Transportation authorities make decisions on billions of dollars' worth of infrastructure expansion projects annually to improve railways. These decisions are not reached quickly and much consideration is required. The net effect of those decisions will greatly affect users' travel time and safety. The approaches advocated in this thesis can perform decision making quickly.

Keywords

Analytical Models, Bi-Objective, Capacity Analysis, Capacity Expansion, Multi Objective, Railways, Sensitivity Analysis, Track Duplication

List of Abbreviations

AUD	Australian Dollar
CPLEX	Optimisation Software
DD	Directional Distribution
ECD	Euclidean Distance
ECM	E-Constraint Method
hr	Hour
HSR	High Speed Rail
km	Kilometre
KPI	Key Performance Indicator
LP	Linear Programming
MILP	Mix Integer Linear Programming
MO	Multi-Objective
NLP	Non Linear Programming
obj	Objective
PD	Proportional Distribution
PF	Percentage Flow
RAI	Rah Ahane Iran (National Railway Network)
SOT	Section Occupation Time
SRT	Sectional Running Time
TEU	Twenty Foot Equivalent Unit
UIC	Capacity Method
WSM	Weighted Sum Method

Nomenclature

$\boldsymbol{i, c, s, t}$	Train, corridor, section, time period index
$\boldsymbol{C, I, S, P}$	Set of corridors, train types, sections, and periods
$\vec{\boldsymbol{x}}_{\boldsymbol{i}}^{\boldsymbol{c}}, \overleftarrow{\boldsymbol{x}}_{\boldsymbol{i}}^{\boldsymbol{c}}$	# of trains, forwards/backwards on corridor c
$\vec{\boldsymbol{y}}_{\boldsymbol{i}}^{\boldsymbol{s}}, \overleftarrow{\boldsymbol{y}}_{\boldsymbol{i}}^{\boldsymbol{s}}$	# of trains, forwards/backwards on section s
$\vec{\boldsymbol{y}}_{\boldsymbol{i,p}}^{\boldsymbol{s}}, \overleftarrow{\boldsymbol{y}}_{\boldsymbol{i,p}}^{\boldsymbol{s}}$	# of trains of type *i* in each section of *s* in time period *p*
$\boldsymbol{T}$	Duration of time period for capacity analysis
$\vec{\boldsymbol{T}}_{\boldsymbol{i}}^{\boldsymbol{s}}, \overleftarrow{\boldsymbol{T}}_{\boldsymbol{i}}^{\boldsymbol{s}}$	Sectional running time for trains of type i in section s in each direction respectively
$\boldsymbol{\mu}_{\boldsymbol{i}}^{\boldsymbol{c}}, \boldsymbol{\eta}_{\boldsymbol{i}}^{\boldsymbol{c}}, \boldsymbol{\sigma}_{\boldsymbol{c}}$	Directional and proportional train mix, flow percentages across corridor c
$\boldsymbol{\Omega}_{\boldsymbol{c}}$	Sections present in each corridor
$\boldsymbol{\tau}_{\boldsymbol{s}}, \boldsymbol{\tau}_{\boldsymbol{s,p}}$	Number of tracks on section s and in time period *p* respectively
L	Length of section or corridor
$\vec{\boldsymbol{x}}_{\boldsymbol{i,p}}^{\boldsymbol{c}}, \overleftarrow{\boldsymbol{x}}_{\boldsymbol{i,p}}^{\boldsymbol{c}}$	# trains of type *i* on corridor *c*, in period *p*
$\boldsymbol{Ct}_{\boldsymbol{s}}^{\parallel}, \boldsymbol{Ct}_{\boldsymbol{s,p}}^{\parallel}$	Cost of track duplication in section *s*, and time period *p* respectively
$\parallel_{\boldsymbol{s}}, \parallel_{\boldsymbol{s}}^{\boldsymbol{max}}$	# of duplicated track in section s and the upper limit respectively
$\parallel_{\boldsymbol{s,p}}, \parallel_{\boldsymbol{s,p}}^{\boldsymbol{max}}$	# of duplicated track in section *s* in period *p* and the upper limit respectively
$\boldsymbol{B}$	Available budget
$\boldsymbol{\eta}^{\boldsymbol{c}}, \boldsymbol{\eta}_{\boldsymbol{i}}^{\boldsymbol{c}}$	Proportional distribution on corridor c for train type *i* specifically
$\boldsymbol{\eta}_{\boldsymbol{i,p}}^{\boldsymbol{c}}$	Proportional train *i* in period *p*, on corridor *c*
$\mathbb{A}$	Absolute capacity of railway network
$\boldsymbol{A}_{\boldsymbol{p}}$	Absolute capacity in period *p*
$\mathbb{A}_{\boldsymbol{p}}$	Intermediate requirement of absolute capacity
$\boldsymbol{A}^{\boldsymbol{I}}, \boldsymbol{A}^{\boldsymbol{F}}$	Initial and final value of capacity
$\mathbb{A}_{\boldsymbol{T}}, \mathbb{A}_{\boldsymbol{c}}, \mathbb{A}_{\boldsymbol{f}}, \mathbb{A}_{\boldsymbol{p}}$	Absolute capacity variations, i.e. train type, corridor
$\boldsymbol{e}_{\boldsymbol{p}}$	Current expenditure in period *p*
$\widehat{\boldsymbol{P}}$	Minimum period for expansion plan
$\boldsymbol{\varphi}_{\boldsymbol{p}}, \boldsymbol{\varphi}_{\boldsymbol{j}}(\boldsymbol{x})$	Different objective function
$\boldsymbol{M}^{+}$	Very big constant number
$\vec{\boldsymbol{M}}_{\boldsymbol{i}}^{\boldsymbol{c}}, \overleftarrow{\boldsymbol{M}}_{\boldsymbol{i}}^{\boldsymbol{c}}$	Payoff per train of type *i* on corridor *c*
$\boldsymbol{x}^{*}$	A particular solution of an optimisation problem
$\boldsymbol{X}$	The set of all possible solutions to an optimisation problem
$\boldsymbol{c}_{\boldsymbol{i}}$	Value of preference
$\boldsymbol{I}_{\boldsymbol{T}}$	A set of train
$\boldsymbol{\Upsilon}, \boldsymbol{\Psi}$	Two different capacity metric
$\boldsymbol{\gamma}_{\boldsymbol{i}}^{\boldsymbol{c}}, \boldsymbol{\psi}_{\boldsymbol{i}}^{\boldsymbol{c}}$	Capacity metric for train type *i* on corridor *c*
$\boldsymbol{\varepsilon}$	Current limitation on objective function value within ECM
$\boldsymbol{\delta}$	Incremental change to an objective function value in ECM
$\boldsymbol{w}$	Objective function weighting in WSM

List of publications from this thesis

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Statement of Original Authorship

The work contained in this thesis has not been previously submitted to meet requirements for an award at this or any other higher education institution. To the best of my knowledge and belief, the thesis contains no material previously published or written by another person except where due reference is made.

QUT Verified Signature

Signature:

Date: 09/03/2015

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Table of Contents

Abstract iv

Keywords v

List of Abbreviations vi

Nomenclature vii

List of publications from this thesis viii

Statement of Original Authorship ix

Acknowledgements x

List of Figures xiii

List of Tables xiv

CHAPTER 1 INTRODUCTION 15
1.1 BACKGROUND 15
1.2 CASE STUDY 17
1.3 RESEARCH AIMS 19
1.4 SIGNIFICANCE AND ORIGINALITY 20
1.5 THESIS OUTLINE 21

CHAPTER 2 LITERATURE REVIEW AND ANALYSIS 23
2.1 NETWORK DESIGN PROBLEM (NDP) 23
2.2 RAILWAY NETWORK IMPROVEMENT 24
2.2.1 Reliability 24
2.2.2 Accessibility 25
2.3 RAILWAY NETWORK TYPES 26
2.3.1 Railway Freight Network 26
2.3.2 High-Speed Rail (HSR) 28
2.4 RAILWAY CAPACITY ANALYSIS 29
2.4.1 UIC Method 30
2.4.2 Train Scheduling Techniques 31
2.5 RAILWAY CAPACITY EXPANSION 32
2.6 MULTI OBJECTIVE OPTIMISATION 34
2.7 DISCUSSION AND KNOWLEDGE GAP 35

CHAPTER 3 UNDERLYING MODEL FOR RAILWAY CAPACITY ANALYSIS 37
3.1 BASE MODEL 37
3.2 ALTERNATIVE MEASURES OF CAPACITY 39
3.3 PRELIMINARY SENSITIVITY ANALYSIS OF CAPACITY 39
3.4 CONCLUSION 41

CHAPTER 4 CAPACITY EXPANSION MODELS 42
4.1 SENSITIVITY ANALYSIS OF TRAIN SPEEDS 42
4.2 TRACK DUPLICATION 47
4.2.1 Static Case 48
4.2.2 Time Varying Cases 49
4.3 NUMERICAL INVESTIGATIONS 50
4.3.1 Manual Duplication 50
4.3.2 Automated Duplication 53
4.3.3 Time Varying Expansions 60
4.4 DISCUSSION 62

CHAPTER 5 BI-CRITERIA MODELS FOR CAPACITY DETERMINATION 63
5.1 MULTI OBJECTIVE THEORY 63
5.2 MODEL FORMULATIONS 66
5.2.1 Competition Between Two Train Services 66

5.2.2 Competition Between Two Corridors 67
5.2.3 Competition Between Two Train Types 67
5.2.4 Competition Between Two Different Metrics 67
5.3 SOLUTION TECHNIQUES 68
5.3.1 E-constraint Method 68
5.3.2 Weighted Sum Method 68
5.4 NUMERICAL INVESTIGATION 69
5.4.1 Competition Between Two Train Services 69
5.4.2 Competition Between Two Corridors 70
5.4.3 Competition Between Two Train Types 72
5.4.4 Max Capacity versus Min Cost 75
5.5 CONCLUSION 79

CHAPTER 6 CONCLUSIONS 81
6.1 MAIN FINDINGS 81
6.1.1 Sensitivity Analysis 81
6.1.2 Capacity expansion models 82
6.1.3 Regulating Competition 83
6.2 SKILLS OBTAINED 84
6.3 FUTURE RESEARCH 84

REFERENCES 85

APPENDIX 1 : CPLEX CODE (BASE MODEL CODE) 94

APPENDIX 2 : CPLEX CODE (STATIC EXPANSION MODEL CODE) 96

APPENDIX 3 : CPLEX CODE (DYNAMIC EXPANSION MODEL CODE) 99

APPENDIX 4 : CPLEX CODE (MULTI-OBJECTIVE SCRIPTS CODE) 102

List of Figures

Figure 1-1: Network diagram for the RAI 18
Figure 4-1: Incremental speed change versus absolute capacity.....45
Figure 4-2: Relative difference 45
Figure 4-3: Incremental speed change versus absolute difference 46
Figure 4-4: Normalised value 46
Figure 4-5: Upgraded network for the RAI (Bold–Duplicated, Dashed–Next longest section) 51
Figure 4-6: Expanded network (unequal flow).....54
Figure 4-7: Comparison of percentage flow on corridor c for different number of $\| \mathbf{s}$.....55
Figure 4-8: Expanded network (equal flow).....56
Figure 4-9: Absolute capacity versus budget increment (limit 1,2,3) 59
Figure 4-10: Absolute capacity by budget increment equal percentage flow 60
Figure 4-11: Absolute capacity by budget increment no restriction on percentage flow 60
Figure 4-12: The plan of expansion activity over time for (limit 2) 61
Figure 5-1: Domination theory.....64
Figure 5-2: Solution quality changes when moving along the Pareto frontier 65
Figure 5-3: The relationship between solution point in the decision and objective space.....65
Figure 5-4: Pareto frontier and associated Euclidean distance.....70
Figure 5-5: Pareto frontier and associated Euclidean distance.....71
Figure 5-6: The Pareto frontiers for each train types' pairing.....73
Figure 5-7: Euclidean distances for all solution points and for all train type pairings 74
Figure 5-8: The Pareto frontier and utopia point line 75
Figure 5-9: Pareto frontier for limit 1, 2, 3 respectively for 20, 500, 1000 solution points respectively76
Figure 5-10: Euclidean distance for limit 1, 2, 3 respectively for 20,500, and 1000 solution points 78

List of Tables

Table 1-1: Network data....19
Table 1-2: Section information (i.e. lengths)....19
Table 3-1: Sensitivity analysis of absolute capacity....40
Table 4-1: Sensitivity analysis of capacity by a single incremental speed change....44
Table 4-2: Absolute capacity increase for mix of trains....44
Table 4-3: Sensitivity analysis of absolute capacity....50
Table 4-4: Duplication of all sections with an additional single track....50
Table 4-5: Sensitivity analysis of duplicated bottleneck sections....51
Table 4-6: Ratio of Table 4-5 to Table 4-3....52
Table 4-7: Ratio of Table 4-4 to Table 4-5....52
Table 4-8: Sensitivity analysis of further duplicated bottleneck sections....52
Table 4-9: Ratio of Table 4-8 to Table 4-5....52
Table 4-10: Ratio of Table 4-4 to Table 4-8....52
Table 4-11: Result of expansion model (No flow constraint)....54
Table 4-12: Expansion results when flow is equal across all the corridors (i.e. PF = 1/9)....55
Table 4-13: Absolute capacity for incremental budget changes (limit 1)....57
Table 4-14: Absolute capacity for incremental budget changes (limit 2)....57
Table 4-15: Absolute capacity for incremental budget changes (limit 3)....58
Table 4-16: Absolute capacity for incremental budget changes(limit 3)....58
Table 4-17: The plan of expansion activity for both equal percentage flow and no restriction on percentage flow....61
Table 4-18: The plan of expansion activity considering inflation....62
Table 5-1: The normalised Pareto solutions....69
Table 5-2: The normalised Pareto solutions....71
Table 5-3: The normalised Pareto solutions....72
Table 5-4: The optimal capacity for different bi objective scenarios....72

Chapter 1 Introduction

1.1 BACKGROUND

Railways are very important in our society and their operation and expansion are incredibly important. They are important, as they allow passengers and freight to be transported across vast distances; in other words they provide freedom of movement, which did not exist in the past. Rail networks are complex systems and consist of many different train types, corridors, and services. Rail lines carry 21 billion passengers and 10 billion tons of freight worldwide annually (IBM, 2014). The history of rail shows that it can be used as an efficient transportation mode to move goods and passengers. Building efficient transportation systems, such as railways, can improve both local and global economies by allowing more goods to be transported, reduce greenhouse gas emissions, make highways safer and reduce road congestion. Congestion causes much lost time, and has a significant financial and environmental cost. The railway industry is faced with four main challenges, which are capacity and congestion, operational efficiency and reliability, structural and competition issues, and safety and security (IBM, 2014).

There is a definite need for intelligent transportation systems that meet the needs of increased population sizes. For example, the population size in Australia is increasing greatly (The World Health Organisation, 2014). The importance of railways as a principal transportation mode has been demonstrated and there is a need for more research that facilitates the development of more advanced and efficient rail systems.

Railway capacity analysis is considered in this thesis. Capacity analysis is necessary to identify the performance of railway networks. If efficient railways are to be designed, constructed and operated, then their performance must be identifiable and their capacity must be transparent. If railways are utilised more efficiently, then more passengers and freight can be transported long distances. By virtue of this, railways become more desirable as a primary mode of transportation, and hence roads and other modes will be relied upon less, and will be less congested. Congestion causes much lost time, and has a significant financial and environmental cost. This necessary task remains difficult, because the definition of capacity is variable, i.e. it differs from country to country. Its calculation is also technically difficult for a variety of reasons. Some approaches have limitations, and there are some aspects of the problem that have not previously been examined or included. Railway capacity can be defined in many different ways. The most common definition of railway capacity involves counting the number of trains that can traverse through the network in a specified time. This thesis has considered the concept of absolute capacity. Absolute capacity determines an upper bound on the capacity of a railway network.

Absolute capacity does not consider the possibility of delays, i.e. on critical sections, from loading/unloading goods and passengers, maintenance, and possible collision conflicts of trains. Hence, absolute capacity does not necessarily determine the real capacity of a railway network and it is a theoretical value. Actual capacity, however, does include delays.

It is important to define both absolute and actual capacity. Comparing these two measures is one way of quantifying the interaction effects between train services across the network. These measures of capacity can also be used to calculate the congestion on railway networks, and can identify the free capacity available for additional train services. Free capacity is identified by comparing what is currently done (i.e. measured) with what you think can be done as determined by an independent capacity analysis. The advantage of absolute capacity is that it can be used for planning purposes as an upper bound of real capacity. Absolute capacity is more robust than actual capacity, which can vary due to its inherently statistical nature.

Railway capacity is not a single unique number. It depends on the mix of trains. Different train types can have substantially different operating characteristics, including dispatching priority. Hence, a capacity model can only be used to identify whether the infrastructure can support future traffic load. The aim of a capacity model is to realise whether there is sufficient capacity in a network. An understanding of how the mix of trains (i.e. the traffic) interacts is necessary for efficient planning, and for the expansion of infrastructure that will provide additional capacity.

Individual corridors can be analysed separately, as if they were not part of a larger, more complex network. The total capacity of all individual corridors may be compared to the capacity of the whole network, in order to identify negative interaction effects and congestion levels. Absolute capacity depends on the number and mix of trains that traverse each corridor. Corridors consist of one or more sections and the possibility that the same section occurs in different corridors is very likely in complex railway networks. Hence, these "shared" sections will be more saturated, i.e. heavily occupied, which can become bottlenecks on the network, and will affect the overall system capacity and the capacity of individual corridors. To increase the capacity of a railway network, bottleneck sections should first be duplicated, i.e. parallel tracks should be added. In theory, some of the existing capacity models that have been developed in previous years should be able to be extended, so that they identify which sections should first be duplicated.

Three approaches can be used to determine railway capacity, namely simulation, analytical/mathematical modelling, and empirical/case based analysis. Simulation models need to be encoded and this takes a significant amount of time and hundreds or thousands of simulation runs are necessary to accumulate enough information about the performance of a system. A mathematical model can define all possible solutions and courses of action. Analytical models are easy to apply. They are superior to current techniques that are used and are sufficiently

accurate for high level planning purposes. In comparison, empirical approaches only analyse a small subset of possible solutions. An empirical approach is not an optimisation approach. An empirical approach only looks at the relative merits of one solution against another. Those solutions are pre-defined beforehand and must be set up specifically - again this takes time.

Track duplications are considered as a means of expanding railway networks. In practice, the cost of duplicating tracks varies significantly. For example it can depend on material costs, the extent of earthworks, the length of sections, workers' salaries, number and position of signals and numbers of sleepers per km. Railways are expensive as they are a form of linear infrastructure. As their length is great, (i.e. excessive) more raw materials and labour are required, for instance, than in other types of construction (i.e. buildings, etc). Railway construction costs differ from country to country and are constantly changing; for instance, there are different labour costs in each country, fluctuations in economic situation, and international events. This makes planning activities difficult. Construction involves different labour and transportation costs and also, construction requires raw materials, which must often be bought at fluctuating prices. Materials must be transported long distances to the site of construction activities. Railway construction also involves earthworks, which are notoriously difficult to plan and are very time consuming activities to perform.

1.2 CASE STUDY

Railway capacity analysis is an important practical task. To verify the validity of the proposed models a real life case study has been selected. This thesis considers a simplified version of Iran's railway network that passes through Tehran and the ports in the south of Iran. RAI is the name of this national rail system. It links the population centres in the west and east of Iran, and has single and double track lines. Iran's most populated cities are close to the capital, Tehran. Iran has a large population of about 80 million people, and constitutes a large area of land geographically. There is much travel by train from the west to the east of the country, between the different religious centres (i.e. cities). Also, the main ports are located in the north and south, near to the sea, on the border of Iran. According to the Tehran Times (2012), 33 million tons of goods and 29 million passengers are transported annually by this railway. In 2011, this network accounted for nine percent and eleven percent of all transportation in Iran. Figure 1-1 gives a visual representation of the main corridors, where passengers and freight are transported and the network data is shown in Table 1-1 and Table 1-2.

This network data was extracted from the official RAI railway network map. Unfortunately this map is not entirely to scale. However, it is sufficiently close to real life values. More accurate values can be obtained from the Jim Ferguson Railway and Tramway Lists (2014), however, some inaccuracies exist in that information too, regarding the names and

positions of railway stations and other infrastructure items, which have been updated in recent years. This network has 38 sections and approximately nine major corridors, whose length varies from 1500-3500 km (Jim Ferguson Railway and Tramway Lists 2014). The Iranian rail network is the preferred mode of transport as the Iranian road network has been recognised as one of the most dangerous networks in the world, with many accidents per year (Bijan, 2013).

RAI has extensive infrastructure expansion plans/objectives to provide improved access and to meet future demands in developing population centres. Many new railway lines have been planned and will be constructed in the near future. The cost of infrastructure expansion over the next 5 to 10 years will be billions of dollars. Figure 1-1 shows the main corridors where passengers and freight are transported and the network data is shown in Table 1-1 and 1-2. The case study data (i.e. the section lengths) as shown in Tables 1-1 and 1-2 has been scaled by 100. This means that the actual distances are 100 times bigger in real life. Hence, the absolute capacity values should be 100 times smaller than reported here. The numerical investigations of the RAI network have shown that corridor A-C, A-D, and C-D have the highest capacity and this is clearly obvious from the network diagram. For example, those corridors have a greater number of smaller sections; this allows more traffic to flow. In practice, those three corridors constitute the paths with the highest passenger and freight demand.

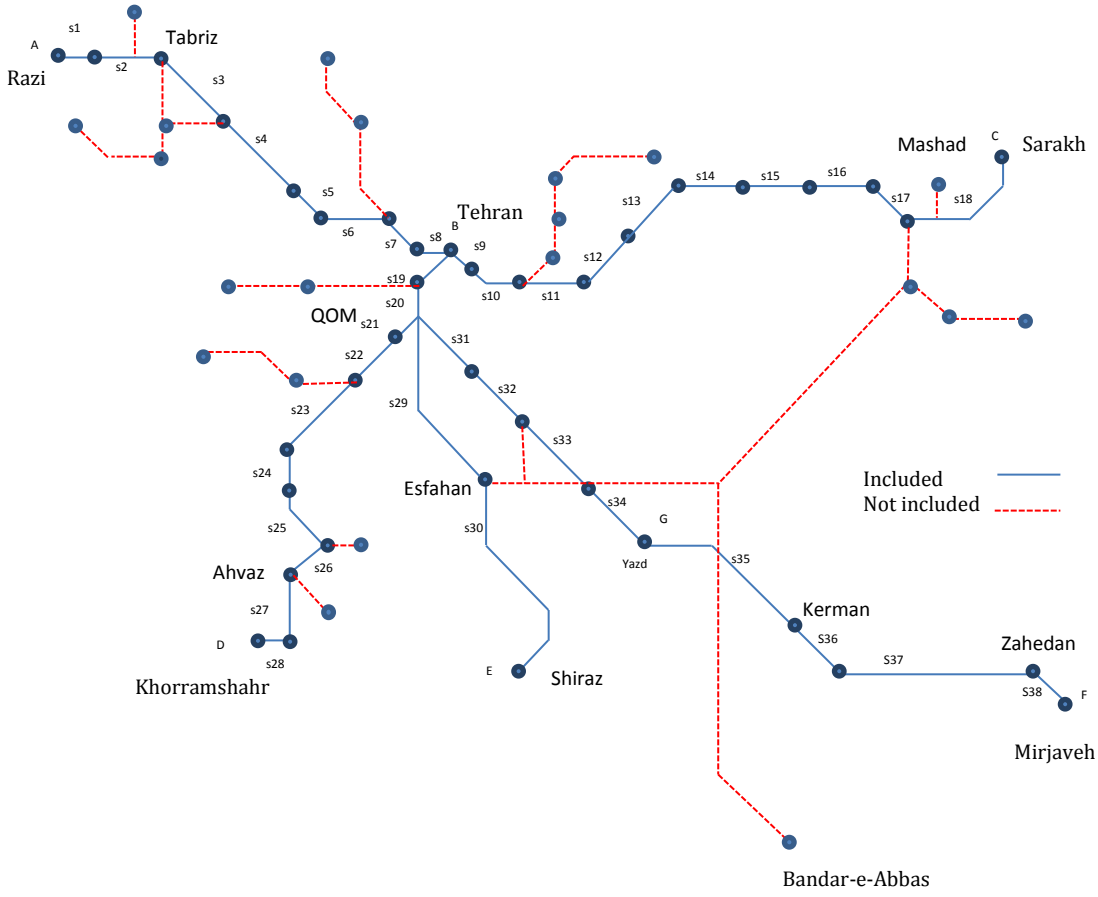

Figure 1-1: Network diagram for the RAI

Table 1-1: Network data

#	Corridor	# Sections	Length(km)
1	A-C	20	25.5
2	A-D	18	22.95
3	A-E	16	23.2
4	A-F	18	31.65
5	C-D	22	26.25
6	C-E	20	26.5
7	C-F	22	34.95
8	D-E	14	19.95
9	D-F	16	28.4

Table 1-2: Section information (i.e. lengths)

#	L	#	L	#	L	#	L
1	0.75	12	1.5	23	2	34	2.1
2	1.9	13	1.1	24	1.2	35	4.2
3	2.1	14	1.5	25	1.4	36	1.5
4	2.2	15	1.5	26	1	37	4.5
5	1	16	1.5	27	1.5	38	1
6	1.5	17	1.5	28	0.65		
7	1	18	2.8	29	4.4		
8	0.65	19	1.3	30	5.7		
9	0.5	20	0.7	31	1.65		
10	1	21	0.7	32	1.5		
11	1.5	22	1.4	33	2.1		

1.3 RESEARCH AIMS

The methods used to evaluate railway capacity include analytical models, simulation, and empirical analysis. This thesis specifically focuses upon the development of a set of comprehensive analytical mathematical models for railway capacity expansion planning and bi objective capacity identification. Analytical models may allow railway planners to measure the performance of railways and to assess the impact of changes to the railway network more quickly.

Capacity expansion model(s) are important and necessary as they determine how to improve system capacity, i.e. they provide optimal cost effective expansion plans. Traditional capacity models only determine what the current system capacity is (Burdett & Kozan, 2006). All railways, however, need to be expanded to meet new and increasing demands. The capacity expansion models fall into two categories, namely, static and time varying. The static expansion model can determine what expansions are necessary to maximise the absolute capacity of a railway network and what is necessary to achieve a specified improvement. The time varying models determine when changes can and should be made. The purpose of these models is to simplify the planning process, which is currently empirical and case based. For example, the current process requires a large amount of manual effort to set up a number of appropriate courses of action. These are then evaluated (i.e. by simulation) to ascertain possible/likely performance. In contrast, the mathematical models can inherently compare all possible courses of action and can select the best from the set of feasible courses of action.

The other main goal of this study is to examine a bi-objective capacity model to address the performance of a railway network by considering the competition between different train types and services, i.e. passenger and freight.

The capacity of the railway network is affected by factors such as train speed, commercial stops, train heterogeneity, distance between railway signals, sectional running times, directional and proportional distribution of trains, and timetable robustness (Abril et al., 2008;

Burdett & Kozan, 2006). Hence, in order to analyse the effect of these factors on network capacity, it is essential to know how railway parameters influence theoretical capacity, for example, by applying a sensitivity analysis. This thesis demonstrates various sensitivity analyses of capacity.

In summary, this Master's project had the following objectives:

1. Identification of improvements and modifications to capacity models
2. Development of a static capacity expansion model for planning purposes.
3. Development of a time varying capacity expansion model for planning expansions
4. Development of bi-objective capacity models for railway capacity analysis
5. Development of sensitivity analysis techniques for railway capacity analysis

1.4 SIGNIFICANCE AND ORIGINALITY

Transportation authorities make decisions on infrastructure expansion projects that cost billions of dollars annually. The net effect of those decisions will greatly affect users' travel time and safety. These decisions are not reached quickly and much consideration is required. This thesis and the models presented within are very useful, because they can significantly reduce the time and effort of decision making activities that planners and managers need to perform. The models can provide huge benefits because they can also identify the optimal course of action from within the set of all possibilities, whereas existing manual techniques can only analyse a small subset of possible solutions.

The aim of this thesis is to provide usable analytical models and methods that will allow planners in both government and industry to maximise the capacity of railway networks. This thesis has identified that analytical capacity models can be extended for the task of optimally expanding railway networks which is novel. The other originality of this thesis is identifying that introduced model can be used to perform a sensitivity analysis of capacity for competing railway capacity metrics.

An additional benefit of the capacity models that were introduced in this thesis is that they can identify what infrastructure expansion and spending is necessary to obtain specified levels of capacity. In that situation, the capacity model also provides a plan. The novelty of the proposed bi-objective models for railway capacity analysis is that two different objective criteria can be included in the decision making process. In contrast, the original model considers only a single objective, but that is not realistic or useful in practical situations. The bi-objective models are novel and original because they can equitably regulate the competition for rail infrastructure for many different types of capacity and performance criteria.

The capacity of railway networks has typically been identified using simulation in previous research. In this thesis, and in contrast, an analytical model has been utilised to automate and speed up the capacity analysis process.

The models developed in this thesis were implemented using the ILOG CPLEX optimisation software, after first being translated to the OPLStudio modelling language. The solution approach for the proposed bi-objective capacity analysis, however, requires the repeated solution of the optimisation model and this is not a standard task in CPLEX because of the iterative nature of the solution process. Programming languages such as C++ are well suited to this task, however they require special libraries and compilation to interface with CPLEX or similar optimisation tools. In this thesis, programming languages were avoided and an underutilised aspect of the CPLEX framework, namely, CPLEX scripting was investigated instead. CPLEX scripting was successfully implemented and no computational issues were encountered with the scripting functionality. This is quite important for practitioners and industry professionals. The use of the CPLEX scripting facility means that the entire solution and analysis process can be kept within a single software package.

1.5 THESIS OUTLINE

This thesis consists of seven chapters; each chapter presents a section of the work that has been completed during a course of one year, and details of these chapters have been summarised below:

1. Chapter 1 represents a background and the scope of research. This chapter consists of research aims, significance and originality, and the thesis outline. The case study that has been used in this thesis is also discussed.
2. Chapter 2 analyses the strengths and weaknesses of past research for railway capacity analysis, railway capacity expansion, and also addresses other topics.
3. Chapter 3 characterises the underlying core “base” capacity model. This chapter is divided into three main subheadings, core capacity model, alternative measure of capacity, and preliminary sensitivity analysis of capacity.
4. Chapter 4 considers railway capacity expansion. Train speed alteration was first considered as a means to improve capacity. More general capacity expansion models were then proposed, involving the selection of which sections to duplicate. The models are then applied to the case study.
5. Chapter 5 considers the development of bi-objective capacity models. It first reviews the multi-objective theory. A variety of models that regulate competition between different capacity metrics are then developed. This chapter

also presents two different solution techniques to solve the proposed bi-objective models. Finally, the numerical results are reported.

6. Chapter 6 reports the conclusions and discusses future directions.

Chapter 2 Literature Review and Analysis

This chapter examines the literature on various topics that are related to this thesis. It consists of seven sections. Section 2.1 begins with a broad overview of network design, as railway expansion is a form of network design problem. Network design problems (NDP) are generally considered in one of three different ways.

In practice, railway network improvements are judged in many different ways. In Section 2.2, two well-known improvement objectives, reliability and accessibility, are discussed. These are reviewed as they are important objectives, even though capacity increases are focussed upon in this thesis.

In practice, there are many different types of railway network. In Section 2.3, different railway network types are discussed. Those networks have different demands placed upon them and they face different challenges. Their goals and objectives are different and that is important to examine, as this thesis considers multi-objective capacity modelling.

Next, railway capacity analysis techniques are reviewed in Section 2.4. In this section, UIC and scheduling techniques are reviewed because they are two prominent techniques. In the Section 2.5, the topic of capacity expansion within railway networks is reviewed. In Section 2.6, multi-objective optimisation models are discussed. Finally, the key findings of this chapter are summarised.

2.1 NETWORK DESIGN PROBLEM (NDP)

The network design problem (NDP) is fundamental and occurs in many industries. This thesis will focus upon the railway network. NDP occurs in many applications in transportation, logistics, and production line (Yaghini and Akhavan, 2012). Most NDPs are solved by operations research and decision making techniques. NDPs can be divided into strategic, tactical and operational techniques (Yaghini and Akhavan, 2012). These three techniques are discussed below:

Strategic planning problem (SPP) considers how to improve an existing system and does not consider how to create a new system. SPP has been applied to multimodal networks, for example (SteadieSeifi et al., 2013,Alumur and Kara, 2008, Alumur et al., 2012, Gelareh et al., 2010). According to UNECE (2009) multimodal transportation is defined as the transportation of goods by a sequence of at least two different modes of transportation. SPP models include single-

allocation, multi-allocation, r-allocation, hierarchical allocation hub-and-spoke. The following solution techniques have been used: exact, approximation, heuristics, meta-heuristics, and hybrid heuristics.

Operational planning problems (OPP) aim to find the best choice of services and associated transportation modes, best routes and allocation of resources to demand (SteadieSeifi et al., 2013). The main characteristics of OPPs are that they deal with stochasticity and dynamicity. Operation research scholars offer OPP models that include 1) Resource management and the distribution of all resources throughout the network, 2) Itinerary replanning that focuses on real time optimisation of schedules, modal routes, and relevant response to operational disturbance. OPP and SPP solution methodologies in operation research techniques are the same for exact, approximation, heuristics, meta-heuristics, and hybrid heuristics.

Tactical planning problems (TPP) aim to optimally utilise existing facilities, allocate capacities, plan frequencies, and direct or consolidate loads. TPP focus on system costs, operation times, network structure, and customer requirements. This planning problem has offered different models with various solutions (SteadieSeifi et al., 2013).

Although, these literature studies all illustrate the importance of operation research and decision making in NDP and introducing solutions, there are still other opportunities for future research to introduce a comprehensive, efficient and reliable network. This research will focus on SPP, because the aim is to improve existing railway networks through an analytical approach.

2.2 RAILWAY NETWORK IMPROVEMENT

This section considers two measures of railway network improvement, namely reliability and accessibility. These two are discussed below.

2.2.1 Reliability

The attraction of having efficient, safe, and reliable transportation systems is that they provide a better service for passenger and freight services and they increase the motivation for the usage of railway. Mathematical and theoretical models are suggested to improve transportation networks (Daunoras et al., 2008, Junevičius and Bogdevičius, 2009, Junevičius and Bogdevičius, 2007, Beasley and Christofides, 1997). There are four concepts of reliability namely reliability of analysis mainly contains four aspects: travel time reliability, connectivity reliability, capacity reliability and performance reliability (Shariat, and Mohaymany and Babaei, 2010).

Wakabayashi and Iida (1992), Bell and Iida (1997), and Sumalee and Watling (2003) all describe methods to analyse the connectivity reliability of transport networks. Connectivity reliability in these models considers the probability of whether a pair of nodes in a network is still

connected or not. In general, the probabilistic and binary state of each link is assumed by all of the literature, which means if a link is operating it is equal to 1, and if not operate in the model will equal to 0.

Travel time reliability has been defined as the probability that a trip between a taken Origin-destination (OD) pair can be completed within the scheduled time (Bell, 1999). Chen et al. (2002a), examined that this measure is useful for evaluating network performance under normal routine flow variations. Another important approach was by Asakura (1999), who defined travel time reliability as a function of the state of degradation. In particular, he claimed that travel time reliability is a function of the ratio of travel times under degraded and non-degraded states (Asakura, 1999). The travel time budget is another aspect of network reliability that has been recently introduced by Lo et al.(2006), and Siu and Lo (2008).

Chen et al., (1999) defined a new measure of network reliability called capacity reliability. Capacity reliability of a network was identified as a new challenge. The probability of a network that can accommodate a certain traffic demand at a required service level is a critical concern (Chen et al., 1999). Chen et al.(2002b) presented the only method in computing capacity reliability, which was a basis for the Monte Carlo simulation. Another measure for reliability in a network is behavioural-related reliability. Yin et al. (2004), considered this measure between each OD pair for different classes of commuters, as the total misuse of commuters on the basis of minimum expected disutility.

2.2.2 Accessibility

Accessibility is defined as the potential for interaction and exchange (Hansen, 1959). This means accessibility provides access to obtain goods, services and activities. The measurement units of accessibility can be trips and generalised costs. Accessibility considered in different modes includes: automobile, truck, transit, cycling, and walking. Consumer benefits of accessibility were considered by maximum transport choice and cost efficiency (Litman, 2003).

Meersman and Voorde (1999) proposed relationships between modal choice and prices with the goal of defining the system's accessibility. Michel et al.(2001) then worked on evaluative analysis of infrastructure projects and transport policies. The work has been continued by geographers and planners who focused on the question of the optimal location of intermodal platforms. They defined measures of spatial accessibility.

Thomas et al (2002) performed a comprehensive review of publications, which have occurred since the early 1970s. Although quantitative analysis and measurement of accessibility is not a new topic, it remains a challenge today for spatial data analysis (Fotheringham et al., 2000). Furthermore, there is little if any literature that considers the supply of transportation capacity and multimodal approaches (Thomas et al., 2002). A question concerning whether the

methods used for passengers can be applied in freight transport is investigated (Thomas et al., 2002). They considered different inputs into the accessibility measures, consisting of distance, direct costs and time congestion as well as total population and economic activity, as weighting factors when looking at accessibility analysis being a strong tool for integrating network characteristics. While accessibility is a critical component of transportation network, it is not the focus of this research. However, the accessibility of railways has been indirectly improved by capacity expansion models in this thesis.

2.3 RAILWAY NETWORK TYPES

Railways are used for different purposes. For example, predominately railways are used for the movement of passengers and freight. These two types can be facilitated by different networks. In this subsection, freight networks are discussed in greater detail, as are high-speed rail networks, as they are a controversial topic and have the potential to greatly increase railway capacity.

2.3.1 Railway Freight Network

Logistics and transportation are vital for companies and governments nation-wide, particularly when the cost of a product is largely due to transportation costs (Crainic and Laporte, 1997). Freight transportation is an important process, both between cities and within cities, with road transport being the most developed to date. Consequently, the extensive overuse of road networks in any freight system causes issues including: traffic congestion, increased energy consumption, and negative environmental and safety impacts (Yamada et al., 2009, Pazour et al., 2010, Caris et al., 2013, Maia and do Couto, 2013, Janic, 2007, Woodburn, 2003). Scholars have attempted to develop efficient multimodal and intermodal transport systems to alleviate these issues (Yaghini and Akhavan, 2012, Zhang et al., 2013, Maia and do Couto, 2013, Priemus, 1999, Taniguchi and Shimamoto, 2004). The usage of at least two different modes of transportation to transport goods from origin to destination is defined as a multimodal transportation; the unit of goods can be a box, container, or a swap body (SteadieSeifi et al., 2013). Intermodal transportation is a type of multimodal transportation, however it deals with containerised freight (Crainic and Kim, 2006). Transport networks significantly support the economics of cities, regions, and countries, and reduce negative impacts on the environment and energy consumption. The development of freight systems in developing countries is important, but they are typically a road based system. However, most developed countries need to improve their existing transport system by using new technology and theory. This research focuses on a key sector of the rail freight system.

Freight transportation networks can be improved in both developed and developing countries. According to Maia and do Couto (2013) freight is a much more complex topic, hence it is considerably harder to model, and more difficult to use when collecting relevant data. Congestion in interstate and village transportation networks is extensive and wastes significant amounts of fuel, and billions of dollars per year in lost productivity, or stalled cargo (Officials, 2007b). For example Australian road networks also experience significant highway congestion due to excessive truck volumes, and this significantly impacts upon the Australian economy. Infrastructure Australia conducted a nation-wide call for submissions to identify 'priority' infrastructure projects in 2009 (Infrastructure Partnerships Australia, 2009). Therefore, the government prioritised infrastructure projects that need to be financed are they are estimated to be of the value of 169 billion dollars (Australian Government, 2008). Road and rail projects accounted for more than half of the critical projects, totalling approximately 90% of the total listed (Infrastructure Partnerships Australia, 2009).

There is a rapid growth in freight transportation, due to increased demand. The Queensland Government claims that demand will increase from 871 million tonnes in 2010-2011 to 1,741 million tonnes within 12 years (Informa Australia, 2014). Queensland's exports in coal have first rank in freight movement with 84.2 per cent of its outward international trade in 2012-13, also metal ores and scrap comprised 7.6 per cent. There is a sharp growth in demand for Queensland's mining commodities from China, India and other economies, that causes expansion of freight requirements significantly (Informa Australia, 2014).

The high demands of freight transportation for domestic and industrial users force governments and those in the goods supply chain (logistics) to adopt new technologies in order to address increasing highway and rail congestion. Commuter rail patronage has also increased sharply, creating further congestion on urban railways. Freight train operations may be limited by the types of 'curfews' already in place, in the major cities of Sydney and Brisbane (Australian Government, 2011). These problems call for universities and researchers to take action in order to remedy the situation.

Extensive research that focuses upon environmental, economic, and social issues in freight transportation has been performed in recent years, namely Apivatanagul and Regan, 2010, Caris et al., 2013, Chang et al., 2008, Contreras and Fernández, 2012, Current et al., 1990, Ferreira, 1997, Lin et al., 2012, Magnanti and Wong, 1984, Pazour et al., 2010, Racunica and Wynter, 2005, Yaghini and Akhavan, 2012, Yamada et al., 2009, Zhang et al., 2013). Marin and Almeron (1996) developed models for planning rail freight networks. Trains are assigned to services and cars are assigned to trains. Xu, He, Song, Li (2009) considered transport assignment in a constrained stochastic flow network. The decision problem is to assign freight transport to different routes in order to maximise reliability and minimise cost, and subject to capacitated arcs. The model was solved using neural networks, genetic algorithms and stochastic simulation.

High-speed rail systems are systems where trains have the ability to travel beyond 150 Km/h (Pazour et al., 2010). HSR for passenger transportation has received much attention in the literature; however in contrast freight transportation has received little or no attention. The opportunities for a high-speed rail freight transportation network still exists in countries like Japan, China, Europe, and USA (de Rus and Nombela, 2007, Pazour et al., 2010). Australian high-speed rail passenger transportation has been investigated (Hensher, 1997, Givoni, 2006, Gunn et al., 1992), but a high-speed rail network has not yet been researched for freight.

In summary, the increasing demand in the freight transportation network could be handled by a separate network to passenger services, however, it is so expensive that it is not viable. Hence, freight trains must use the same network as passenger trains. Freight trains usually traverse at off-peak times because there are fewer passenger trains on the network (i.e. late at night and in the middle of day) and because passenger trains have priority at other times.

2.3.2 High-Speed Rail (HSR)

The attractions of proposing high-speed rail in railway transportation are reduced travel times and the potential of increased numbers of trains. Congestion has been a considerable concern on the interstate transportation networks, as well as costing billions of dollars per year in lost productivity, stalled cargo, or wasted fuel in freight transportation (Officials, 2007b). The European Union, China, and Korea are the most famous countries where the high-speed train is used (Ryder, 2012). The Department for Transport (2011) in the UK claims a high-speed line between London and Birmingham would cost £32 billion to build, and generate benefits of around £44 billion in revenue and savings in travel times, also allowing "the UK's key urban economies to improve their productivity, make new business, and access more directly the economic strength of London and the South East" (Ryder, 2012). High-speed rail is said to have increased the amount of tourists visiting the city of Lleida (a city in Spain) by 15% and attracted new industries (Ryder, 2012). HSR technology is defined as a solution to congested roads and rail (de Rus and Nombela, 2007). Pazour et al (2008) introduced a planning problem that chooses whether HSR is used on specific arcs of a transportation network. The objective is to minimise the total time to accomplish a set of trips. Each arc has a capacity limitation. The mathematical formulation for this problem is non-linear. Their approach shows that adding an extra section in the HSR network could be beneficial.

Although HSR is very effective, it does not come free. For example, the costs of HSR are huge. HSR costs are from 9-40 million euros with an average of 18, and greater value over difficult terrain. A high-speed rail network with average speeds of 140-150 mph connecting 33 major cities in USA would cost well over $500 billion. In Australia, the cost of HSR infrastructure approximately is 30 million dollars / km. These prices are twice the cost of the

Interstate Highway System; such a true high-speed rail system would provide less than 1/10th the mobility offered by the interstate systems. It is claimed that these costs include only the projected capital costs (O'Toole, 2010). Levinson et al. (1997) claimed that high-speed rail requires huge amounts of expenses compared with expanding the existing air service, and is marginally more expensive than auto travel. Givoni (2006) also stated that high-speed trains required the high investment of infrastructure and its economic development benefits could not be justified.

Chang et al. (2000) considered the planning of passenger train services on a future high speed rail line. The problem considered a multi-objective of minimal total operating cost (i.e. for the rail operator) and minimal total travel time loss (i.e. for passengers). The solution consisted of a plan of stops, a frequency of travel for each service, and the fleet size. Passenger volumes were taken into account. This study stated that HSR operation requires much more coordinated planning than traditional rail systems in order to provide high-volume, high-frequency passenger services. Hence, in this thesis, the scope of our research is increasing the capacity of a railway network by focusing on the existing system, not by changing the rail with high-speed rail.

2.4 RAILWAY CAPACITY ANALYSIS

The capacity of railway networks has been discussed extensively in the literature and there is an abundance of research on the topic. In this sub-section, these approaches will be discussed. Capacity is highly dependent on infrastructure, traffic, and operating parameters (Abril, et al., 2008). The integration, coordination, and planning of the different key components is typically concentrated upon. For instance, Burdett and Kozan (2004) discussed the details behind the application of train timetabling in order to identify the capacity of a railway network. In that research, interference delays were ignored and this considerably reduced the complexity of the timetabling problem. Burdett and Kozan (2006) followed up their previous research from (2004) and developed analytical capacity determination models. In particular, they identified how the mix of different train types affects the absolute capacity of a railway network. Their proposed mathematical optimisation model takes into account the percentage mix of trains, the direction of travel, dwelling times, train speeds, and the presence and position of signals.

Pachl and White (2004) considered capacity management of railways and proposed blocking time concepts. This is just another way of measuring total train occupancy time on each section. This paper provides a precursor to train scheduling via a machine scheduling approach.

Mussone and Wolfler Calvo (2013) proposed an analytical model, which is able to calculate the capacity of a railway system without decomposing it into sub-problems. An advantage of their approach is that it systematically analyses the whole system and identifies all underlying bottlenecks. Their approach has been applied to a real railway network in Germany.

Landex (2009) evaluated the single track operation of railway networks by using the UIC 406 capacity method. This method is suitable for estimating the capacity consumption when an unscheduled single track is operating. The scheduled capacity consumption can be easily indicated for network capacity statements on maps by the UIC 406 capacity method. Landex (2009) claimed that it is important to identify where to divide the railway lines into line sections and how the conflicts between trains at crossing stations and junctions needs to be analysed.

The challenge of capacity allocation on a railway network was addressed by Gibson (2003). This research identified the key features of rail timetables and track access rights and introduced three basic methodologies for allocating capacity on the rail network, such as administered, cost-based, and market/value-base. Lusby, Larsen, Ehrgott, and Ryan (2011) provided a broad overview of the different techniques for routing trains in railway networks in order to maximise capacity utilisation. In summary, solving these types of problems, results in large formulations. Furthermore, these approaches are inflexible because additional train paths cannot be included easily.

Sectional running time (SRT) is an important component that affects the capacity of a railway network. Kozan and Burdett (2005), Vromans, Dekker, and Kroon (2006), and Harrod (2009) have shown that SRT has a significant impact on capacity analysis, for instance, a slight change in SRT can have a remarkable effect on the amount of capacity.

De Kort et al (2003) proposed an approach to identify railway capacity, which is also based upon the logic of bottleneck analysis. The main idea behind their approach is the utilisation of a generic building block. This allows complex railway networks to be simplified for the analysis process. Though useful, this approach has a number of limitations. For example, it does not explicitly distinguish between trains of different types. It was, however, reported that this aspect can be incorporated by imposing a distribution on the different train types and weighting the result appropriately. The approach was applied to a case study of a high speed railway line in the Netherlands.

Lu et al (2013) considered the development of a framework for evaluating the performance of railway networks. They proposed a “quality of service” framework that includes attributes like punctuality, resilience, energy/resource usage, journey time, etc. An empirical analysis of train delays was performed to demonstrate the evaluation of a number of key performance indicators (KPIs).

2.4.1 UIC Method

This is another prominent capacity identification technique, as observed by the number of articles in recent years that use it. The UIC method is applied to an existing schedule. It has been described as a compression technique, because it seeks to compress the schedule as much as

it can, after which the schedule's final completion time is used to specify the capacity. A weakness of the UIC is that a timetable is always required. This is also an advantage because it also takes into account the way a particular railway system is actually used. In essence, the Burdett and Kozan (2006) method and the UIC method are similar because both result in the saturation of critical sections of rail.

2.4.2 Train Scheduling Techniques

Train scheduling is briefly discussed in this section because it can also be used to identify railway capacity. By scheduling trains with a makespan objective (e.g. where makespan is defined as the length of a schedule), the tracks can be utilised optimally. Comparing the number of scheduled trains to the makespan time duration gives the capacity. It provides the actual capacity as opposed to absolute capacity because it describes a detailed plan of actual train movements over time, rather than a rough estimate. In other words, it explicitly handles train interactions whereas absolute capacity models do not. Although this method is most accurate, this is one of the most challenging and difficult problems in railway research. It is also a complex and time consuming task in practice. There are many noteworthy articles that have focussed upon this method: Jovanović and Harker (1991), Kwon et al. (1998), Kraay and Harker (1995), Boysen et al. (2011), and Peng et al. (2011), Adenso-Diaz et al. (1999), Billionnet (2003), Brannlund et al. (1998), Cai and Goh (1994), Cai et al. (1998), Carey (1994a), Carey (1994b), Carey and Carville (2003), Carey and Crawford (2007), Carey and Lockwood (1992), Chiang et al. (1998), Cordeau et al. (1998), D'Ariano et al. (2007), Dessouky et al. (2006), Dorfman and Medanic (2004), Goverde (2007), Kroon and Peeters (2003), Kroon (1997), Lindner (2000), Mazzarello and Ottaviani (2007), Odijk (1996), Rodriguez (2007), Sahin (1999), Stella et al. (2006), Tornquist and Persson (2007), Zwaneveld et al. (1996), Zwaneveld et al. (2001).

Burdett and Kozan (2009) extended a discrete sequencing approach for train scheduling. They addressed this phenomena through two advanced features of scheduling namely, by incorporating essential composite perturbation operations and by restricting unnecessary multiple overtaking. Their model addressed multiple overtaking conflicts, which can occur when two trains travelling in the same direction pass each other successively.

Burdett and Kozan (2009) proposed a sequencing approach for train scheduling on parallel lines by crossover points. Compound buffers were introduced and are beneficial because the movement between lines is made possible without the need to make additional routing decisions. They analysed different parallel track topology and operating protocols. Their research is followed by creating completely new train schedules that are more accurate and efficient than permitted by current techniques (Burdett, & Kozan, 2010). Their research significantly addressed

the representation and construction of accurate train schedules by a hybrid job shop approach. They claimed that acceleration and deceleration significantly affects the quality of a schedule.

2.5 RAILWAY CAPACITY EXPANSION

Capacity expansion is an important topic for various types of system. Capacity expansion in transportation networks is necessary for many reasons. For example, Americans lose 3.7 billion hours and 2.3 billion gallons of fuel while sitting in congested transportation modes, and waste 9.4 USD billion annually as a result of airline delays (The American Association of State Highway and Transportation Officials, 2007). The American Association of State Highway and Transportation Officials (2007) stated that imports and exports will provide approximately 35 percent of the country's GDP by 2020 and 60 percent in 2030. Consequently, having an improved and expanded transportation network is vital.

Railways are a significant mode of transportation. They handle a significant amount of traffic and need to be expanded often. Railways are reported as essential for intermodal and bulk freight transportation, particularly for items such as automobiles, coal, and ore. It is also been highlighted that expansion of a rail line represents a permanent, high-cost investment that will be stranded if business needs change, and/or high cost and risk of expansion limits the railroads' ability to 'scale up' capacity to meet shifting demand. Hence, bottlenecks occur across the country, mainly in and out of ports, around cities, and near the intersections of different railroads (The American Association of State Highway and Transportation Officials, 2007). The importance of rail transportation is obvious, however future demands are great and the optimal investment of funds for railway network expansions is necessary. Maggi, Masser and Nijkamp (1992) looked at missing rail networks in Europe and subsequent lack of capacity that arises from their absence. That article provides significant motivation for railway network expansion activities. The article of Lai, Shih, and Jong (2010) is noteworthy, as they suggested a capacity model to evaluate network capacity and to help planners by introducing possible expansion option to determine the optimal network investment. The model assumes the entire network is double tracked. They ignore meetings between trains and assume that they occur within sections. Also, the impacts and causes of heterogeneity with freight and passenger traffic are determined by using dispatch simulation software. The impact of different types of heterogeneity on implications for capacity planning was studied to introduce more effective planning and efficient rail operations (Dingler, Lai, & Barkan, 2009). The paper of Shih, Dick, Sogin, and Barkan (2014) is also noteworthy because they compared different capacity expansion strategies for single track railway lines. For example, they considered the effect of placing additional sidings and the extension of existing sidings. They also considered track duplications in a limited way.

This paper chose the best expansion strategy by performing an efficiency and reliability analysis using simulations.

For systems other than railways, the recent thesis by Singh et al. (2012) is notable. They developed a generic mixed integer linear programming (MILP) capacity expansion planning model to identify infrastructure improvements for optimising a bulk material supply chain. It was necessary to use meta-heuristics because that model was computationally intractable.

Some support tools have been introduced for capacity management. Krueger (1999) introduced a practical parametric capacity model to increase the efficiency of track movements. This model identified bottleneck areas, and other places where congestion occurs. Another decision support tool was introduced to allocate capital investment, toward determining the best possible way for optimal capacity planning (Lai & Barkan, 2011). This decision support tool can determine the optimal investment plan by maximising the return on investment from capacity expansion plans. Therefore, it will enhance the ability of stakeholders to provide reliable service to customers.

This thesis builds upon the analysis performed in Yaghini, Nikoo, Ahadi (2014). In that paper the impact of different train types were investigated for the Tehran-Zanjan corridor in Iran. They found that capacity tends to increase nonlinearly and the mix of train types reduces the railway line capacity. In contrast, the entire Iranian network is considered in this thesis, and also capacity increase via infrastructure expansion has been considered.

A number of articles have previously investigated train speed as a means of expanding capacity (Dingler, et al., 2009; Fröidh, Sipilä, Warg, 2014; Harrod, 2009). Buri and Tzieropoulos (2009) investigated variables that affect the speed of trains and assessed the relative importance of those with respect to railway capacity. They reported that train speed and signal position have a great impact on capacity, and this was shown in the new signalling and management system for the European rail networks (ERTMS) (Buri & Tzieropoulos, 2009). Increasing the power of trains and making trains more homogenous were found to be promising possibilities as infrastructure expansions were deemed perhaps too expensive. Fröidh, et al. (2014) considered the imbalance in trains' speeds on Scandinavian rail networks that causes reductions in capacity utilisation. A timetable analysis and some simulations were performed. In those analyses, trains speed increments were considered.

Goverde et al (2013) have assessed a Dutch railway corridor with different signalling configurations. An assessment of signalling and train dispatching systems was made. The methodology used in the analysis is an extension of existing capacity assessment methods, such as the UIC 406 compression technique, and a new capacity determination approach was not proposed.

2.6 MULTI OBJECTIVE OPTIMISATION

The capacity of railways depends on many different parameters and criteria. Railway capacity analysis under these circumstances can be facilitated by multi-objective optimisation models. In this section, research involving multi-objective optimisation has been reviewed.

Zhou and Zhong (2005) stated the potential of using multi-objective scheduling methods for generating Pareto solutions and for railroad timetable planning application purposes. The objective functions are minimising the expected waiting times for high speed trains; as well as minimising the total travel times of high speed and medium speed trains. Their proposed model has the ability to optimise and balance travel time and the waiting time of existing trains. They also considered both acceleration and deceleration. These results demonstrate the importance and benefits of a multi objective scheduling models. In the concept of scheduling Ghoseiri, Szidarovszky, and Asgharpour (2004) also introduced a multi objective train scheduling model. Their focus was on the passenger train scheduling problem on a railroad network. The results of their proposed model was minimising the fuel consumption cost and minimising the total passenger travel time, which satisfies both company and passengers respectively.

Railway traffic management is an important criterion for dispatchers. Railway conflict detection and resolution are of most concern to dispatchers. Corman, D'Ariano, Pacciarelli, and Pranzo (2012) introduced a bi-objective model, in which the objectives minimise the consecutive delays between trains and maximise the total value of satisfied connections. They claimed that small changes in a limited set of connections can have a serious impact on delay time.

Effective planning of railway networks and train services is considered in literature. Chang et al. (2000) introduced a multi-objective model for passenger train service planning. The model presented is for minimising the total operating cost of the operator as well as minimising the total travel time loss of the passenger. This model can generate a best compromise train service plan, including the optimal stop-schedule plan, service frequency, and fleet size. It is claimed that the best planning outcome is the optimal plan not constrained by specific stop-schedules. Their research also conducted an empirical study on Taiwan's HSR line.

Ghoseiri et al (2004) developed a multi-criteria optimisation approach to scheduling trains. The competing objectives were fuel consumption and total passenger time. The Pareto frontier was determined using the e-constraint method and then a "distance" based method was utilised to solve the multi-objective decision problem. Twenty-one modest-sized test cases were solved.

The articles by Kim and De Weck (2005, 2006) are generic and do not consider topics in railways. Those papers are, however, noteworthy for the development of multi-objective optimisation techniques. In their 2005 paper, a method was developed for determining the Pareto

front for bi-objective optimisation problems. Their approach is an adaptive version of the WSM, and is labelled AWS. It approximates a Pareto front by gradually increasing the number of solutions on the front. It concentrates computational effort where it is needed. In the 2006 paper, multi-objectives were considered. A mesh of Pareto front patches was identified and refined.

Chang et al (2000) considered the planning of passenger train services on a future high speed rail line. The problem considered the multi-objectives of minimal total operating cost (i.e. for the rail operator) and minimal total travel time loss (i.e. for passengers). The solution consisted of a plan of stops, a frequency of travel for each service, and the fleet size. Passenger volumes are taken into account.

Goverde et al. (2013) proposed a thoughtful assessment of railway line capacity consumption of different railway signalling systems under scheduled and disturbed conditions. They introduced a new concept of dynamic infrastructure occupation to assess infrastructure capacity under disturbed conditions as a complement to the established capacity indicator of scheduled infrastructure occupation. Infrastructure occupation has been calculated by a dynamic timetable compression method. Their main innovation is the quantification of the scheduled and dynamic infrastructure occupation, and the possibility to reduce delay propagation. Their methodology is able to extend the usual capacity assessment method in order to evaluate disturbed traffic conditions.

2.7 DISCUSSION AND KNOWLEDGE GAP

A variety of techniques have been used for capacity analysis. One of those techniques, called the UIC method, utilises and analyses an existing timetable. However obtaining a timetable is not always easy, and using a single timetable to base decisions upon is not necessarily valid. As capacity expansion does not involve an existing timetable, the UIC method is also not applicable. Therefore the research performed in this thesis is not based upon the analysis of a timetable. In retrospect it is unlikely that one could be provided for the case study of the RAI network.

It should be noted that train scheduling was not considered as a capacity analysis technique. It can be used, but that problem is NP-hard and cannot be solved to optimality; also the result is questionable. The result is the actual and not the absolute capacity. It is not easy to duplicate sections in train scheduling algorithms because this requires additional routing decision variables. A multi-objective analysis is also difficult because the Pareto frontier cannot be identified, and obtained solutions may not be Pareto optimal. The capacity models in contrast are far superior for higher level planning.

Absolute capacity determination has been focussed upon in this thesis. It is believed that it is more important for planning purposes, as it can identify an upper bound that is less open to disagreement and variation. Interaction effects and other operational conditions are variable by nature and clearly affect the real capacity. No information is available on those conditions in the RAI and hence the actual capacity is not computed.

An analysis of the literature has shown that the Burdett and Kozan (2006) capacity identification approach is quite prominent. The main reason for this is that it is conceptually easy to use. It provides an analysis of the structural properties (i.e. absolute capacity) of the network where the term structural property refers to the layout and the rail infrastructure present, like stations, sidings and signals. These are static and arguably affect the capacity most significantly. Absolute capacity is reduced, according to how trains are operated on the system. The Burdett and Kozan (2006) approach can also include that aspect, by incorporating empirical and stochastic information, and that is another of its virtues.

An analysis of the Burdett and Kozan (2006) approach has revealed that many advanced features can be incorporated and that it has the potential to incorporate many other important features. For these reasons, it has been adopted in this thesis as the starting point for all new work. The model is advantageous because it just requires the solution of a simple linear programming model. That model has one main constraint (i.e. section saturation). The rest are a percentage of mixed constraints, which can be added or removed at the user's discretion (i.e. sometimes they are not necessary). A multi-objective approach is easy to apply as it just adds additional objective functions. Capacity expansion is also easy, because only an additional decision variable is integrated.

This thesis has focused upon improving the capacity of existing railway networks. It is suggested that improving existing rail networks by traditional means is better than investing in HSR. In addition, exploiting existing infrastructure with new technology (such as tilt trains) is even better. Furthermore, building entirely new infrastructure in peripheral areas where extra capacity is not needed should be avoided.

Chapter 3 Underlying Model for Railway Capacity Analysis

The literature review and the preliminary sensitivity analysis in Chapter 2 indicated that a capacity analysis of railway studies demands more than the current models, which can provide an optimal and beneficial movement of rail on railway networks. Chapter 2 has also shown that capacity analyses are performed by different techniques and approaches. Simulation, analytical and empirical approaches are discussed and different models are explained. In Chapter 2 also, different ways of capacity expansion are reviewed, namely increasing speed and changing a normal train to a high-speed train. The advantages and disadvantages of different expansion approaches are highlighted.

In Chapter 2, the Burdett and Kozan (2006) capacity model was identified as a good model for railway capacity analysis. Hence, it has been chosen as the starting point for the aspirations of this thesis. In this chapter, that underlying model is first described. Alternative capacity metrics that could be used in this underlying model are discussed. A sensitivity analysis of that model is then examined and conclusions are made.

3.1 BASE MODEL

This thesis focuses on analytical capacity analysis. Chapter 2 examines Burdett and Kozan (2006) addressing the question of how the mix of different trains can be traversed to achieve maximum capacity in a railway network. The model developed a technique and methodology for maximising the capacity of a rail network. The research defined various operational conditions, such as the number of different trains, the distance, the direction of traverse, the length of arcs, the speed, and the number of nods, the dwelling time, and the signals of railways. The Burdett and Kozan (2006) model is discussed in this section as it will be extended in later chapters of this thesis. This model identifies the absolute capacity as opposed to real operational capacity. Hence it provides an analysis of the structural properties of a railway network.

The requirements of the model are: A list of the sections and corridors, the length of each sections, a list of the sections present in each corridor, the number of train types, the sectional running time, number of tracks in each section, the duration of the capacity planning period, the

specified percentage mix of trains required. The variables used in the model are summarised in the Nomenclature, however they are also defined here as they occur.

The objective of this model is to determine the number of trains that traverse all corridors, in both the forward and reverse direction, over a specified time period T.

Maximize: $\boldsymbol{AbsCap} = \sum_{c \in C} \sum_{i \in I} (\vec{x}_i^c + \overleftarrow{x}_i^c)$ (3-1)

In that equation $\vec{x}_i^c, \overleftarrow{x}_i^c$ defines the number of trains of type *i* which traverse corridor *c* respectively.

Primary constraint:

$\sum_{i \in I} \left(\vec{y}_i^s \vec{T}_i^s + \overleftarrow{y}_i^s \overleftarrow{T}_i^s \right) \leq T \times \tau_s \quad \forall s \in S$ [Section saturation] (3-2)

$\vec{y}_i^s = \sum_{c \in C | s \in \Omega_c} (\vec{x}_i^c)$ and $\overleftarrow{y}_i^s = \sum_{c \in C | s \in \Omega_c} (\overleftarrow{x}_i^c)$ [Section usage] (3-3)

Constraint (3-2) ensures that the flow through each section of the network must be less than or equal to the saturation limit. It should be noted that equation (3-3) is a mechanism to translate train numbers across different corridors to a compatible number across sections.

Train mix incorporation:

$\vec{x}_i^c + \overleftarrow{x}_i^c = \eta_i^c \sum_{j \in I} \left(\vec{x}_j^c + \overleftarrow{x}_j^c \right) \quad \forall i \in I, c \in C$ [Proportional mix] (3-4)

$\vec{x}_i^c = \mu_i^c (\vec{x}_i^c + \overleftarrow{x}_i^c) \quad \forall i \in I, c \in C$ [Directional mix] (3-5)

$\sum_{i \in I} (\vec{x}_i^c + \overleftarrow{x}_i^c) = \sigma_c \sum_{c' \in C} \sum_{i \in I} (\vec{x}_i^{c'} + \overleftarrow{x}_i^{c'}) \quad \forall c \in C$ [Mix across corridors] (3-6)

Constraint (3-4) ensures that the proportional mix of trains in each corridor is satisfied. Constraint (3-5) regulates the flow of trains in each direction. Similarly, constraint (3-6) regulates the flow of traffic between the different corridors.

Absolute capacity is not a single unique number and varies for different mixes of trains. The capacity model deals with this by utilising directional and proportional distributions that describe and regulate the percentage mix of trains on each corridor and in each direction. Both freight and passenger train services are included in the percentage mix. In complex networks there is high probability that a section will be part of several different corridors. These shared sections will be more saturated, i.e. heavily occupied.

The complete mathematical formulation is as follows:

Maximize: $\boldsymbol{AbsCap} = \sum_{c \in C} \sum_{i \in I} (\vec{x}_i^c + \overleftarrow{x}_i^c)$ (3-7)

Subject To:

$\sum_{i \in I} \left(\vec{y}_i^s \vec{T}_i^s + \overleftarrow{y}_i^s \overleftarrow{T}_i^s \right) \leq T \times \tau_s \quad \forall s \in S$ [Section saturation] (3-8)

$\vec{x}_i^c + \overleftarrow{x}_i^c = \eta_i^c \sum_{j \in I} \left(\vec{x}_j^c + \overleftarrow{x}_j^c \right) \quad \forall i \in I, c \in C$ [Proportional mix] (3-9)

$\vec{x}_i^c = \mu_i^c (\vec{x}_i^c + \overleftarrow{x}_i^c) \quad \forall i \in I, c \in C$ [Directional mix] (3-10)

$\sum_{i \in I} (\vec{x}_i^c + \overleftarrow{x}_i^c) = \sigma_c \sum_{c' \in C} \sum_{i \in I} (\vec{x}_i^{c'} + \overleftarrow{x}_i^{c'}) \quad \forall c \in C$ [Mix across corridors] (3-11)

$$\vec{x}_i^c, \overleftarrow{x}_i^c \geq 0 \;\; \forall i \in I, c \in C \qquad \text{[Positivity requirement]} \qquad (3\text{-}12)$$

$$\vec{y}_i^s = \sum_{c \in C | s \in \Omega_c} (\vec{x}_i^c) \text{ and } \overleftarrow{y}_i^s = \sum_{c \in C | s \in \Omega_c} (\overleftarrow{x}_i^c) \qquad \text{[Section usage]} \qquad (3\text{-}13)$$

The last constraint ensures that the decision variables, i.e. the number of trains of each type traversing each corridor, are positive.

Constraint (3-11) regulates competition across corridors, however it is necessary, and perhaps more realistic, that competition must be regulated between specific corridors with common sections. Hence in this thesis a new idea is put forth. For example a new parameter $\sigma_c^{c,c'}$ should be defined, which describes the % of trains that should use corridor c in comparison to the combined total that use corridor c and c'. Hence $\sigma_c^{c,c'} + \sigma_{c'}^{c,c'} = 1$ and the following constraint should be added:

$$\sum_{i \in I} (\vec{x}_i^c + \overleftarrow{x}_i^c) = \sigma_{c,c'} \sum_{i \in I} \left(\vec{x}_i^c + \overleftarrow{x}_i^c + \vec{x}_i^{c'} + \overleftarrow{x}_i^{c'} \right) \qquad (3\text{-}14)$$

3.2 ALTERNATIVE MEASURES OF CAPACITY

In this section, alternative measures of capacity are reviewed and formulated. Traditionally, capacity is measured in terms of the number of trains that traverse each corridor in a given time. However, in practice, there are different alternative metrics that are equally important and could be used in capacity analysis. In most scenarios the absolute capacity is defined as follows:

$$A = \sum_{c \in C} \sum_{i \in I} \left(\vec{x}_i^c \vec{M}_i^c + \overleftarrow{x}_i^c \overleftarrow{M}_i^c \right) \qquad (3\text{-}15)$$

Equation 3.15 assumes that there is a payoff of $\vec{M}_i^c, \overleftarrow{M}_i^c$ per train. In simple terms, the total payoff (i.e. the capacity) is increased as the number of trains is increased. Payoff means different things, in different scenarios, to different people at different times. For instance, the gross (total) output of freight could be measured. In that situation, each train type consists of a different number of wagons and each wagon carries a load of goods of an associated weight. In other scenarios, performance is based upon the number of containers or TEUs. Furthermore, in other scenarios it is the value of the goods transported by trains. In summary, the payoff type could be a count (i.e. number), weight (i.e. tonnage), or monetary value (i.e. AUD). Capacity can also be viewed in terms of network performance.

3.3 PRELIMINARY SENSITIVITY ANALYSIS OF CAPACITY

In this section, a sensitivity analysis has been performed. In that analysis, the model has been applied (i.e. solved) many times. It is important to note that three train types were

considered, whose speeds were 80, 100 and 120 km/h respectively. The analysis was also performed for a time period of one day, i.e. 1440 minutes. For three train types, and assuming increments of 0.01 (i.e. 1%), there is $C_{100}^{102} = \frac{102 \times 101}{2} = 5151$ possible ways of changing η^c such that $\eta_1^c + \eta_2^c + \eta_3^c = 1$. If there were only two train types, then the actual number of possibilities reduces considerably, i.e. to a value of 101. In general, the actual number is given by the following formula $C_N^{N+|I|-1} = \frac{(N+|I|-1)!}{(|I|-1)!(N)!}$ where $N = 100$ and $|I|$ is the number of train types. This formula and other related ones have been discussed in Burdett and Kozan (2003). It should also be noted that $\eta_i^c = 0$ is allowed here. This formula is derived by realising that there are 100 values that can be assigned to the $|I|$ groups plus $|I| - 1$ values of zero percent are possible.

In summary, the number of possible proportional distributions is very large and it is impractical to evaluate each one, i.e. by solving the optimisation model, although it is possible to do so. The question of what to do with all of these results and what they mean then arises. In conclusion, it is best to take an alternative strategy. In this thesis, it is proposed that individual corridors and individual train types are analysed. The proportional distribution and the percentage flow parameters can easily be altered to facilitate this. These results are shown in Table 3-1. These restrictions were then discarded and multiple train types were considered as was the capacity of the entire network. In the table below, "Mix" refers to an arbitrarily selected mix of trains, which is shown in Appendix B.

Table 3-1: Sensitivity analysis of absolute capacity

i	$\mathbb{A}$ (net)	$\mathbb{A}$ (individual corridors)										Utilisation (reduction)
		1	2	3	4	5	6	7	8	9	Sum	
1 (LB)	1641	685.71	872.73	336.84	426.67	685.71	336.84	426.67	338.64	426.67	4536.48	0.36(0.64)
2	2051.2	875.14	1090.91	421.05	533.33	857.14	421.05	533.33	421.05	533.33	5686.33	0.36(0.64)
3 (UB)	2461.5	1028.6	1309.1	505.26	640	1028.6	505.26	640	505.26	640	6802.08	0.36(0.64)
Mix	2095.5	945.38	1051.5	346.55	439.86	824.84	383.65	588.24	405.83	438.96	5424.81	0.39(0.61)

In Table 3-1, the results for train type 1 can be viewed as lower bounds on capacity, for instance for individual corridors and for the whole network, as that train is slowest. Furthermore, train type 3 results can be viewed as upper bounds, because it is fastest. Any proportional mix of trains must result in a value of capacity between the aforementioned limits.

The utilisation value (and the associated value of reduction) in the last column describes the relationship between the network's capacity as a whole with the total capacity of all the individual corridors together. Hence in this example, approximately one third of that sum is achieved. This means that the structure of the network does not allow individual corridors to be utilised fully and in isolation. In other words, they must share the network's lines with trains from other corridors. The extent of the interaction effects is quite high for the specified mix of trains. If the corridors were defined in different ways, then the utilisation and reduction values could be used to make a comparison with other alternatives.

3.4 CONCLUSION

In this chapter, the base model that used in this thesis is discussed and reviewed. In the base model, Burdett and Kozan (2006), it is shown that the capacity of the railway network is affected by factors such as train speed, train heterogeneity, distance, sectional running times, directional and proportional distribution of trains. In particular, they identified how the mix of different train types affects the absolute capacity of a railway network.

The base model takes into account the percentage mix of trains, the direction of travel, dwelling times, train speeds, and the presence and position of signals. The Burdett and Kozan (2006) capacity identification approach is quite prominent. The main reason for this is that it is conceptually easy to use. It provides an analysis of the structural properties (i.e. absolute capacity) of the network, where the term structural property refers to the layout and the rail infrastructure present, like stations, sidings and signals. These are static and arguably affect the capacity most significantly.

Absolute capacity is reduced, according to how trains are operated on the system. The Burdett and Kozan (2006) approach can also include that aspect by incorporating empirical and stochastic information and that is another virtue. An alternative measure of capacity is discussed in this chapter. It is illustrated that capacity could be measured by different metrics, namely, number of trains, number of wagons, number of pieces of cargo, number of passengers.

In this chapter, it is shown that sensitivity analysis is difficult to do. A complete sensitivity analysis is time consuming computationally. As a result, it can be concluded that proposing an automated method, by introducing a capacity expansion model, could be advantageous The next chapter examines the capacity expansion models.

Chapter 4 Capacity Expansion Models

This chapter considers the expansion of capacity in railway networks. Capacity expansion is an important topic in railways that is of increasing importance in our congested and overpopulated societies. For this reason, it is under consideration in this thesis. Capacity expansion model(s) are important and necessary as they determine how to improve system capacity. The original base model only determined what the current system capacity is. All railways however need to be expanded to meet new and increasing demands. There are many ways to expand the capacity of a railway network and there are pros and cons to each of these (Lai & Barkan, 2011; Lusby, et al., 2011; Salido, Barber, & Ingolotti, 2008). A comprehensive, generic and automated approach is needed, but to this researcher's knowledge is yet to be developed.

This chapter consists of four sections. The first section analyses how train speed alterations can expand capacity. In the second section, track duplication methods are devised. The static and time varying expansion models are also investigated. Numerical investigations are examined in third section. Finally, the originality of Chapter 4 is discussed. The CPLEX code for the models developed in this chapter has been provided in the Appendices.

4.1 SENSITIVITY ANALYSIS OF TRAIN SPEEDS

In this section, the effect of train speed alterations has been investigated as a preliminary capacity expansion technique. Increasing train speeds is a relatively straightforward approach to increase capacity and does not require construction activities or other great changes to the topology of the network. Theoretically, capacity should increase when faster trains are used because travel times will be reduced and track infrastructure will be occupied for less time by individual trains (Dingler, et al., 2009). However to increase capacity, it is necessary to purchase a considerable number of trains. The cost of buying many trains is very expensive, particularly if an entirely new fleet of trains is required. The total cost is likely to be in the millions of dollars. However, comparatively speaking, other capacity expansion options will be equally, if not more expensive. From a practical point of view, it is unknown whether such an approach is cost effective or not, and this is something that should be investigated further.

A number of articles have previously investigated train speed (Dingler et al. (2009), Froidh et al (2014), Harrod (2009)). In those articles, the analysis was performed using simulation. In this paper, and in contrast, an analytical model has been chosen to automate and

speed up the process of this type of analysis. Buri and Tzieropoulos (2009) investigated variables that determine the speed of trains and assessed the relative importance of those with respect to railway capacity. Increasing the power of trains and making trains more homogenous were found to be a promising possibility as infrastructure expansions were deemed perhaps too expensive. Froidh, Sipila and Warg (2014) considered the imbalance in trains' speeds on Scandinavian rail networks that causes reductions in capacity utilisation. A timetable analysis and some simulations were performed. In those analyses, trains' speed increments were considered.

Increasing train speeds is conceptually a simple and straight forward method to expand railway capacity, for example in comparison to other more extensive and elaborate alternatives. In this thesis, an analytical capacity model has been investigated as a means of performing a sensitivity analysis of train speeds. The results of this sensitivity analysis can help improve the operation of this railway system and help it cope with additional demands in the future. To test the new approaches, a case study of the Rah Ahane Iran (RAI) national railway network has been selected. The absolute capacity levels for this railway network have been determined and the analysis shows that increasing trains speeds may not be entirely cost effective in all circumstances.

In this capacity model, three different types of trains are considered. Their speeds are 80, 100, and 120 km/hr respectively. In order to conduct a sensitivity analysis, the model is run in two different ways with a time period of 1440 minutes (one day time period). The first case assumes that there is only one train type that is used and its speed is changed incrementally (i.e. the speed is increasing by one km/hr). As each train type category is differentiated by speed and there is 20 km/hr difference, the model is solved 20 times. The train speed increments do not exceed the speed of the next train type. For example, the speed of train type one can be increased from 80 to 99 km/hr but no more than 100 km/hr as it becomes equivalent to a train of type two. The second case assumes all trains types are used and their speeds are changed incrementally and simultaneously. For example, the first one is changed from 80 to 81, second one from 100 to 101, and finally the third one from 120 to 121. The model is run 20 times, the same as before.

The results of the 20 incremental changes were collected and tabulated in Table 4-1 and Table 4-2. Some graphs were plotted to visualise the rate of increasing capacity. Figures 4-1, 4-2, 4-3, and 4.4 show that increases in capacity are primarily linear when only a single train type is used. If there is a mix of train types, the increase in capacity is non-linear. There is some curvature but it is not great.

The difference between successive applications of the model over 20 times were compared and analysed. These differences are the absolute increase in the capacity. A relative measure is also computed, which describes the difference in comparison to the starting capacity. Also, the absolute difference per unit of time is computed because it is a normalised/scaled value

and shows the increase to the system for any time period rather that specific one. The results are as follows:

Table 4-1: Sensitivity analysis of capacity by a single incremental speed change

Type	$\mathbb{A}$ (LB)	$\mathbb{A}$ (UB)	Relative difference	Abs Difference	Normalized value
1	2594.5	3243.11	1.25	32.43	0.023
2	3243.1	3891.73	1.00	32.44	0.023
3	3891.7	4540.35	0.83	32.43	0.023

Table 4-2: Absolute capacity increase for mix of trains

Incremental Change	Type 1	Type 2	Type 3	All 3types
0	3338.4	3338.4	3338.4	3338.4
1	3346.15	3353.56	3348.3	3371.36
2	3353.77	3368.54	3358.1	3404.31
3	3361.27	3383.36	3367.81	3437.24
4	3368.66	3398.02	3377.42	3470.16
5	3375.93	3412.53	3386.94	3503.07
6	3383.1	3426.89	3396.38	3535.97
7	3390.17	3441.09	3405.73	3568.86
8	3397.13	3455.15	3415	3601.75
9	3403.99	3469.07	3424.18	3634.62
10	3410.75	3482.84	3433.28	3667.49
11	3417.41	3496.48	3442.29	3700.35
12	3423.98	3509.97	3451.23	3733.2
13	3430.46	3523.33	3460.09	3766.04
14	3436.85	3536.55	3468.86	3798.88
15	3443.14	3549.64	3477.57	3831.71
16	3449.35	3562.6	3486.19	3864.53
17	3455.48	3575.44	3494.74	3897.35
18	3461.52	3588.15	3503.22	3930.16
19	3467.48	3600.73	3511.62	3962.96
20	3473.36	3613.19	3519.95	3995.76

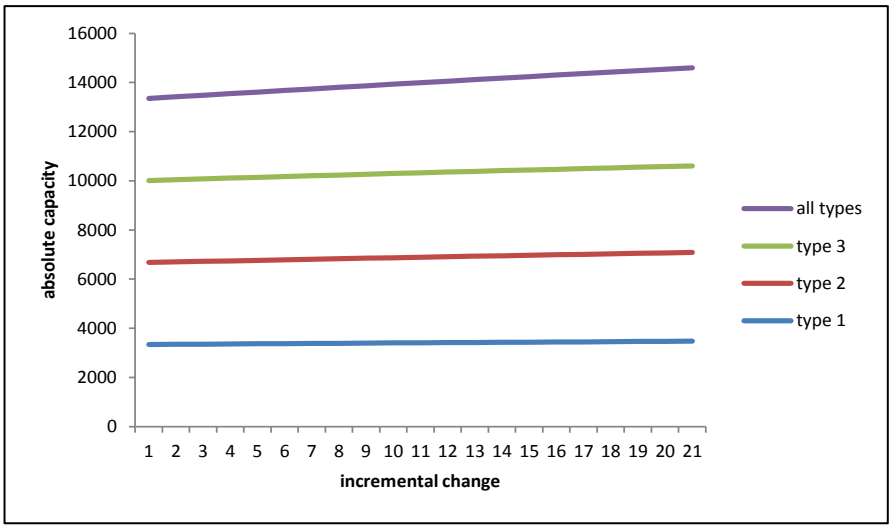

Figure 4-1: Incremental speed change versus absolute capacity

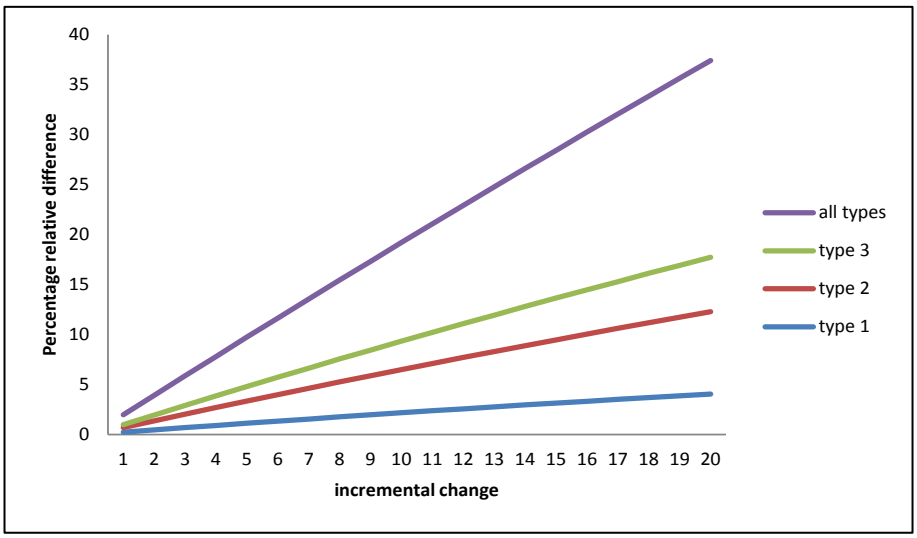

Figure 4-2: Relative difference

In Table 4-2, the results for each train type can be viewed. In this table, bounds are given for the absolute capacity. For example, the lower bound occurs when its speed is at its normal level. The upper bound occurs when its speed is increased to the upper limit (i.e. initial speed plus 20 km/hr). The relative increase in capacity and absolute difference per unit of time are also shown in the table. The absolute difference per unit of time is a normalised value and this means that duration of the time period does not affect it. There is no change in the results for the different trains' speeds, i.e. the same relative and absolute differences occur. Hence, this table includes one set of values for each train type. The overall effect of changing train speed is a linear increase in the capacity. In Table 4-1 only one type of train was analysed, however, in Table 4-2 a mix of trains is assumed. The particular mix of trains are as follows: PD = [[0.0,0.44,0.56], [0.31,0.45,0.24], [0.86,0.14,0.0], [0.87,0.1,0.03], [0.39,0.26,0.35], [0.49,0.36,0.15], [0.0,0.44,0.56], [0.31,0.45,0.24], [0.86,0.14,0.0]]. In contrast, Table 4-1 shows the non-linear relationship between successive solutions; hence all the values are shown in the table.

In Figures 4-1 and 4-2, values for absolute capacity and percentage relative difference are linearly increasing. The lines in Figure 4-1 are quite flat looking, but in fact they are not flat; the increase is just small. This is because a mix of trains has been considered and only one of the trains has its speed increased incrementally. The first train type has the lowest line in both figures, as its speed is lowest compared with the other trains. The slope of the last line is higher as changes are happening for all three train types. The last line is roughly the sum of the other three ones. In conclusion, significant improvements in absolute capacity can be achieved with a relatively small increasing speed.

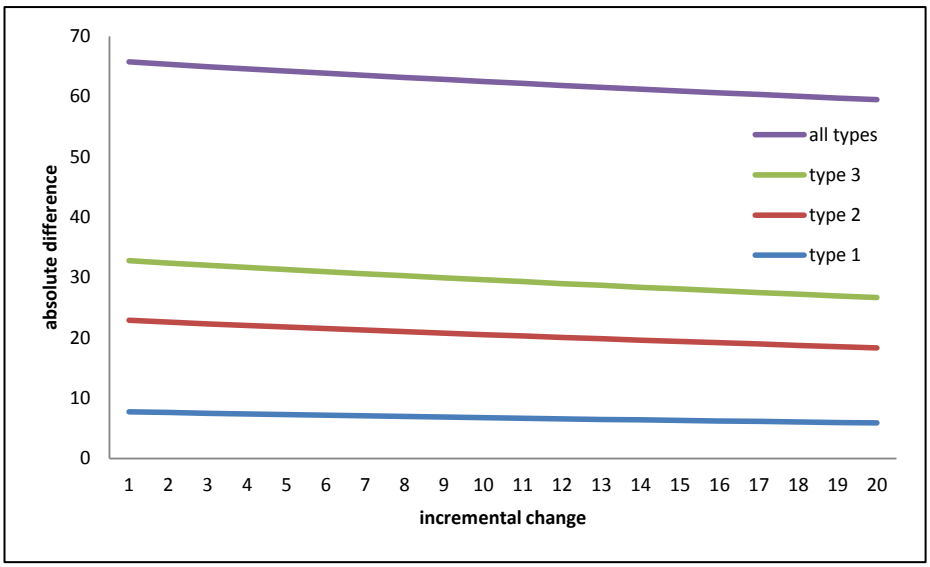

Figure 4-3: Incremental speed change versus absolute difference

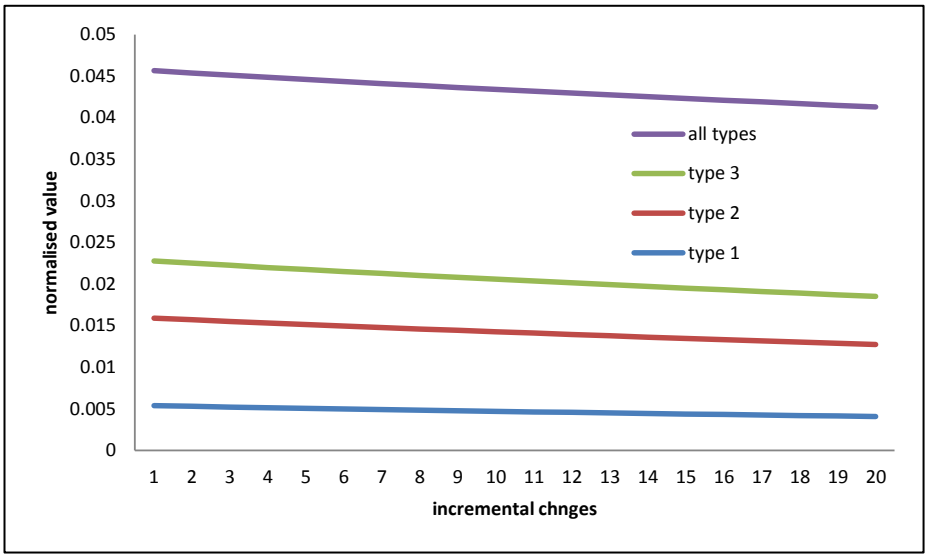

Figure 4-4: Normalised value

In Figures 4-3 and 4-4, the values for absolute difference and normalised value are linearly decreasing. While the absolute value of capacity is increasing, it is not increasing as rapidly, for example, as the trains' speed is increased to higher and higher levels. Hence, there is

limiting phenomena occurring and it is less effective to increase capacity in this way. The reason for the decrease is the interaction effect associated with the specified mix of trains.

4.2 TRACK DUPLICATION

Planners can have different restrictions when considering how to improve the capacity of a railway network. Some limitations include those that are budgetary, time, and spatial. The first restriction occurs as a result of having a specific budget available to improve the capacity of a railway network (i.e. from government, or third party stakeholders). The second restriction occurs, for instance, if there is a limit on the total time available to perform the necessary improvements. The unavailability of land or the presence of limited space is another limitation that can greatly affect capacity planning.

Previous capacity models can be used to assist planners to plan the construction of railway networks. In theory, capacity model(s) can be extended so that they identify what infrastructure expansions and spending is necessary to obtain specified levels of capacity. In that situation the capacity model also provides a plan.

The capacity models focus on upgrading existing railway network components as the construction of a new railway line has a high cost and in some cases is not beneficial. Track duplication is one way to reduce congestion in railway networks. It increases the flow of trains on specific corridors. If every section was duplicated, then trains moving in each direction could be assigned to dedicated lines. Hence uni-directional flow can be realised on each track, instead of bi-directional. Corridors have higher capacity when traffic flow is unidirectional as there is no lost time from passing activities. Duplicating all tracks is essentially the same as building another railway line, but may not be beneficial in terms of cost as each section must be upgraded with additional tracks. From a capacity viewpoint, it is necessary to improve some and not all sections.

In this thesis, capacity expansion models are categorised as either static or time varying. A static model determines what is necessary to achieve a significant improvement or else an improvement of a specified level. A time varying model determines when changes can and should be made. The purpose of these models is to simplify the planning process, which is currently empirical and case based. For example the current process requires a large amount of manual effort to set up a number of appropriate courses of action. These are then evaluated (i.e. by simulation) to ascertain possible/likely performance. In contrast the mathematical models can inherently compare all possible courses of action and can select the best from the set of feasible courses of action.

4.2.1 **Static Case**

The static model can determine the absolute capacity of a railway network. This allows the free capacity to be determined. It can determine the maximum flow of trains on each corridor in a specific time period.

In order to expand network capacity, track duplication has been considered. For example, duplicating the number of tracks that occur between adjacent locations on the network can significantly increase the capacity of the rail network. Upon reflection, it is easy to include track duplication in the capacity model by adding the following constraints:

$$\sum_{i\in I}\left(\vec{y}_i^s\vec{T}_i^s + \overleftarrow{y}_i^s\overleftarrow{T}_i^s\right) \leq (\tau_s + \|_s)T \quad \forall s \in S \qquad \text{[Section saturation]} \qquad (4\text{-}1)$$

$$\sum_{s\in S}\left(Ct_s^{\|}.\|_s\right) \leq B \qquad \text{[Limitation on spending - budget]} \qquad (4\text{-}2)$$

$$0 \leq \|_s \leq \|_s^{max} \quad \forall s \in S \qquad \text{[Limit on track numbers]} \qquad (4\text{-}3)$$

$$\sum_{s\in S}(\|_s) \leq N \qquad \text{[Limit on number of duplications]} \qquad (4\text{-}4)$$

In the above modifications, the number of parallel tracks added to each section is defined by $\|_s$. Also, the cost of a single duplicated track is $Ct_s^{\|}$. The maximum number of tracks that can be added to section s is defined as $\|_s^{max}$. Constraint (4-1) is replaced with constraint (3-8) as it includes the effect of track duplications on capacity. For example in the normal situation where we have a single section of track, there are a total of T units of time for trains to use that section. However, if we duplicate that section of rail, twice as many trains can traverse the same section and capacity can, in theory, be doubled. Thus in total there are 2T units of time available. In general $(\tau_s + \|_s)$ is the total number of tracks available and a multiplier for the increase in capacity. Constraint (4-2) is added to enforce a specified limitation on the total budget (i.e. B) for duplicating tracks. Maximising the capacity is still the objective function here. An assumption is also made for the sectional running times to be the same on parallel tracks. This assumption must be made, otherwise additional decisions concerning which parallel track individual trains are assigned to, are needed. From a practical perspective, track duplication cannot be separated from budgetary considerations. This is because the problem would become unbounded, and sections would be duplicated without limit. In the event that a budget is not provided, it is necessary to restrict the number of duplications, and to do this, constraint (4-4) should be added. It can be viewed as an optional constraint and does not need to be added otherwise.

Another decision-making problem can also be formulated whereby the sections that should be upgraded in order to achieve a specified level of absolute capacity is to be determined. In that problem, a budget could be provided, but it is not entirely necessary. This variant problem has an alternative objective, which is cost minimisation.

4.2.2 Time Varying Cases

Capacity increases could be required over time. A time varying expansion model can be introduced to help planners to increase the capacity in certain period by considering the optimal cost and time. In order to identify a plan of capacity expansions over time, the following model is formulated:

Minimize: $\hat{P}$		(4-5)
Subject To:		
$\sum_{i\in I}\left(\vec{y}_{i,p}^{s}\vec{T}_{i}^{s}+\overleftarrow{y}_{i,p}^{s}\overleftarrow{T}_{i}^{s}\right)\leq T\times\tau_{s,p}$ $\forall s\in S,\forall p\in P$	[Section saturation]	(4-6)
$\vec{y}_{i,p}^{s}=\sum_{c\in C\mid s\in\Omega_c}\left(\vec{x}_{i,p}^{c}\right)$ & $\overleftarrow{y}_{i,p}^{s}=\sum_{c\in C\mid s\in\Omega_c}\left(\overleftarrow{x}_{i,p}^{c}\right)$	[Section usage]	(4-7)
$\vec{x}_{i,p}^{c}+\overleftarrow{x}_{i,p}^{c}=\eta_{i,p}^{c}\sum_{j\in I}\left(\vec{x}_{j,p}^{c}+\overleftarrow{x}_{j,p}^{c}\right)\ \ \forall i\in I,c\in C,\ \forall p\in P$	[Proportional mix]	(4-8)
$\vec{x}_{i,p}^{c}=\mu_{i,p}^{c}\left(\vec{x}_{i,p}^{c}+\overleftarrow{x}_{i,p}^{c}\right)\ \forall i\in I,c\in C,\forall p\in P$	[Directional mix]	(4-9)
$\sum_{i\in I}\left(\vec{x}_{i,p}^{c}+\overleftarrow{x}_{i,p}^{c}\right)=\sigma_c\sum_{c'\in C}\sum_{i\in I}\left(\vec{x}_{i,p}^{c'}+\overleftarrow{x}_{i,p}^{c'}\right)$ $\forall c\in C,\forall p\in P$	[Mix across corridors]	(4-10)
$\vec{x}_{i,p}^{c},\overleftarrow{x}_{i,p}^{c}\geq 0\ \ \forall i\in I,c\in C,\forall p\in P$	[Positivity requirement]	(4-11)
$0\leq \|_{s,p}\leq \|_{s}^{\max}\ ;\ \tau_{s,p}\leq \|_{s}^{\max}+\tau_{s,0}\ \forall s\in S,\forall p\in P$	[Limit on track numbers]	(4-12)
$\tau_{s,p}=\tau_{s,p-1}\,;\tau_{s,p}=\tau_{s,p-1}+\|_{s,p-1}$ $\forall s\in S,\forall p\in P\mid p>1$	[Current track numbers]	(4-13)
$A_p=\sum_{c\in C}\sum_{i\in I}\left(\vec{x}_{i,p}^{c}+\overleftarrow{x}_{i,p}^{c}\right)\geq \mathbb{A}_p,\forall p\in P$	[intermediate requirement]	(4-14)
$e_p=\sum_{s\in S}\left(Ct_{s,p}^{\|}.\|_{s,p}\right)\ \forall p\in P$	[Current Expenditure]	(4-15)
$1\leq\hat{P}\leq\bar{P}$	[bound on the period]	(4-16)
$\sum_{p=1\ldots np}e_p\leq B_p\ ,\ \forall p\in P$	[limit on spending]	(4-17)
$e_{p+1}\leq\left(1-\varphi_p\right)M^{+},\forall p\in P\mid p<P$	[no further spending]	(4-18)
$A_p\geq\varphi_p\,.A_f\,,\forall p\in P$	[min requirement]	(4-19)
$A_p\geq A_{p-1}\,,\forall p\in P\mid p>1$	[increasing capacity]	(4-20)
$p<\hat{P}+\left(\varphi_p\right)M^{+}\ \forall p\in P$	[binary condition]	(4-21)
$p\geq\hat{P}+\left(\varphi_p-1\right)M^{+}\ \forall p\in P$	[binary condition]	(4-22)
$\varphi_p\in\{0,1\}\ \ \forall p\in P$	[binary parameter]	(4-23)

The primary purpose of this model is to create an expansion plan of minimal duration. The output of this model is a list of which sections should be duplicated in each period of time. Furthermore, this model ensures that the capacity of the system has been increased to a specified level. The decisions that are made are limited by the budget that has been provided. The parameters in this model are the same as previously defined above, except that a subscript for the time period p has been introduced. The planning period is limited. The duration of each period is, for example, one year however any other suitable period of time could be selected. Half of the constraints (i.e. equations 4-6, 4-11) have already been defined and these focus upon the determination of network capacity. The remaining constraints deal purely with the expansion decisions.

Equation (4-17) ensures that the total spending across all periods of time must be less than the specified budgetary limit. Equation (4-14) ensures that the absolute capacity in each period is greater than the last period, i.e. it is strictly increasing. The model meets the specified minimum requirement A_f for absolute capacity and furthermore that there will be no further spending after time period$\widehat{P}$. The binary variable φ_p has been introduced in order to identify whether each period p for instance occurs before or after $\widehat{P}$. For example if $p \geq \widehat{P}$ then $\varphi_p = 1$. Similarly if $p < \widehat{P}$, then $\varphi_p = 0$. This binary variable is used within the constraints 4-19, 4-20, 4-21, 4-22, and 4-23. Constraints 4-21 and 4-22 describe the relationship between the index p, $\widehat{P}$, and φ_p .

4.3 NUMERICAL INVESTIGATIONS

4.3.1 Manual Duplication

Track duplications are considered in this section. Table 4-4 shows that duplicating all sections do not change the extent of the observed interaction effects between corridors, except for the case where a proportional mix of trains were selected. If the results in Table 4-5 were divided by those in Table 4-4, then the ratio is approximately 2 for all cases. Hence, it can be concluded that if capacity is to be doubled, then in theory an additional track could be built in each section parallel to the existing one.

Table 4-3: Sensitivity analysis of absolute capacity

i	$\mathbb{A}$ (net)	$\mathbb{A}$ (individual corridors)										Utilisation (reduction)
		1	2	3	4	5	6	7	8	9	Sum	
1 (LB)	1641	685.71	872.73	336.84	426.67	685.71	336.84	426.67	338.64	426.67	4536.48	0.36(0.64)
2	2051.2	875.14	1090.91	421.05	533.33	857.14	421.05	533.33	421.05	533.33	5686.33	0.36(0.64)
3 (UB)	2461.5	1028.6	1309.1	505.26	640	1028.6	505.26	640	505.26	640	6802.08	0.36(0.64)
Mix	2095.5	945.38	1051.5	346.55	439.86	824.84	383.65	588.24	405.83	438.96	5424.81	0.39(0.61)

Table 4-4: Duplication of all sections with an additional single track

i	$\mathbb{A}$ (net)	$\mathbb{A}$ (individual corridors)										Utilisation (reduction)
		1	2	3	4	5	6	7	8	9	Sum	
1 (LB)	3282	1371.4	1745.5	673.7	853.33	1371.4	673.7	853.33	673.7	853.33	9069.39	0.36(0.64)
2	4102.4	1714.3	2181.8	842.11	1066.7	1714.3	842.11	1066.7	842.11	1066.7	11336.8	0.36(0.64)
3 (UB)	4922.9	2057.1	2618.2	1010.5	1280	2057.1	1010.5	1280	1010.5	1280	13603.9	0.36(0.64)
Mix	4190.9	1890.8	2103	693.1	879.7	1649.7	767.3	1176.5	811.67	877.91	10849.7	0.39(0.61)

Duplicating all sections of rail is however quite extreme and costly. Hence, it would be best to duplicate fewer sections if possible, but still obtain comparable levels of capacity if not the same level. Therefore, the bottleneck sections on each serial link have been duplicated first and the model has been re-solved to see what effect occurs. This course of action was taken because of the existing theory on bottlenecks. For example, it is the bottleneck section that dictates the capacity of a single line and altering other sections is superfluous until issues on that section are resolved. For example, in the RAI network, there are six separate linear segments. In

each of those, the following sections are longest: s4, s18, s19, s23, s30, s37. These bottleneck sections were duplicated and the resulting absolute values are shown in table 4-5 below. Figure 4-5 shows the changes to the network.

The values in Table 4-5 lie between those in Table 4-4 and 4-5. For some corridors, larger improvements have occurred, but in others, there has been less. To identify the exact difference, the ratios of the values in Table 4-6 with those in Table 4-4 and 4-5 have been computed (i.e. individually). Upon closer inspection, the reason for the presence or absence of improvement is purely based upon the distribution of the section lengths in each linear segment. In some segments the duplication of the bottleneck has shifted the bottleneck outside of that linear segment to another part of the network, whereas in other cases the bottleneck still occurs within the linear segment. Table 4-6 shows that duplicating the bottleneck sections in each linear segment can increase capacity by approximately 1 to 1.3 times. However, the values in Table 4-7 are still somewhat higher than those in Table 4-4, i.e. 1.5 to 1.91 times greater. Hence, some other sections still need to be duplicated in order to obtain those higher levels of capacity.

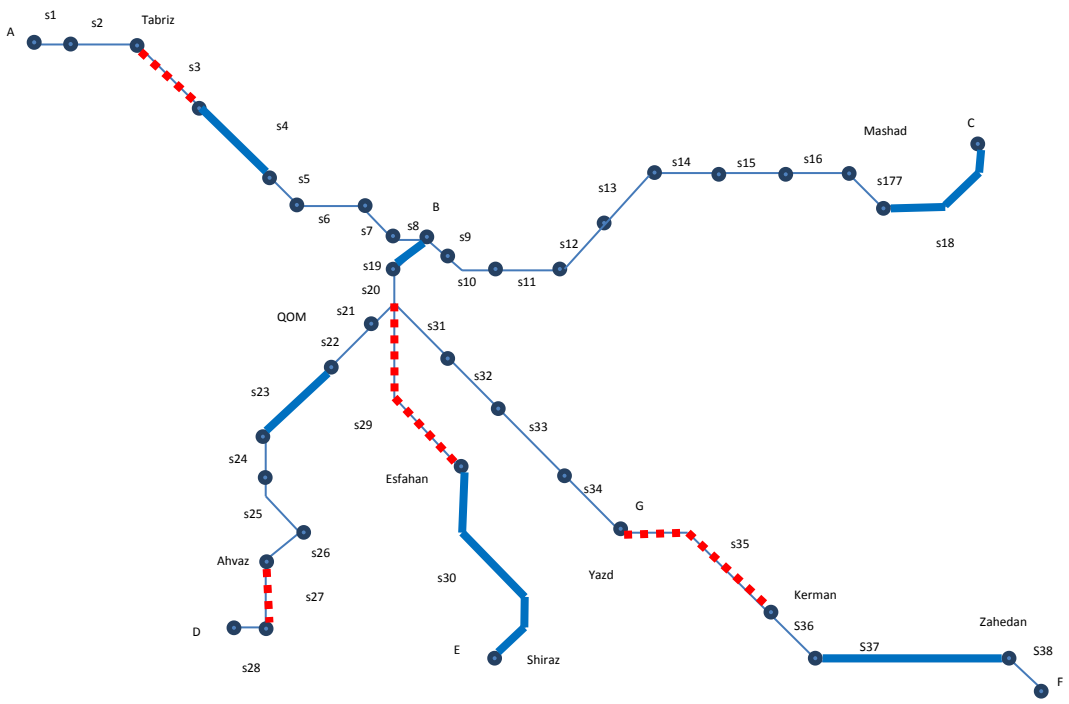

Figure 4-5: Upgraded network for the RAI (Bold–Duplicated, Dashed–Next longest section)

Table 4-5: Sensitivity analysis of duplicated bottleneck sections

i	𝔸 (net)	𝔸 (individual corridors)										Utilisation (reduction)
		1	2	3	4	5	6	7	8	9	Sum	
1 (LB)	2183.9	914.3	914.3	436.4	457.1	1280	436.4	457.1	436.4	457.1	5789.1	0.38(0.62)
2	2729.9	1142.9	1142.9	545.5	571.4	1600	545.5	571.4	545.5	571.4	7236.5	0.38(0.62)
3 (UB)	3275.8	1371.4	1371.4	654.6	685.7	1920	654.6	685.71	654.6	685.7	8683.7	0.38(0.62)
Mix	2787.6	1260.5	1101.5	448.9	471.3	1539.7	497	630.3	525.7	470.3	6945.2	0.4(0.6)

Table 4-6: Ratio of Table 4-5 to Table 4-3

i	$\mathbb{A}$ (net)	$\mathbb{A}$ (individual corridors)									
		1	2	3	4	5	6	7	8	9	Sum
1 (LB)	1.33	1.33	1.05	1.3	1.07	1.87	1.3	1.07	1.29	1.07	1.28
2	1.33	1.31	1.05	1.3	1.07	1.87	1.3	1.07	1.3	1.07	1.27
3 (UB)	1.33	1.33	1.05	1.3	1.07	1.87	1.3	1.07	1.3	1.07	1.28
Mix	1.33	1.33	1.05	1.3	1.07	1.87	1.3	1.07	1.3	1.07	1.28

Table 4-7: Ratio of Table 4-4 to Table 4-5

i	$\mathbb{A}$ (net)	$\mathbb{A}$ (individual corridors)									
		1	2	3	4	5	6	7	8	9	Sum
1 (LB)	1.5	1.5	1.91	1.54	1.87	1.07	1.54	1.87	1.54	1.87	1.57
2	1.5	1.5	1.91	1.54	1.87	1.07	1.54	1.87	1.54	1.87	1.57
3 (UB)	1.5	1.5	1.91	1.54	1.87	1.07	1.54	1.87	1.54	1.87	1.57
Mix	1.5	1.5	1.91	1.54	1.87	1.07	1.54	1.87	1.54	1.87	1.56

Given the aforementioned success at increasing capacity by duplicating the bottleneck sections, the next logical step is to duplicate the next set of bottleneck sections on each linear segment. In other words, an iterative process of track duplications is proposed in this thesis, in order to increase capacity. Those sections are s3, s27, s29, and s35. The model was resolved and the results are shown in Table 4-8 below.

Table 4-8: Sensitivity analysis of further duplicated bottleneck sections

i	$\mathbb{A}$ (net)	$\mathbb{A}$ (individual corridors)										Utilisation (reduction)
		1	2	3	4	5	6	7	8	9	Sum	
1 (LB)	2594.5	1010.5	1010.5	673.7	853.3	1280	673.7	853.33	673.7	853.33	7882.1	0.33(0.67)
2	3243.1	1263.2	1263.2	842.1	1066.7	1600	842.1	1066.7	842.1	1066.7	9852.8	0.33(0.67)
3 (UB)	3891.7	1515.8	1515.8	1010.5	1280	1920	1010.5	1280	1010.5	1280	11823.1	0.33(0.67)
Mix	3338.4	1393.2	1271.5	693.1	879.7	1539.7	767.3	1176.5	811.67	877.91	9410.6	0.35(0.65)

The values in Table 4-5 lie between those in Table 4-3 and 4-4. For some corridors, larger improvements have occurred, and in some of them the capacity is exactly equal. To identify the exact difference, the ratios of the values in Table 4-5 with those in Table 4-3 and 4-4 have been computed (i.e. individually).

Table 4-9: Ratio of Table 4-8 to Table 4-5

i	$\mathbb{A}$ (net)	$\mathbb{A}$ (individual corridors)									
		1	2	3	4	5	6	7	8	9	Sum
1 (LB)	1.19	1.11	1.11	1.54	1.87	1.00	1.54	1.87	1.54	1.87	1.36
2	1.19	1.11	1.11	1.54	1.87	1.00	1.54	1.87	1.54	1.87	1.36
3 (UB)	1.19	1.11	1.11	1.54	1.87	1.00	1.54	1.87	1.54	1.87	1.36
Mix	1.20	1.11	1.15	1.54	1.87	1.00	1.54	1.87	1.54	1.87	1.35

Table 4-10: Ratio of Table 4-4 to Table 4-8

i	$\mathbb{A}$ (net)	$\mathbb{A}$ (individual corridors)									
		1	2	3	4	5	6	7	8	9	Sum
1 (LB)	1.26	1.36	1.73	1.00	1.00	1.07	1.00	1.00	1.00	1.00	1.15
2	1.26	1.36	1.73	1.00	1.00	1.07	1.00	1.00	1.00	1.00	1.15
3 (UB)	1.26	1.36	1.73	1.00	1.00	1.07	1.00	1.00	1.00	1.00	1.15
Mix	1.26	1.36	1.65	1.00	1.00	1.07	1.00	1.00	1.00	1.00	1.15

Tables 4-9 and 4-10 show that the latest set of duplications has increased capacity, by approximately 1 to 1.87 times. Furthermore, the values in Table 4-5 are quite close to the values that are shown in Table 4-8. Some further duplication is warranted in order to reach the capacity

values in Table 4-5 but it is difficult to identify with certainty where sections should be duplicated. Hence, a mathematical model that identifies these positions would be highly beneficial.

4.3.2 Automated Duplication

In this section, the basic capacity model has been taken and the aforementioned modifications for expanding capacity via track duplications have been added. The model has been applied to the case study and the results are presented in Table 4-11. In Table 4-11 when the restriction of a single duplication (i.e. $\|_s^{max} = 1$) per section is enforced, the absolute capacity cannot be increased beyond 4190.9 trains. Upon inspection, the reason for this is that in order to increase capacity further, the same sections should be duplicated again whereas the other unduplicated sections should not. When the restriction of a single duplication per section is relaxed to two, then the absolute capacity does increase. In comparison, the increase is quite significant because the capacity changes from 4190.9 to 6081.9 trains. After 46 track duplications, absolute capacity does not increase any further. In the third column when the maximum duplication is 3, this allows the absolute capacity to be increased even further. These increases stop after 66 track duplications; capacity does not increase from additional duplications. The increase in capacity for incremental changes to the total number of duplications is not uniform. Furthermore, some incremental changes do not result in an increase to capacity. Sometimes multiple duplications must be performed in order for the absolute capacity to be increased.

In the manual approach, it is assumed that duplicating bottleneck sections is the best way to increase capacity. However, Table 4-11 shows that capacity can be increased further by choosing different sections. For example, for the specified mix of trains, the manual approach was able to increase capacity to 2787.6 trains while the expansion model was able to increase it to 3271.6 trains. This is a significant increase.

Table 4-11: Result of expansion model (No flow constraint)

$\|_s^{max} = 1$				$\|_s^{max} = 2$				$\|_s^{max} = 3$			
#	A	#	A	#	A	#	A	#	A	#	A
0	2095.5	20	4136.1	0	2095.5	20	4803	0	2095.5	20	4918.7
1	2504.6	21	4190.9	1	2504.6	21	4886.1	1	2504.6	21	4990
2	2669.2	22	.	2	2669.2	22	4991.6	2	2669.2	22	5066.3
3	2799.1	23	.	3	2799.1	23	5095.3	3	2799.1	23	5176.3
4	2935.6	24	.	4	2935.6	24	5205	4	2935.6	24	5248.7
5	3060	25	.	5	3060	25	5276.1	5	3060	25	5358.7
6	3271.6	26	.	6	3271.6	26	5307.5	6	3271.6	26	5430
7	3331.9	27	.	7	3331.9	27	5307.5	7	3331.9	27	5498
8	3472.4	28	.	8	3472.4	28	5370.6	8	3472,4	28	5555.4
9	3532.4	29	.	9	3556.8	29	5422.5	9	3556.8	29	5721.8
10	3674.1	30	.	10	3674.1	30	5503.9	10	3674.1	30	5803.3
11	3730.6	31	.	11	3776.7	31	5611.2	11	3781.7	31	5871.1
12	3816.2	32	.	12	3847.8	32	5705.6	12	3847.8	32	6060.3
13	3895.5	33	.	13	3970.9	33	5786.7	13	4001.6	33	6105.1
14	4060.5	34	.	14	4089.2	34	5881.1	14	4089.2	34	6284.1
15	4115.3	35	.	15	4244.1	35	5923.2	15	4244.1	35	6362.1
16	4115.3	36	.	16	4351.3	36	5923.2	16	4351.3	36	6433.4
17	4115.3	37	.	17	4501.1	37	5980.8	17	4583.7	37	6475.5
18	4115.3	38	.	18	4608.3	38	6081.9	18	4661.7	38	6651.1
19	4115.3	39	.	19	4689.8	39	.	19	4828.8	39	.

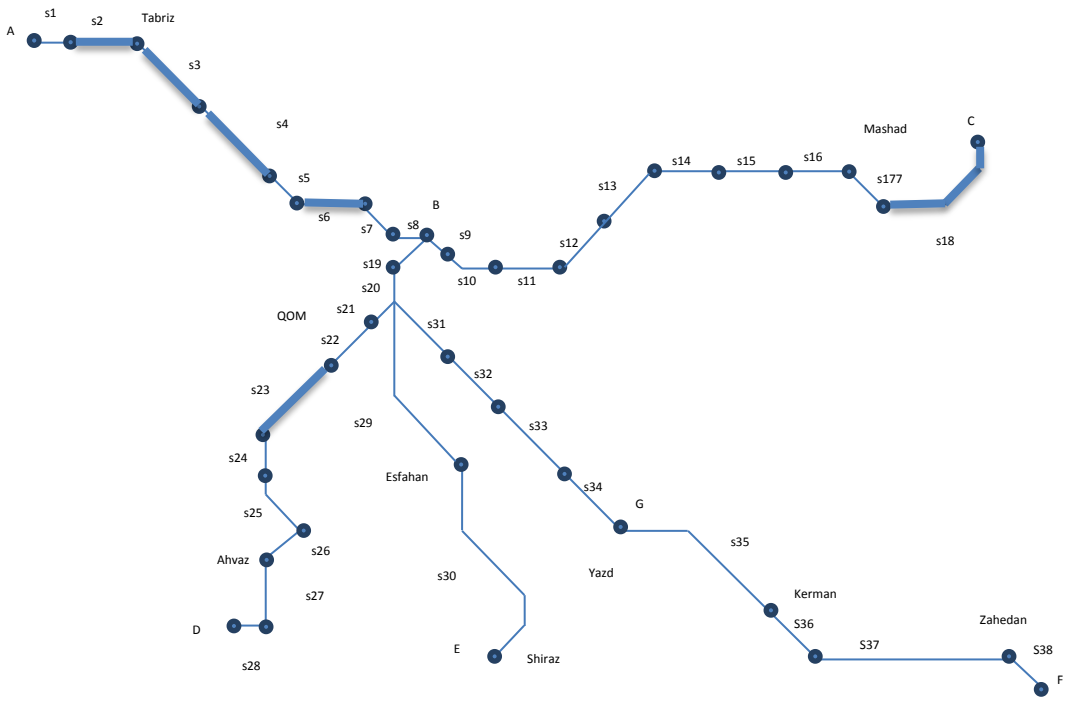

Figure 4-6: Expanded network (unequal flow)

The results in Table 4-11 have different flows on each corridor. The model has chosen different flows each time in order to maximise capacity for the current set of restrictions. This means that the model has been biased towards different corridors as supposed to other ones (see Figure 4-6). Hence some corridors have not been given any flow at all; in practice this is unacceptable. Consequently, minimum flows should be defined for each corridor and the analysis must be re-performed. This is not to say that the results in Table 4-11 are of no value. These results provide an upper bound of the effect of duplication and provide higher levels of capacity than would have been obtained otherwise.

When there is no restriction on corridor flows, the model maximises capacity by giving preference to corridors that have a higher capacity. This is clearly shown in Figure 4-7 because none of the lines are flat, i.e. each line varies and the percentages are vastly different between corridors. There is no special pattern to the assignment of flow to different corridors.

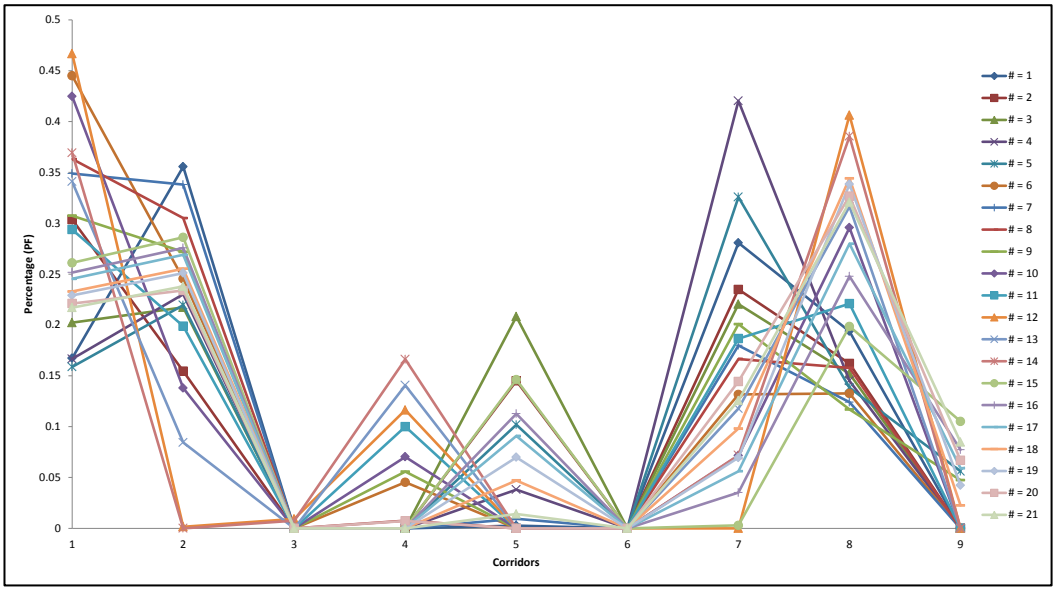

Figure 4-7: Comparison of percentage flow on corridor c for different number of $\|_s$

Table 4-12: Expansion results when flow is equal across all the corridors (i.e. PF = 1/9)

$\|_s^{max} = 1$		$\|_s^{max} = 2$		$\|_s^{max} = 3$			
#	**A**	**#**	**A**	**#**	**A**	**#**	**A**
0	1132.32	0	1132.32	0	1132.32	19	3558.44
1	1436.37	1	1436.37	1	1436.37	20	3558.44
2	1466.87	2	1466.87	2	1466.87	21	3649.22
3	1538.96	3	1538.96	3	1538.96	22	3649.22
4	1954.94	4	1954.94	4	1954.94	23	3649.22
5	2248.12	5	2248.12	5	2248.12	24	3649.22
6	2264.64	6	2264.64	6	2264.64	25	3649.22
7	.	7	2355.17	7	2355.17	26	3649.22
8	.	8	2490.91	8	2490.91	27	3909.88
9	.	9	2555.5	9	2555.5	28	3917.36
10	.	10	2603.08	10	2603.08	29	4151.51
11	.	11	2872.73	11	2872.73	30	4309.1
12	.	12	2933.73	12	2933.73	31	4309.1
13	.	13	3077.93	13	3077.93	32	4309.1
14	.	14	3077.93	14	3077.93	33	4400.6
15	.	15	3077.93	15	3077.93	34	4496.23
16	.	16	3297.24	16	3297.24	35	4529.27
17	.	17	3321.21	17	3321.21	36	.
18	.	18	3396.96	18	3396.96	37	.

In Table 4-12, the percentage flow for each corridor is assumed equal. Duplicating the specific sections shown in Figure 4-8 allows a greater flow of trains. In some linear segments, two sections were duplicated while some linear segments were not duplicated at all. Due to the imbalance in the lengths of the different sections within different corridors, this is clearly necessary. For example, some of the biggest sections were duplicated in the corridor A-E in order to be able to compete with the flow of other linear segments like A-C. These results are not

immediately obvious or transparent. Hence, choosing correct sections for duplication can be quite difficult.

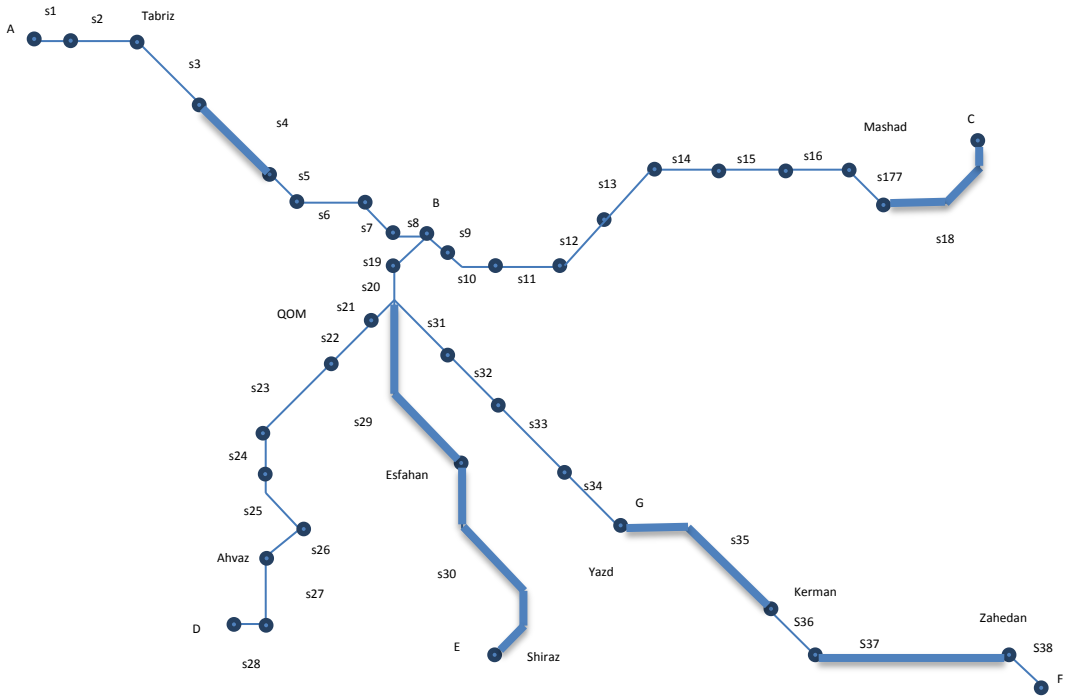

Figure 4-8: Expanded network (equal flow)

The sensitivity analysis in the preceding tables was based upon the incremental change to the number of track duplications. However it is necessary to investigate incremental changes to the budget. The model was run again for the two different percentage flows. For instance the percentage flow was not included initially, and then it was introduced so that the flow is the same on each corridor. The results for different limits of the number of duplications in each section were also analysed. The results are shown in Tables 4.12, 4.13, 4.14, 4.15 and these are also summarised graphically in Figure 4-9. The model was run 200 times. However, the tables are truncated to the point where no further change occurred.

Table 4-13: Absolute capacity for incremental budget changes (limit 1)

No %flow restriction						Equal %flow			
#	A	#	A	#	A	#	A	#	A
0	2095.46	21	3600.93	42	4115.32	0	1132.32	21	1954.94
1	2095.46	22	3600.93	43	4115.32	1	1132.32	22	2248.12
2	2260.11	23	3600.93	44	4115.32	2	1132.32	23	2248.12
3	2504.55	24	3631.49	45	4115.32	3	1132.32	24	2264.64
4	2504.55	25	3669.63	46	4136.12	4	1132.32	25	2264.64
5	2669.2	26	3669.63	47	4136.12	5	1132.32	26	2264.64
6	2669.2	27	3811.02	48	4190.92	6	1436.37	27	2264.64
7	2716.24	28	3895.46	49	.	7	1436.37	28	2264.64
8	2716.24	29	3895.46	50	.	8	1436.37	29	2264.64
9	2785.69	30	3895.46	51	.	9	1436.37	30	2264.64
10	2815.1	31	3895.46	52	.	10	1436.37	31	2264.64
11	3059.97	32	3895.46	53	.	11	1466.87	32	2264.64
12	3059.97	33	3895.46	54	.	12	1466.87	33	2264.64
13	3271.64	34	3964.16	55	.	13	1466.87	34	2264.64
14	3318.69	35	3964.16	56	.	14	1466.87	35	2264.64
15	3318.69	36	3964.16	57	.	15	1538.96	36	2264.64
16	3318.69	37	4060.51	58	.	16	1538.96	37	2264.64
17	3417.55	38	4060.51	59	.	17	1538.96	38	2264.64
18	3554.03	39	4115.32	60	.	18	1538.96	39	2264.64
19	3554.03	40	4115.32	61	.	19	1954.94	40	2264.64
20	3600.93	41	4115.32	62	.	20	1954.94	41	2264.64

Table 4-14: Absolute capacity for incremental budget changes (limit 2)

No %flow restriction										equal %flow					
#	A	#	A	#	A	#	A	#	A	#	A	#	A	#	A
0	2095.46	21	3700.44	42	4854.23	63	5526.81	84	5993.77	0	1132.32	21	1954.94	42	2933.73
1	2095.46	22	3700.44	43	4881.12	64	5554.52	85	6044.69	1	1132.32	22	2248.12	43	2933.73
2	2260.11	23	3930.22	44	4933.07	65	5632.8	86	6063.26	2	1132.32	23	2248.12	44	2933.73
3	2504.55	24	4016.4	45	5027.47	66	5648.61	87	6080.54	3	1132.32	24	2264.64	45	2933.73
4	2504.55	25	4024.31	46	5113.65	67	5672.96	88	6103.43	4	1132.32	25	2264.64	46	3077.93
5	2669.2	26	4073.78	47	5121.55	68	5688.77	89	6103.43	5	1132.32	26	2264.64	47	3077.93
6	2669.2	27	4094.72	48	5121.55	69	5709.86	90	6143.11	6	1436.37	27	2264.64	48	3077.93
7	2716.24	28	4141.62	49	5161.72	70	5709.86	91	6161.68	7	1436.37	28	2264.64	49	3077.93
8	2716.24	29	4144.79	50	5172.14	71	5786.01	92	6173.1	8	1436.37	29	2264.64	50	3077.93
9	2785.69	30	4252.99	51	5190.71	72	5786.01	93	6210.44	9	1436.37	30	2355.17	51	3077.93
10	2815.1	31	4492.51	52	5212.3	73	5826.17	94	6229.01	10	1436.37	31	2355.17	52	3077.93
11	3059.97	32	4578.68	53	5230.87	74	5846.21	95	6246.22	11	1466.87	32	2490.91	53	3077.93
12	3059.97	33	4586.59	54	5311.21	75	5881.13	96	6269.17	12	1466.87	33	2490.91	54	3077.93
13	3271.64	34	4586.59	55	5370.63	76	5881.13	97	6269.17	13	1466.87	34	2555.5	55	3297.24
14	3318.69	35	4626.75	56	5389.2	77	5896.73	98	6286.39	14	1466.87	35	2603.08	56	3321.21
15	3323.41	36	4637.18	57	5416.08	78	5921.29	99	6286.39	15	1538.96	36	2603.08	57	3321.21
16	3323.41	37	4655.75	58	5429.36	79	5951.69	100	6286.39	16	1538.96	37	2872.73	58	3396.96
17	3417.55	38	4677.34	59	5456.25	80	5951.69	101	6286.39	17	1538.96	38	2872.73	59	3396.96
18	3554.03	39	4695.91	60	5456.25	81	5991.86	102	6286.39	18	1538.96	39	2872.73	60	3396.96
19	3554.03	40	4776.24	61	5486.65	82	5991.86	103	6286.39	19	1954.94	40	2872.73	61	3396.96
20	3600.93	41	4835.67	62	5491.48	83	5991.86	104	6286.39	20	1954.94	41	2872.73	62	3396.96

Table 4-15: Absolute capacity for incremental budget changes (limit 3)

No %flow restriction													
#	A	#	A	#	A	#	A	#	A	#	A	#	A
0	2095.46	21	3700.44	42	5188.71	63	6179.47	84	7089.5	105	7640.49	106	8079.36
1	2095.46	22	3700.44	43	5188.71	64	6297.39	85	7108.07	106	7640.49	107	8097.92
2	2260.11	23	3930.22	44	5188.71	65	6328.66	86	7201.87	107	7682.57	108	8210.9
3	2504.55	24	4016.4	45	5263.28	66	6536.61	87	7201.87	108	7682.57	109	8210.9
4	2504.55	25	4024.31	46	5281.85	67	6536.61	88	7201.87	109	7709.92	110	8210.9
5	2669.2	26	4073.78	47	5307.69	68	6674.81	89	7201.87	110	7728.49	111	8210.9
6	2669.2	27	4094.72	48	5434.84	69	6674.81	90	7244.42	111	7766.36	112	8210.9
7	2716.24	28	4141.62	49	5546.94	70	6674.81	91	7262.99	112	7766.36	113	8230.63
8	2716.24	29	4144.79	50	5655.5	71	6674.81	92	7345.63	113	7770.94	114	8230.63
9	2785.69	30	4252.99	51	5793.7	72	6695.85	93	7345.63	114	7790.93	115	8230.63
10	2815.1	31	4492.51	52	5793.7	73	6743.51	94	7345.96	115	7859.59	116	8242.45
11	3059.97	32	4578.68	53	5793.7	74	6827.33	95	7370.24	116	7859.59	117	8355.42
12	3059.97	33	4586.59	54	5793.7	75	6845.9	96	7370.24	117	7935.02	118	8355.42
13	3271.64	34	4586.59	55	5814.73	76	6969.34	97	7434.39	118	7935.02	119	8355.42
14	3318.69	35	4626.75	56	5814.73	77	6969.34	98	7434.39	119	7935.02	120	8355.42
15	3323.41	36	4637.18	57	6016.23	78	6969.34	99	7434.39	120	7935.02	121	8355.42
16	3323.41	37	4704.91	58	6102.41	79	6969.34	100	7584.04	121	7961.44	122	8381.85
17	3417.55	38	4818.04	59	6110.32	80	6969.34	101	7602.61	122	7961.44	123	8381.85
18	3554.03	39	4850.56	60	6110.32	81	7011.42	102	7640.49	123	8014.18	124	8381.85
19	3554.03	40	5062.66	61	6110.32	82	7038.04	103	7640.49	124	8023.45	125	8381.85
20	3600.93	41	5062.66	62	6179.47	83	7044.74	104	7640.49	125	8023.45	126	8381.85

Table 4-16: Absolute capacity for incremental budget changes(limit 3)

equal PF													
#	A	#	A	#	A	#	A	#	A	#	A	#	A
0	1132.32	16	1538.96	32	2490.91	48	3077.93	64	3558.44	80	4151.51	96	4529.27
1	1132.32	17	1538.96	33	2490.91	49	3077.93	65	3558.44	81	4309.1	97	4529.27
2	1132.32	18	1538.96	34	2555.5	50	3077.93	66	3649.22	82	4309.1	98	4529.27
3	1132.32	19	1954.94	35	2603.08	51	3077.93	67	3649.22	83	4309.1	99	4529.27
4	1132.32	20	1954.94	36	2603.08	52	3077.93	68	3649.22	84	4309.1	100	4529.27
5	1132.32	21	1954.94	37	2872.73	53	3077.93	69	3649.22	85	4309.1	101	4529.27
6	1436.37	22	2248.12	38	2872.73	54	3077.93	70	3649.22	86	4309.1	102	4529.27
7	1436.37	23	2248.12	39	2872.73	55	3297.24	71	3649.22	87	4309.1	103	4529.27
8	1436.37	24	2264.64	40	2872.73	56	3321.21	72	3649.22	88	4400.6	104	4529.27
9	1436.37	25	2264.64	41	2872.73	57	3321.21	73	3649.22	89	4400.6		
10	1436.37	26	2264.64	42	2933.73	58	3396.96	74	3649.22	90	4400.6		
11	1466.87	27	2264.64	43	2933.73	59	3396.96	75	3909.88	91	4400.6		
12	1466.87	28	2264.64	44	2933.73	60	3396.96	76	3909.88	92	4400.6		
13	1466.87	29	2264.64	45	2933.73	61	3396.96	77	3909.88	93	4496.23		
14	1466.87	30	2355.17	46	3077.93	62	3396.96	78	3917.36	94	4496.23		
15	1538.96	31	2355.17	47	3077.93	63	3558.44	79	3917.36	95	4529.27		

The curves in Figure 4-9 taper off and become flat and do not increase without limit. In the first chart the increase is quite shallow. In the second chart the increase is greater, but in the third it is greatest when the duplication limit is most relaxed.

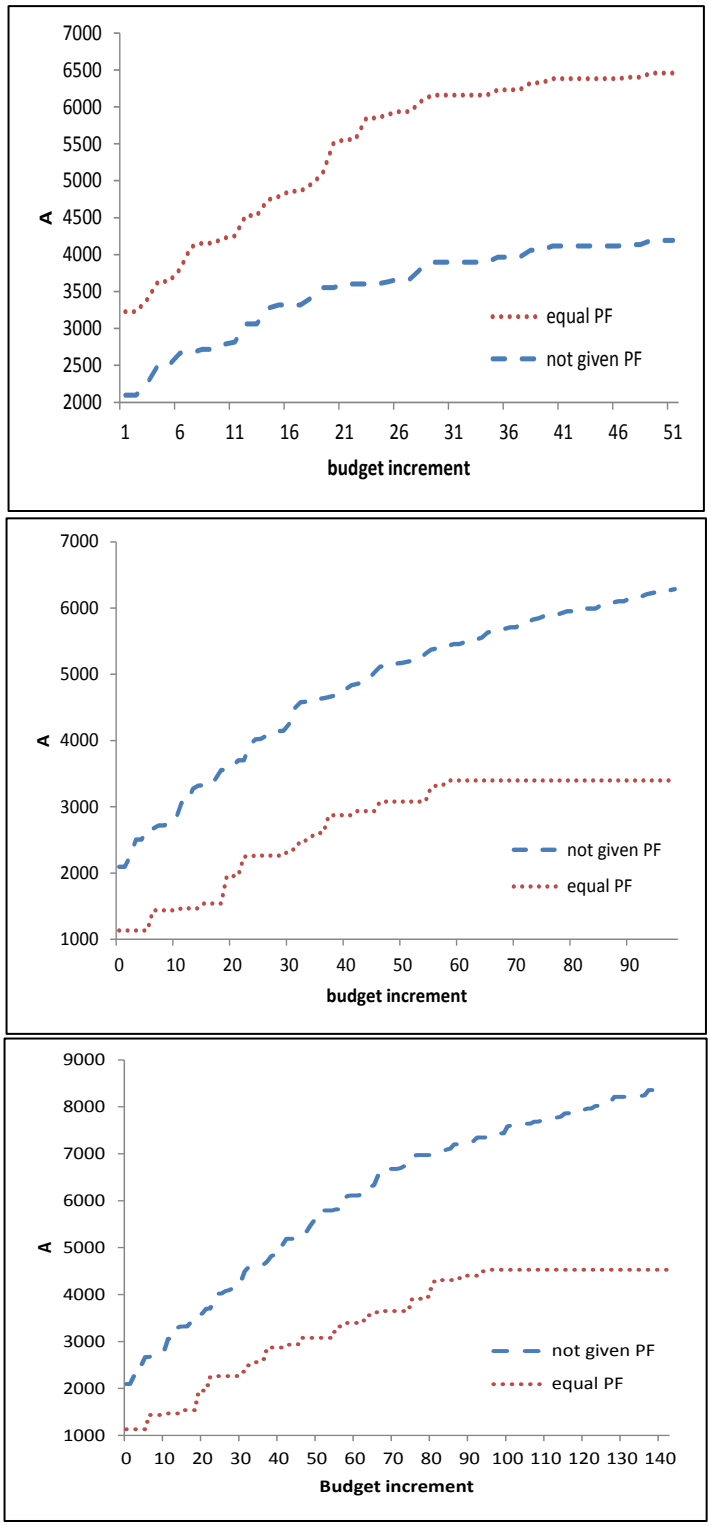

Figure 4-9: Absolute capacity versus budget increment (limit 1,2,3)

Figures 4-10 and 4-11 also show the results for different track duplication limits in each section for both equal percentage flow (i.e. 4-10) and no restriction on percentage flow (i.e. 4-11). It should be noted that in 4-11, the model bypasses bottleneck issues but in contrast the model cannot bypass the bottleneck issues in 4-10. This is clearly demonstrated by the branching of the lines and the common parts that occur beforehand.

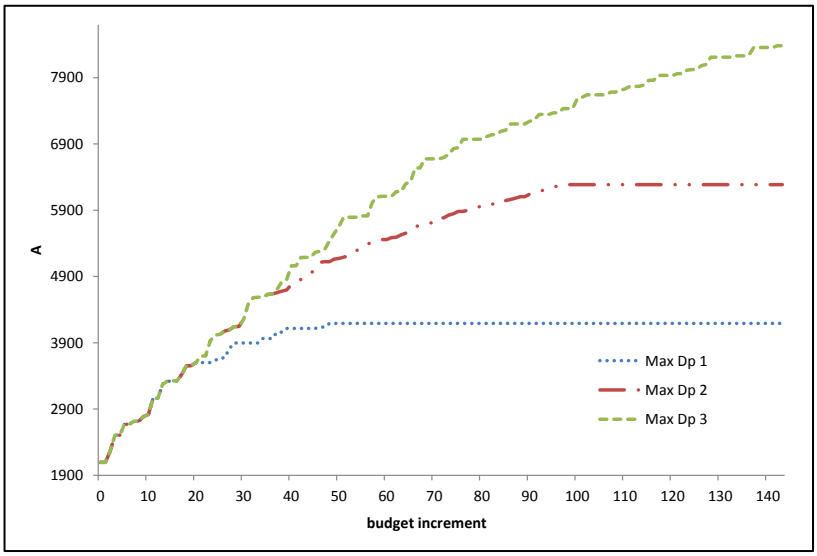

Figure 4-10: Absolute capacity by budget increment equal percentage flow

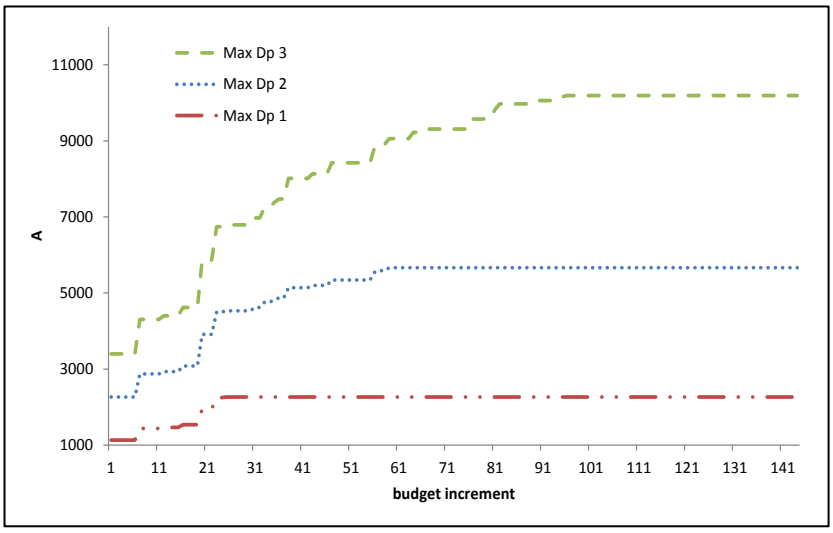

Figure 4-11: Absolute capacity by budget increment no restriction on percentage flow

4.3.3 Time Varying Expansions

In practice it is unlikely that the expansion of the RAI railway network can be performed at once. Hence, the application of the time varying capacity expansion model is necessary. That model was applied in this section. This analysis first assumes that the track duplication cost is \$100k per km of rail, and a budget of say \$100 million is available, with \$10 million available each year (i.e. over a period of ten years). These parameters are indicative of real life values but have been chosen by conjecture. Intermediate capacity requirements are imposed in the following way: $A_p = p\left(\frac{\mathbb{A}^F - \mathbb{A}^I}{10}\right)$. The time varying model was run for the RAI railway network for different limits of the number of duplications in each section. It should be noted that the limitation of maximum one for duplication is impossible as it cannot reach the target capacity

(i.e. 5000). The target capacity was reached in three periods. The solution is shown in Table 4-16. The expenditure in each period (i.e. E) is shown in Tables 4-16 and 4-17. E is defined as the cost per $ 1 000. The section that was duplicated in each period (i.e. S) is also shown. After three periods no more expansion occurs and there is no more spending. It should be noted that the results in Tables 4-16 and 4-17 consider a fixed intermediate absolute capacity over ten periods which are: [2059, 2647.2, 2941.3, 3235.4, 3529.5, 3823.6, 4117.7, 4411.8, 4705.9, 5000].

Table 4-17: The plan of expansion activity for both equal percentage flow and no restriction on percentage flow

	$\|\|_s^{max} = 3$		$\|\|_s^{max} = 4$		$\|\|_s^{max} = 10$	
P	**E**	**S**	**E**	**S**	**E**	**S**
1	**990**	6,12,13,14,15,16,18,23,27,29,35,37	990	2,3,4,6,7,18,23,25,26,29	990	2,3,4,5,6,7,18,19,21,22,23,24,25,26,27
2	**990**	2,3,4,7,11,17,18,19,22,24,25,26,27,30	990	1,2,4,5,6,19,22,23,24,25,27,30	990	3,23,29,30
3	**759**	2,3,5,9,20,22,30,35,37	693	7,8,20,21,27,28,29,30	990	11,12,13,14,15,16,17,20,22,23,29,30

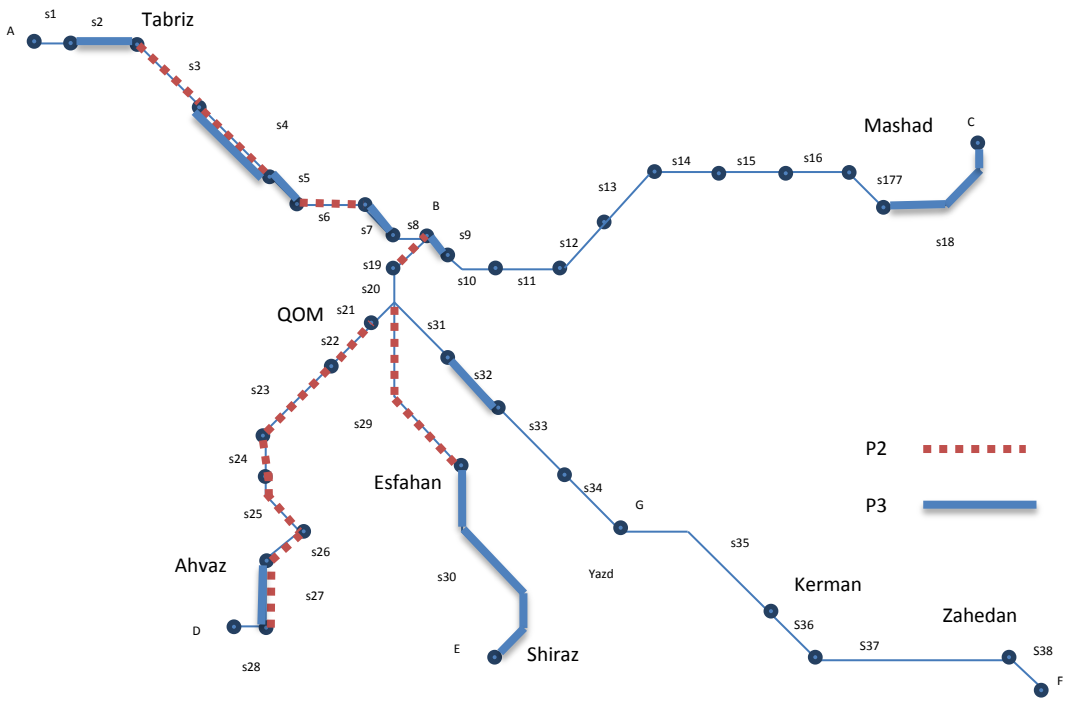

Figure 4-12: The plan of expansion activity over time for (limit 2)

The price of infrastructure changes from year to year, due to inflation and other economic forces. Incorporating inflation in the model is very useful and practical and a more realistic plan can be generated. The time varying model could address this possibility by adding a 3 percent increase to track duplication costs from period to period (i.e.$Ct^{\|}_{s,p+1} = Ct^{\|}_{s,p} * 1.03$). Table 4-18 shows that the target capacity is reached in six periods, and also expenses are increased.

Table 4-18: The plan of expansion activity considering inflation

	$\|\|_s^{max} = 2$		$\|\|_s^{max} = 3$		$\|\|_s^{max} = 10$	
P	**E**	**S**	**E**	**S**	**E**	**S**
1	140	20	0		0	
2	927	2,3,4,18	927	2,3,4,18,	927	2,3,4,18
3	954	4,6,7,18,27	954	4,6,10,21,22,	954	3,5,6,7,22,23
4	981	2,21,23,25	981	5,13,22,23,27,	981	3,4,19,23,25
5	980	3,5,6,10,14,26,28	980	3,7,24,25,26,28,	952	1,2,8,24,26,27
6	943	12,13,17,22,24,27	989	2,19,20,23,24,27,	644	2,4,6
7	826	11,15,16,19,24	-	-	-	-

4.4 DISCUSSION

In this thesis, a sensitivity analysis of train speed has been performed using an analytical capacity model. Changing train speeds is one way to expand the capacity of the railway network. The RAI national network was considered in the case study and the effect of incremental changes to train speeds has been discovered. The sensitivity analysis has shown that increasing the speed of trains can result in increased absolute capacity. However, for some mixes of trains, the benefit of doing so may not be worthwhile because the increase per instant of time actually decreases as the train speed is increased. In future, other forms of capacity expansion could be compared with the results in this thesis to find the most cost effective way of performing capacity expansion.

Capacity expansion of this network was then explored and an iterative process was then proposed as a preliminary technique. While successful, this approach does not guarantee an optimal plan of expansion activities. Hence this thesis has demonstrated a need for some formal model extensions such as track duplication, capacity expansion, etc. These extensions are necessary in order to speed up the analysis process, and to provide an autonomous process, as opposed to a manual process whereby an analyst has to alter various parameters and to define different cases.

A track duplication component for the capacity model has also been proposed. In practice, capacity expansion is normally a manual task performed by consultants and requires the establishment of a set of permissible expansion scenarios. In this thesis I have compared that manual task to my automated model. The new method has been motivated by the difficulty of performing capacity expansion by hand.

Static and time varying models were formulated and applied. The proposed model could be used to help planners increase capacity over a specified time; the model can easily assess different budgets and costs in each period. The static model was introduced to identify the best sections to duplicate immediately to obtain the maximum capacity. The time varying model determines a plan over time and can consider things like inflation. A sensitivity analysis of the model was performed for specified limitation on the maximum number of duplications in each section. The expansion model can include other costs such as the cost of earth works and it is quite robust and generic.

Chapter 5 Bi-Criteria Models for Capacity Determination

Railway capacity analysis and railway performance analysis often deals with multiple conflicting key performance indicators (KPIs). These complexities are the subject of multi objective optimisation. For instance, one may want to maximise the performance of a system while minimising its cost. This chapter considers a variety of different scenarios involving multiple competing objectives. Two different solution techniques were implemented to solve the resulting mathematical models.

In previous analytical approaches, absolute capacity was defined as the number of trains that can traverse in a railway network across different corridors, over specified time. However, this number is not completely indicative of real performance in many situations. For instance, in a freight network, the performance could be evaluated by the total tonnage of goods that are transported over time. If freight is not measured in this way then the value of the freight or the number of units of freight could be used as a substitute. Hence in summary, the payoff per train is a number, a weight, or a sum of money. The absolute capacity could be defined in the following way:

$$\mathbb{A} = \sum_{c \in C} \sum_{i \in I} \left(\vec{x}_i^c \vec{M}_i^c + \overleftarrow{x}_i^c \overleftarrow{M}_i^c \right) \tag{5-1}$$

Where $\vec{M}_i^c$ and $\overleftarrow{M}_i^c$ is the payoff per train of type i on corridor c. The majority of railway networks have to cater for passenger and freight traffic. For these “mixed” railway networks, one alternative is to convert passengers and passenger trains into a financial value, in order to compare them with freight services. This is however quite subjective and political; hence it is not easy to compare a train full of passengers to a train full of goods. Another approach is to use a bi-objective approach or even a multi-objective approach.

Four different bi-objective models are proposed to regulate specific forms of competition. These models consist of two different capacity metrics. Before those models are introduced, the theory behind multi-objective optimisation is reviewed.

5.1 MULTI OBJECTIVE THEORY

The goal of multi objective optimisation is to find solutions that are considered equally good for all objectives. Without loss of generality, let the jth objective function be defined

as $\varphi_j(x)$ where x is the current capacity model solution. That solution consists of the number of each train type on each corridor.

It should be noted that the ideal best and worst point respectively is as follows $(0,0)$ and $(1,1)$ when we are minimising two different objectives simultaneously. When we are maximising both, it is reversed. It should be also noted that these are normalised (i.e. scaled values). The following formula is used to normalise values:

$$\hat{\varphi}_j = \frac{(\varphi_j - \underline{\varphi}_j)}{(\overline{\varphi}_j - \underline{\varphi}_j)} \in [0,1] \tag{5-2}$$

To determine $\underline{\varphi}_j$ and $\overline{\varphi}_j$ a single criteria optimisation must be performed for the jth objective, ignoring all others. Hence for two objectives, four solutions are required:

$$\underline{\varphi}_1 = Min_{x \in X}(\varphi_1(x)) \ \ and \ \overline{\varphi}_1 = Max_{x \in X}(\varphi_1(x)) \tag{5-3}$$

$$\underline{\varphi}_2 = Min_{x \in X}(\varphi_2(x)) \ \ and \ \overline{\varphi}_2 = Max_{x \in X}(\varphi_2(x)) \tag{5-4}$$

In multi objective theory the idea is to identify Pareto optimal solutions. There are several definitions for optimality. The primary definitions are as follows:

Pareto Optimality 1: $x^* \in X$ is Pareto optimal if there is no $x \in X$ such that $\varphi_j(x) \leq \varphi_j(x^*)$ for j=1, 2 with at least one strict equality (i.e. $\varphi_k(x) = \varphi_k(x^*)$. This means that a point is weakly Pareto optimal if there is no other point that improves all of the objective functions simultaneously. In contrast, a point is Pareto optimal if there is no other point that improves at least one objective function without detriment to another function Marler and Arora (2010).

Pareto Optimality 2: $x^* \in X$ is Pareto optimal iff there is no $x \in X$ such that $\varphi_j(x) \leq \varphi_j(x^*)$ for j =1, 2 and $\varphi_k(x) < \varphi_k(x^*)$ for at least one k.

Solution domination is an important concept in multi-objective optimisation. This is shown visually in Figure 5-1.

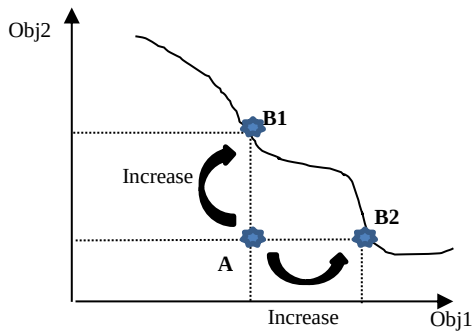

Figure 5-1: Domination theory

Figure 5-1 shows that point A and B1 have the same value for Obj 1. Point A and B2 have the same value for Obj 2. Point A is dominated by both B1 and B2. They are more optimal than A

because one of the objective values is superior but the other one is exactly the same. B1 and B2 are Pareto optimal because there is no other solution that is superior and is feasible. For example you cannot increase the values of either objective function as they are already on the boundary, without decreasing the other objective function value. By definition, a feasible solution B dominates A if B is at least as good as A with respect to every objective and is strictly better than A with respect to at least one objective. This relationship is denoted as B ≳ A or B >>A. Figure 5-2 below shows the main properties of Pareto frontier solutions. For example, the diagrams show that while the value for objective 1 is increasing, the value for objective 2 is decreasing.

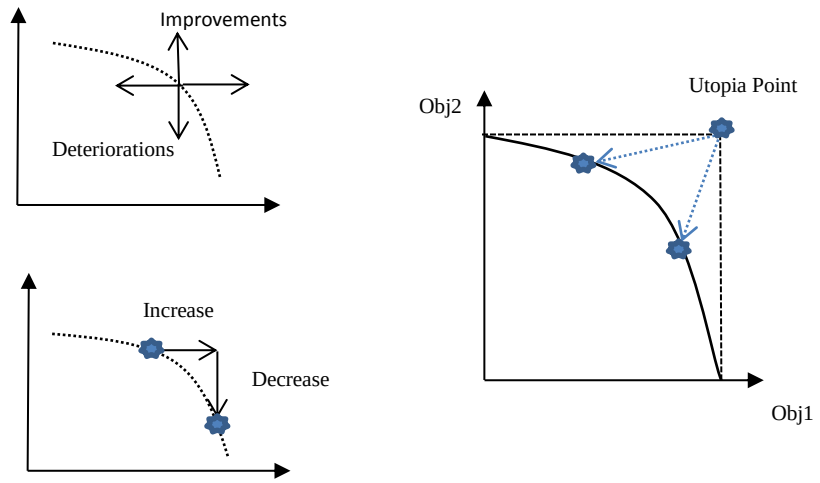

Figure 5-2: Solution quality changes when moving along the Pareto frontier

Figure 5-3 below shows the relationship between points in the decision space and those points in the objective space.

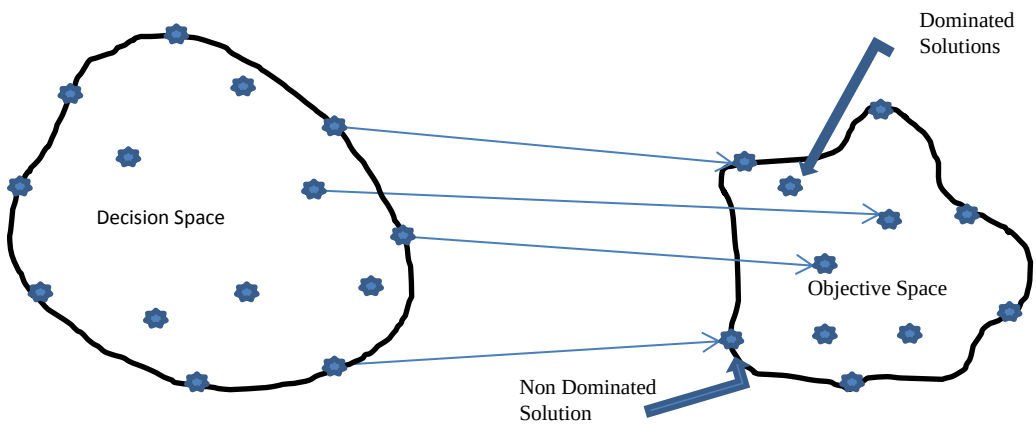

Figure 5-3: The relationship between solution point in the decision and objective space

In simple terms, a solution that is strictly superior (i.e. has superior values for each of the metrics) does not exist. According to Abbass and Sarker (2002), "no other solutions in the search space are superior to Pareto solutions when all objectives are simultaneously considered".

In order to find the most ideal solution on the Pareto frontier the concept of distance has to be defined and evaluated. The distance can be measured in two different ways. The first is from the utopia point. Hence the idea is to choose the point that is closest. The second alternative is to define the distance from the worst point. Hence the idea is to choose the point that is furthest away. There are many different mathematical definitions for distance. The following can be used:

$Min\ \boldsymbol{c_1\widehat{\varphi}_1 + c_2\widehat{\varphi}_2}$	[Distance from the Utopia point]	(5-5)
$Min\sqrt{\boldsymbol{c_1(\widehat{\varphi}_1)^2 + c_2(\widehat{\varphi}_2)^2}}$	[Distance from the Utopia point]	(5-6)
$Max\ \boldsymbol{c_1(1-\widehat{\varphi}_1) + c_2(1-\widehat{\varphi}_2)}$	[Distance from worst point]	(5-7)
$Min\ Max\ \{\boldsymbol{c_1\widehat{\varphi}_1, c_2\widehat{\varphi}_2}\}$	[Distance from the Utopia point]	(5-8)

5.2 MODEL FORMULATIONS

Models are introduced for four different bi-objective scenarios. Each of those models regulates the following forms of competition: train versus train, service versus service, corridor versus corridor, KPI versus KPI.

In order to formulate these models, relevant constraints are necessary from the base model as a starting point. The base model has a set of standard constraints that regulate the mix of trains and the flow of trains across every section of the network over time. The mix regulation constraints are optional, and can be added or removed as desired. However, the objective functions of each bi-objective model are different. The following constraints are necessary:

$\sum_{i\in I}\left(\vec{y}_i^s\vec{T}_i^s + \overleftarrow{y}_i^s\overleftarrow{T}_i^s\right) \leq \boldsymbol{T \times \tau_s} \quad \forall \boldsymbol{s} \in \boldsymbol{S}$	[Section saturation]	(5-9)
$\vec{x}_i^c = (\vec{x}_i^c + \overleftarrow{x}_i^c) \ \ \forall \boldsymbol{i} \in \boldsymbol{I}, \boldsymbol{c} \in \boldsymbol{C}$	[No. of trains on c]	(5-10)
$\vec{x}_i^c = \boldsymbol{\mu}_i^c(\vec{x}_i^c + \overleftarrow{x}_i^c) \ \ \forall \boldsymbol{i} \in \boldsymbol{I}, \boldsymbol{c} \in \boldsymbol{C}$	[Directional mix]	(5-11)
$\vec{x}_i^c, \overleftarrow{x}_i^c \geq \boldsymbol{0} \ \ \forall \boldsymbol{i} \in \boldsymbol{I}, \boldsymbol{c} \in \boldsymbol{C}$	[Positivity requirement]	(5-12)
$\vec{y}_i^s = \sum_{c\in C\|s\in\Omega_c}(\vec{x}_i^c)$ and $\overleftarrow{y}_i^s = \sum_{c\in C\|s\in\Omega_c}(\overleftarrow{x}_i^c)$	[Section usage]	(5-13)

5.2.1 Competition Between Two Train Services

The optimisation model for regulating the competition between two train service types has the following objectives:

$$\boldsymbol{Maximize} \quad \mathbb{A}_p = \sum_{c\in C}\sum_{i\in I_p}(\vec{x}_i^c + \overleftarrow{x}_i^c) \tag{5-14}$$

$$\boldsymbol{Maximize} \quad \mathbb{A}_f = \sum_{c\in C}\sum_{i'\in I_f}\left(\boldsymbol{M}_{i'}^c\vec{x}_{i'}^c + \boldsymbol{M}_{i'}^c\overleftarrow{x}_{i'}^c\right) \tag{5-15}$$

The number of trains of each type is defined as $\mathbb{A}_{\boldsymbol{p}}, \mathbb{A}_{\boldsymbol{f}}$ respectively. Hence the criterion for optimisation and decision making is bi-objective. In other words, the model attempts to maximise the flow of both types.

5.2.2 Competition Between Two Corridors

In practice, it is highly likely that traffic on different corridors will compete for access. Competition occurs because different corridors may overlap. For instance the same section(s) of rail may be used by each. It is assumed here that each corridor can potentially compete with every other corridor. In this thesis, forward and reverse flow of one corridor is considered. The objective of the proposed optimisation model is hence bi-objective. The model attempts to maximise the flow on forward and reverse for a corridor respectively. The objective functions for regulating the competition between two corridors is shown below:

$$\boldsymbol{Maximize} \quad \mathbb{A}_{\boldsymbol{C}} = \sum_{\boldsymbol{c}\in\boldsymbol{C}} \sum_{\boldsymbol{i}\in\boldsymbol{I}} (\vec{\boldsymbol{x}}_{\boldsymbol{i}}^{\boldsymbol{c}} + \overleftarrow{\boldsymbol{x}}_{\boldsymbol{i}}^{\boldsymbol{c}}) \tag{5-16}$$

$$\boldsymbol{Maximize} \quad \mathbb{A}_{\boldsymbol{C}'} = \sum_{\boldsymbol{c}\in\boldsymbol{C}'} \sum_{\boldsymbol{i}\in\boldsymbol{I}} (\vec{\boldsymbol{x}}_{\boldsymbol{i}'}^{\boldsymbol{c}} + \overleftarrow{\boldsymbol{x}}_{\boldsymbol{i}'}^{\boldsymbol{c}}) \tag{5-17}$$

It should be noted that the constraint for this model is same as constraints that are mentioned in section 5.1.1. However, it is necessary to remove the percentage flow constraint. It also is necessary to keep the constraints involving the proportional distributions as they regulate flow within individual corridors and for different train types.

5.2.3 Competition Between Two Train Types

In real life, every train type could compete with every other train type. It is assumed however that trains of a particular type do not compete against other trains of the same type. The proposed capacity optimisation model is hence bi-objective. In the proposed model it is assumed that trains could be assigned to any corridor, and there is no restriction on that. The objective functions for regulating the competition between two train types is as follows:

$$\boldsymbol{Maximize} \quad \mathbb{A}_{\boldsymbol{T}} = \sum_{\boldsymbol{c}\in\boldsymbol{C}} \sum_{\boldsymbol{i}\in\boldsymbol{I}_{\boldsymbol{T}}} (\vec{\boldsymbol{x}}_{\boldsymbol{i}}^{\boldsymbol{c}} + \overleftarrow{\boldsymbol{x}}_{\boldsymbol{i}}^{\boldsymbol{c}}) \tag{5-18}$$

$$\boldsymbol{Maximize} \quad \mathbb{A}_{\boldsymbol{T}'} = \sum_{\boldsymbol{c}\in\boldsymbol{C}} \sum_{\boldsymbol{i}'\in\boldsymbol{I}_{\boldsymbol{T}'}} (\vec{\boldsymbol{x}}_{\boldsymbol{i}'}^{\boldsymbol{c}} + \overleftarrow{\boldsymbol{x}}_{\boldsymbol{i}'}^{\boldsymbol{c}}) \tag{5-19}$$

5.2.4 Competition Between Two Different Metrics

There are many ways to expand railway networks. To identify how to increase the capacity of a railway network is required. To identify how to expand a network, there are competing different objectives which make this task quite difficult. As an example, track duplication (i.e. number of tracks) and cost (i.e. money required for duplications) could be two objectives that compete together to increase the capacity of a railway network. Hence, the

proposed model is bi-objective and the aim is regulating the competition between two objectives with different metrics. The objective functions are shown below:

$$\boldsymbol{Maximize} \quad \boldsymbol{\Upsilon} \tag{5-20}$$
$$\boldsymbol{Maximize} \quad \boldsymbol{\Psi} \tag{5-21}$$

In many (but not all) situations, each metric is calculated in the following way:

$$\boldsymbol{\Upsilon} = \sum_{c\in C}\sum_{i\in I}(\boldsymbol{\gamma}_i^c\overrightarrow{\boldsymbol{x}}_i^c + \boldsymbol{\gamma}_i^c\overleftarrow{\boldsymbol{x}}_i^c) \tag{5-22}$$
$$\boldsymbol{\Psi} = \sum_{c\in C}\sum_{i\in I}(\boldsymbol{\psi}_i^c\overrightarrow{\boldsymbol{x}}_i^c + \boldsymbol{\psi}_i^c\overleftarrow{\boldsymbol{x}}_i^c) \tag{5-23}$$

5.3 SOLUTION TECHNIQUES

A bi-objective problem can be solved by different solution techniques such as weighted sum method (WSM), e-constraint (ECM), parametric method, and adaptive search. In this thesis, the two most common and perhaps prominent techniques are considered, namely ECM and WSM. These two techniques may be good for generating the Pareto frontier. This section will test that hypothesis.

5.3.1 E-constraint Method

The results from this model allow the analyst to be able to identify bounds on the objectives in a sequential manner. The relative prioritised objectives for decision-makers are reflected by the magnitude of ε in this method (Sabri & Beamon, 2000). This technique systematically solves for different ε. Initially set $\varepsilon = \overline{\varphi}_2$, then relax this upper bound. The approach for relaxation is open to interpretation for example one strategy is $\varepsilon = \overline{\varphi}_2, \overline{\varphi}_2 - \delta, \dots, \underline{\varphi}_2$. The model is formulated as follows for this specific solution technique:

$$\boldsymbol{Maximize} \ \boldsymbol{\varphi_1(x)} \tag{5-24}$$

Subject to :

$$\varphi_2(x) \leq \varepsilon \ , \qquad x \in X \tag{5-25}$$

5.3.2 Weighted Sum Method

The most common approach to multi-objective optimisation is weighted sum method (WSM). For a bi-objective optimisation model, it is formulated as follows:

$$\boldsymbol{Z = w\varphi_1(x) + (1 - w)\varphi_2(x)} \tag{5-26}$$

where $0 \leq w \leq 1$. Weighted sum method for bi-criteria converts the objectives to a single objective by multiplying each objective by a weight and then adds all of them together. The model solves for many different values of w and creates a Pareto frontier from those values.

5.4 NUMERICAL INVESTIGATION

In this section, the aforementioned models were solved by the two selected solution techniques, namely ECM and WSM. The results are extracted, tabulated, plotted and analysed. However, preliminary numerical investigations have shown that the WSM technique is unable to identify more than two points on the Pareto frontier. Those two points are equivalent to the two upper bounds. Hence, it cannot facilitate the discovery of the optimal solution on the Pareto frontier curve and due to this limitation, WSM results were not reported. In Chapter 3, it was reported that all section lengths have been scaled by 100. This means that the actual distances are 100 times longer in real life. Hence, the absolute capacity values should be 100 times smaller than reported here. The CPLEX code for the ECM and WSM solution techniques has been provided in the Appendices The CPLEX scripting functionality was used to facilitate the repeated solution of the capacity models.

5.4.1 Competition Between Two Train Services

5.4.1.1 ECM

The capacity model of section 5.2.1 was solved using the ECM technique. This model can find the optimal capacity for the situation where freight and passenger trains simultaneously compete for railway infrastructure. The lower bounds of absolute capacity for both train types are zero and the upper bounds for passenger and freight services are 2461.46 and 1640.98 respectively. Figure 5-4 shows the un-normalised Pareto frontier. In order to identify the optimal solution, the Euclidean distances were calculated for each solution on the frontier. The results are shown in Table 5-1 and Figure 5-4.

Table 5-1: The normalised Pareto solutions

#	A_PAS	A_FRE	Dis	#	A_PAS	A_FRE	Dis	#	A_PAS	A_FRE	Dis
0	0	1	0.70711	7	0.35	0.65	0.52202	14	0.7	0.3	0.538516
1	0.05	0.95	0.67268	8	0.4	0.6	0.5099	15	0.75	0.25	0.559017
2	0.1	0.9	0.64031	9	0.45	0.55	0.50249	16	0.8	0.2	0.583095
3	0.15	0.85	0.61033	10	0.5	0.5	0.5	17	0.85	0.15	0.610328
4	0.2	0.8	0.5831	11	0.55	0.45	0.50249	18	0.9	0.1	0.640312
5	0.25	0.75	0.55902	12	0.6	0.4	0.5099	19	0.95	0.05	0.672681
6	0.3	0.7	0.53852	13	0.65	0.35	0.52202	20	1	0	0.707107

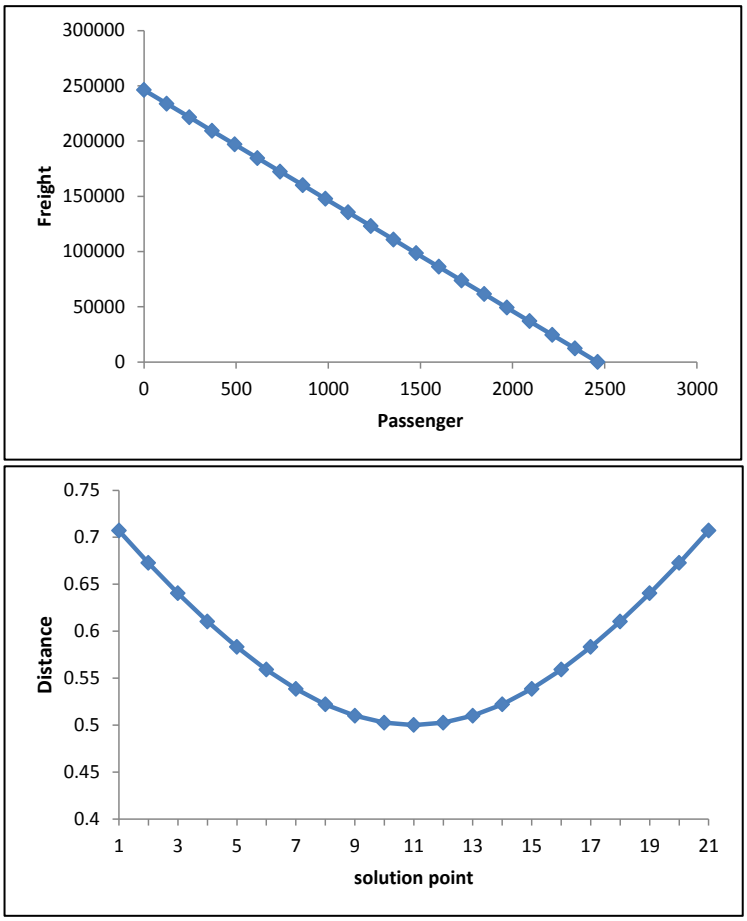

Figure 5-4: Pareto frontier and associated Euclidean distance

The results show that the Pareto frontier is symmetric and it is piecewise linear. It is piecewise linear because it is made of different line segments, each with a very similar gradient. To optimally regulate the competition of both freight and passenger services, the minimum Euclidean distance were identified. The best solution occurred at point ten. The absolute capacities for passenger and freight respectively are 1230.7 and 820.49. The Euclidean distance is 0.5. This is the best solution because a greater number of divisions were evaluated and the same solution was obtained.

5.4.2 Competition Between Two Corridors

The capacity model of section 5.2.2 was solved using ECM to find the optimal capacity for the situation where two corridors simultaneously compete for railway infrastructure. It should be noted, in this model the forward and reverse movement of an individual corridor is considered as two corridors. This model takes into account the acceleration of a train. In other word, the forward and reverse movement of trains could be different, based on different sectional running times for specific sections. This concern could be addressed with this model. The competition of both corridors is regulated by identifying the Euclidean distance. The results are shown below:

Table 5-2: The normalised Pareto solutions

#	A_C	A_C'	Distance	#	A_C	A_C'	Distance	#	A_C	A_C'	Distance
0	0	1	0.707107	7	0.35	0.65	0.522015	14	0.7	0.3	0.538516
1	0.05	0.95	0.672681	8	0.4	0.6	0.509902	15	0.75	0.25	0.559017
2	0.1	0.9	0.640312	9	0.45	0.55	0.502494	16	0.8	0.2	0.583095
3	0.15	0.85	0.610328	10	0.5	0.5	0.5	17	0.85	0.15	0.610328
4	0.2	0.8	0.583095	11	0.55	0.45	0.502494	18	0.9	0.1	0.640312
5	0.25	0.75	0.559017	12	0.6	0.4	0.509902	19	0.95	0.05	0.672681
6	0.3	0.7	0.538516	13	0.65	0.35	0.522015	20	1	0	0.707107

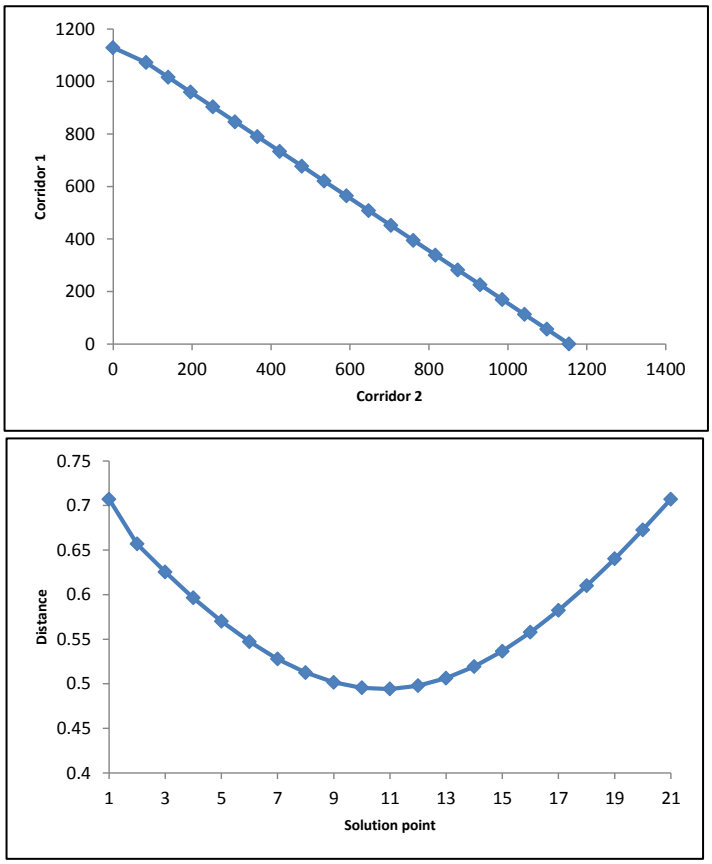

Figure 5-5: Pareto frontier and associated Euclidean distance

The result shows that the Pareto frontier is piecewise linear because it is made of different line segments, each with a very similar gradient. The lower bound for absolute capacity was zero for both corridors and the upper bounds for corridor 1 and corridor 2 respectively were 1196.01 and 979.592. The optimal point is the tenth with a minimum Euclidean distance of 0.5. The results show that although the diagram for C1 versus C2 looks linear, it is somewhat non-linear. Table 5-2 illustrates all normalised results for all 20 solution points.

5.4.3 Competition Between Two Train Types

The capacity model of section 5.2.3 was solved using ECM to find the optimal solution that regulates the competition between any two train types. It should be noted that the four train types have the speed of 60, 80, 100, and 120 respectively. The lower bound for absolute capacity for all train types is zero. The upper bound for trains of type 1, 2, 3, and 4 are 1230.73, 1640.98, 2051.22, and 2461.46 respectively. For all pairings of the four train types, Figure 5-6 shows the resulting Pareto frontiers.

Hence there are six separate frontiers.

The numerical results for train type 2 and 3 however are shown in Table 5-3 as an example.

Table 5-3: The normalised Pareto solutions

#	A[2]	A[3]	Dis	#	A[2]	A[3]	Dis	#	A[2]	A[3]	Dis
0	0	2051.22	0.707107	7	574.341	1333.29	0.522015	14	1148.68	615.366	0.538516
1	82.0488	1948.66	0.672681	8	656.39	1230.73	0.509902	15	1230.73	512.805	0.559017
2	164.098	1846.1	0.640312	9	738.439	1128.17	0.502494	16	1312.78	410.244	0.583095
3	246.146	1743.54	0.610328	10	820.488	1025.61	0.5	17	1394.83	307.683	0.610328
4	328.195	1640.98	0.583095	11	902.536	923.049	0.502494	18	1476.88	205.122	0.640312
5	410.244	1538.41	0.559017	12	984.585	820.488	0.509902	19	1558.93	102.561	0.672681
6	492.293	1435.85	0.538516	13	1066.63	717.927	0.522015	20	1640.98	0	0.707107

The optimal capacity solutions are shown in Table 5-4.

Table 5-4: The optimal capacity for different bi objective scenarios

First Train	Second Train	Capacity First Train	Capacity Second Train	Total Capacity
1	2	615.366	820.488	1435.854
1	3	615.366	1025.61	1640.976
1	4	615.366	1230.73	1846.096
2	3	820.488	1025.61	1846.098
2	4	820.488	1230.73	2051.218
3	4	1025.61	1230.73	2256.34

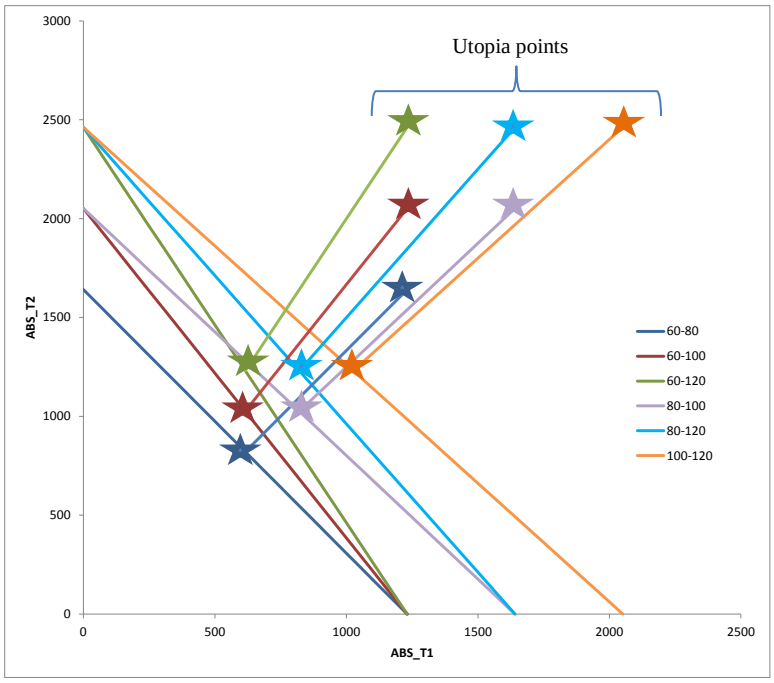

Figure 5-6: The Pareto frontiers for each train types' pairing

The results show that the Pareto frontiers are all symmetric. The Pareto frontiers are linear apart from some minor rounding errors. The linearity that has been observed in the figure occurs because the sectional running times of the two train types are determined by the same formula. In practice, this is not necessarily the case because different trains have different weight, length, size, and coefficient of drag (i.e. aerodynamics) properties, different braking systems, different acceleration, and different engines. Hence, the SRT would be different in real life and the relationship would not necessarily be linear. There are six unique utopia points that are shown in the figure. The minimum distances from the Pareto frontier to the different utopia points are shown as perpendicular lines. In order to identify the optimal solution, the Euclidean distances were also calculated for each solution on the frontier. These are shown in Figure 5-7. The best solution occurs midway at point 10 in all situations and its normalised value is 0.5.

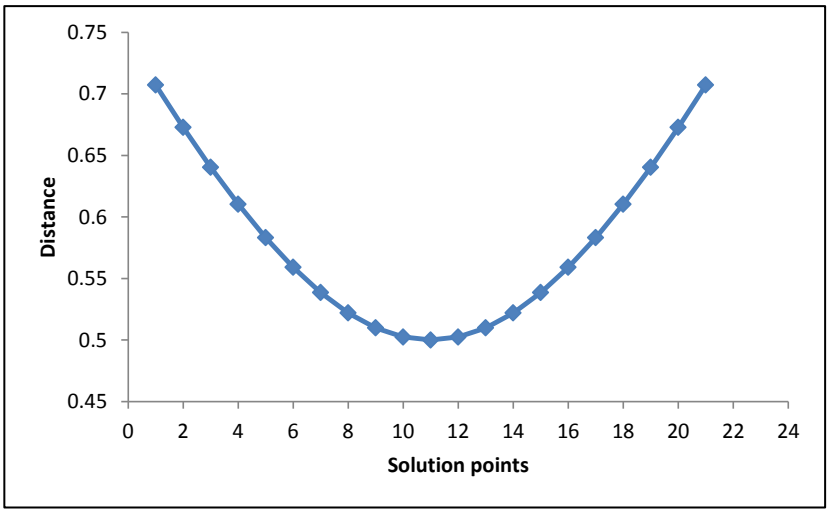

Figure 5-7: Euclidean distances for all solution points and for all train type pairings

The results are not exciting but it can be concluded from the analysis that the full Pareto analysis did not need to be performed, due to the shape and symmetry of the Pareto frontiers. The optimal solution can be exactly quantified and the following theorem provides the proof why.

Theorem: The optimal point occurs in the middle of the Pareto frontier when the sectional running time is given by a specific formula, namely: SRT = DIST / VEL.

Proof: The Pareto frontier is a straight line between points (0, UB_2) and (UB_1,0) and this is shown in Figure 5-8. The Pareto frontier is described by the equation of a straight line, namely the following:

$$y = mx + c \text{ where } m = \frac{-UB_2}{UB_1} \text{ and } c = UB_2 \,.$$

The gradient and intercept is determined by noticing that $y(0) = UB_2$ and $y(UB_1) = 0$. The shortest distance from the utopia point to the Pareto frontier is determined by the intersection of the line perpendicular to the Pareto frontier. The line perpendicular to the Pareto frontier is described by the following equation:

$$y' = m'x + c' \text{ where } m' = -m = \frac{UB_2}{UB_1} \text{ and } c' = 0 \,.$$

The gradient and intercept is determined by noticing that $y'(UB_1) = UB_2$ and $y(0) = 0$. The point of intersection of functions y and y' is identified in the following way:

$$y(x) = y'(x)$$
$$\frac{-UB_2}{UB_1}x + UB_2 = \frac{UB_2}{UB_1}x$$
$$UB_2 = 2x\frac{UB_2}{UB_1}$$
$$x = \frac{1}{2}\frac{UB_2 UB_1}{UB_1} = \frac{1}{2}UB_1$$

$$y\left(\frac{1}{2}\,UB_1\right) = \frac{UB_2}{UB_1} \times \frac{1}{2}\,UB_1 = \frac{1}{2}\,UB_2$$

Hence the optimal point occurs at $\left(\frac{\mathrm{UB}_1}{2}, \frac{\mathrm{UB}_2}{2}\right)$ as required.

In layman's terms, the above proof is explained in the following way:

The SRT of different train types are directly proportional to each other, hence the change in absolute capacity is also directly proportional to the ratio of those values. Hence during ECM, a decrease in one of the train types results in a constant increase in the other train type. Hence the Pareto frontier is exactly a straight line.

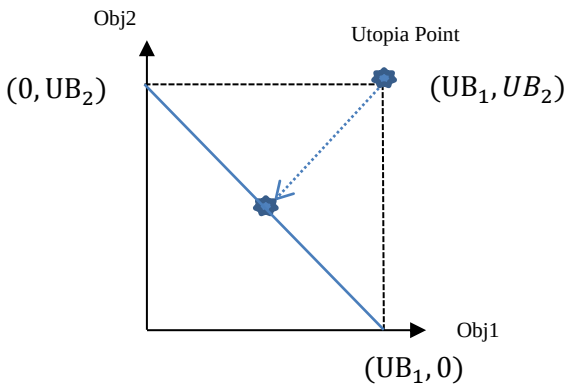

Figure 5-8: The Pareto frontier and utopia point line

5.4.4 Max Capacity versus Min Cost

5.4.4.1 ECM

The capacity model of section 5.2.4 was run to find the optimal expansion plan that simultaneously maximises absolute capacity and minimises spending. The model was run for different duplication limits, i.e. 1, 2 and 3. The numbers of divisions investigated were 20, 500 and 1000. Each of these divisions results in a different solution point. The different number of divisions used in ECM greatly affects the quality of the obtained Pareto frontier. More divisions are generally superior however additional computational efforts are required.

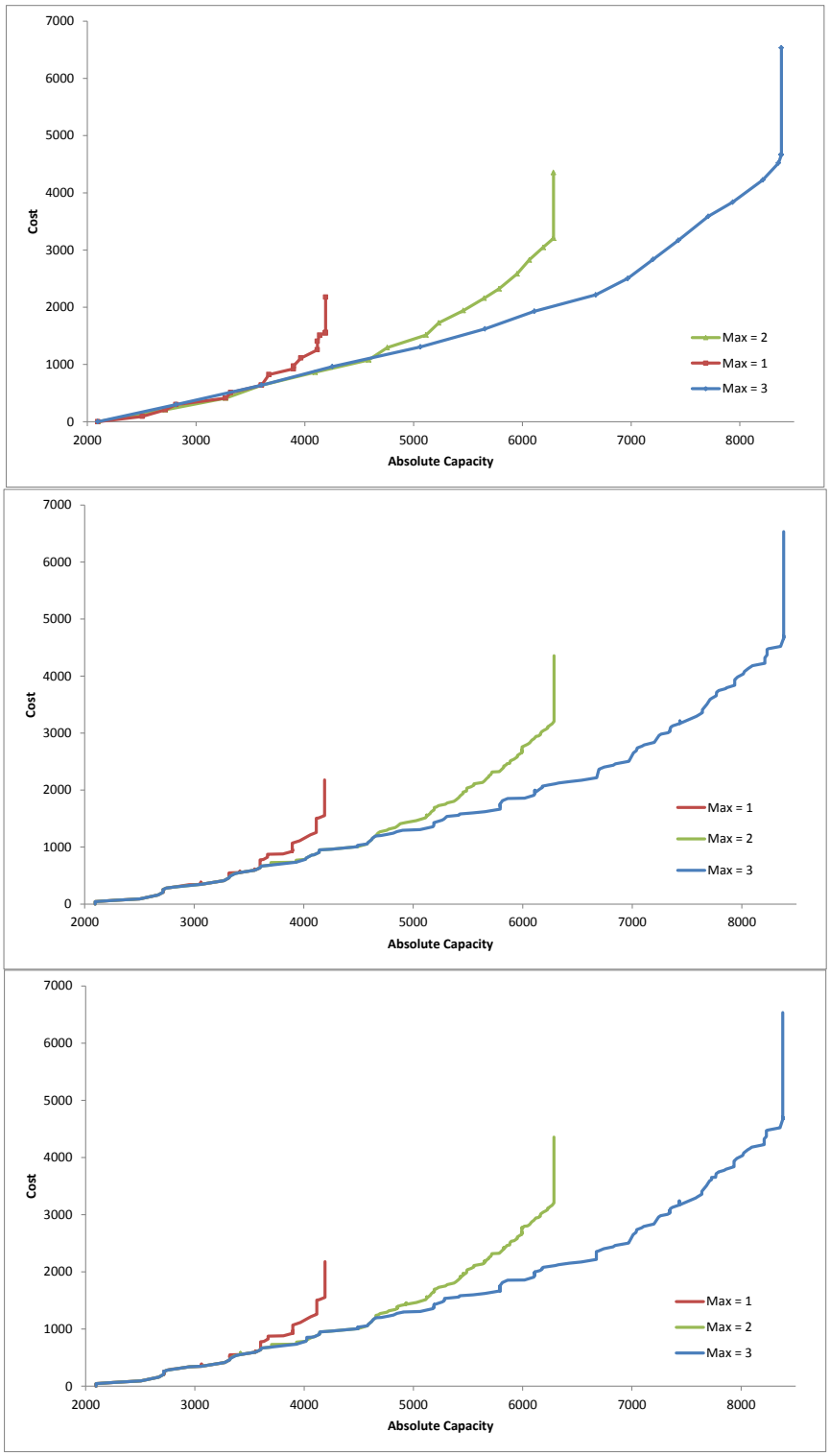

Figure 5-9: Pareto frontier for limit 1, 2, 3 respectively for 20, 500, 1000 solution points respectively

Figure 5-9 shows the Pareto frontier for different numbers of divisions (i.e. 20, 500 and 1000) for one of the objective namely the number of passenger trains. It is difficult to identify how many are required in order to obtain the greatest level of detail. Obviously more are required for greater accuracy and precision. The difference between 500 and 1000 points is not noticeable,

however, twenty points is clearly too few. As the time to perform the analysis is negligible (i.e. less than a minute for 1000 pts) this issue is quite irrelevant and more points can easily be generated as part of an iterative process. An adaptive ECM would certainly be useful.

In each diagram there are three different lines. Each line is associated with a particular track duplication limit. These lines can be compared directly and that is why they are plotted together. More duplication limits can be assessed but the same-shaped functions occur. The overriding relationship is non-linear but the actual curve is made up of many linear segments.

It is clearly obvious that there is a limiting bottleneck in each of the three lines as there are three large vertical segments at the end. The bottlenecks occur at points where increased spending results in no further increase in absolute capacity. For example these bottlenecks occur when the spending level is approximately 2300, 3200, and 4800 respectively. There are also some minor intermediate bottlenecks.

The three lines seem to share a common part before they bifurcate, however upon closer scrutiny this is not the case. For example the lines actually differ when the chart is magnified. The results are however very close there and for all intents and purposes the same.

Relaxation of the duplication limit allows roughly a 2 000 train increase in the absolute capacity between the three bottleneck points. The gradient of the lines are all different. For a limit of one train it is very steep. The third line has the most consistent increase in absolute capacity with respect to increased spending.

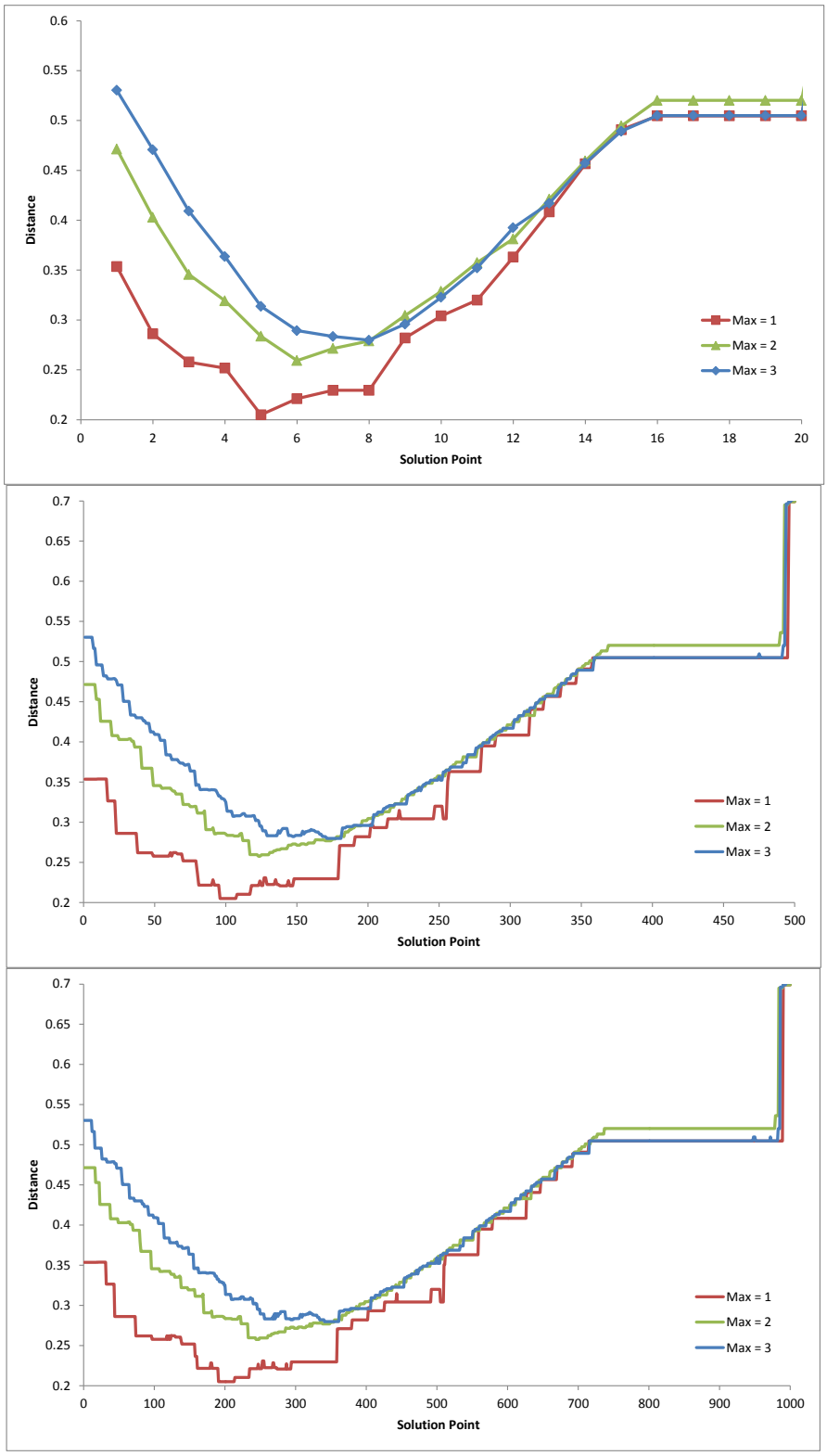

Figure 5-10: Euclidean distance for limit 1, 2, 3 respectively for 20,500, and 1000 solution points

Figure 5-10 shows the Euclidean distances. The number of solution points is again 20, 500, and 1000. In each diagram there are three different lines. These refer to different values of the track duplication limits 1, 2, and 3 respectively. These distance functions are asymmetric and

highly piecewise linear. The precision is greatest after 500 divisions. Twenty divisions are again insufficient. There are many flat segments and near vertical segments. The many small horizontal (i.e. flat) lines show that there is no change whatsoever in the Euclidean distance between adjacent solution points on the Pareto frontier. The vertical lines are not vertical at all. They just appear to be, because of the great difference that occurs in the Euclidean distance between some adjacent points. The figure shows a number of unusual spikes. These spikes in reality do not exist. This is an issue with excel scatter plots. In this figure some overlapping happens between lines. However, it cannot compare the lines directly because they have different upper bounds. Hence Euclidean distances are computed with respect to different utopia points. In this figure, the red line has the smallest Euclidean distance. However, it cannot be concluded that the red line is superior to the other two situations. It should be noted that the minimum distance occurs at different places in each line. At the end of each line there is a very long flat segment where the Euclidean does not change at all. This is quite interesting as the distance to the utopia point remains the same as you move across the Pareto frontier and after you have reached this special bottleneck point.

5.5 CONCLUSION

This chapter focussed on the development of improved capacity models for railways. The competition between train services, train types, railway corridors and alternative capacity metrics has been incorporated and captured by the formulation of various bi- objective mathematical models. This approach is necessary because the mix of trains that use railways is variable, and capacity determination and other planning activities should not be based purely upon a static mix. This bi-objective approach is applicable to real life and is valuable as it performs a sensitivity analysis of capacity for a wide range of parametric values. In order to solve the bi-objective model, two different solution techniques were applied namely ECM and WSM. The ECM technique was found to be preferable because the entire Pareto frontier can be identified. In contrast, WSM is only able to identify a small number of points. It is shown that as the number of divisions is increased, the Pareto frontier is more accurately characterised. The number of divisions that are required is an unknown parameter and further analysis is required in future research. It is clear from the numerical investigations there is a certain level of precision that is necessary and any level above this is not beneficial.

It has been shown that the Pareto analysis is not required when comparing different train types. The reason for this is due to the formula used for calculation of sectional running times. For the other scenarios this is not true. The capacity versus expansion bi-objective approach has been found to be equivalent to the iterative approach proposed in Chapter 4. Bi-objective optimisation provides a better framework because it explores all possible solutions and

automatically finds the best one. However, the other approach is manual and iterative; thus it requires more effort. Obviously the model in Chapter 4 is necessary because it allows specific capacity expansion scenarios to be solved.

Chapter 6 Conclusions

Railway transportation is predominantly used to transport passengers and freight. The demand for passenger and freight transportation is rapidly increasing in many countries for a variety of reasons. In order to optimise the financial and economic prosperity of a nation, efficient railways are required. Building and optimising railway networks, however, is a difficult topic due to the inherent structural complexity of railways and various operational factors. The importance of railway networks is clear and that is why they are considered in this thesis. In this thesis, the focus is upon capacity determination and expansion planning. This chapter summarises the main findings of this research. Furthermore, the required skills and future research directions are reported.

6.1 MAIN FINDINGS

This thesis has shown that the core capacity model of Burdett and Kozan (2006) can be easily extended. For example, in this thesis, that model was able to be extended, so that it can identify the optimal expansion of railway networks. The approach has also been extended for multi-objective capacity analysis. Because traditional capacity models have a single objective, the bi-objective approach is quite novel. Its strength also lies in the fact that it can be used to perform a sensitivity analysis of capacity for competing railway capacity metrics. The proposed bi-objective approach can also identify what infrastructure expansions and spending are most appropriate and equitable. In contrast, the original model considers only a single objective and that is not realistic or useful in practical situations.

In practice, the capacity of railway networks has typically been identified by using simulation. Simulation is useful, however it is not a decision making tool, it is a decision support tool. In this thesis, an analytical model has been developed to automate and speed up the capacity analysis process. Furthermore, the proposed analytical models in this thesis are easy to apply, have minimal data requirements, are easy to implement, and are sufficiently accurate for most planning activities.

6.1.1 Sensitivity Analysis of Proposed Models

In Chapter 3, how to perform a more in depth and general sensitivity analysis of capacity was demonstrated. This was necessary because typically, the solution of the capacity model is a single value and this is not representative of all possibilities in practice. This analysis requires that

the analytical models be solved numerous times to provide a variety of lower and upper bounds for absolute capacity.

A sensitivity analysis of train speed was also performed in Chapter 4. It was found that changing train speeds is a very straight forward and effective approach to expand the capacity of the railway network. The RAI national network was considered in this thesis and the effect of incremental changes to train speeds has been identified. The sensitivity analysis has shown that increasing the speed of trains can easily result in increased absolute capacity. Unfortunately, for some specified mixes of trains, the benefit of doing so may not be as worthwhile because the increase per instant of time actually decreases as the train speed is increased.

6.1.2 Capacity expansion models

In Chapter 4, track duplication was considered as a means of increasing railway network capacity. In particular, track duplication components were added to the capacity model. In practice, capacity expansion is normally a manual task performed by consultants and requires the establishment of a set of permissible expansion scenarios. An iterative process for performing capacity expansion was explored as a preliminary technique. While successful, this approach does not guarantee an optimal plan of expansion activities. Hence, this thesis has demonstrated a real need for formal model extensions. It was also found that the manual process was slow and awkward as various parameters had to be altered and different cases had to be defined specifically. Hence the extensions are very necessary in order to speed up the analysis process, and to provide an autonomous process. That manual task has been compared to the proposed automated model. The new model is superior and it provides better solutions.

When the analytical models were applied to analyse the capacity of the National Iranian Rail network, it was found that corridors A-C, A-D, and C-D have the highest capacity and this is clearly evident from the network diagram. For example, those corridors have a greater number of smaller sections; this allows more traffic to flow. In practice, those three corridors constitute the paths with the highest passenger and freight demand.

A time varying capacity expansion model was also formulated and applied. That proposed model can be used to help planners increase the capacity over a specified time; it can easily assess different budgets and costs in each period. In contrast, the static model was introduced only to identify the best sections to duplicate immediately to obtain the maximum capacity. The time varying model determines a plan over time and can consider things like inflation. The expansion model can include other costs such as the cost of earth works and it has been found to be quite robust and generic. A sensitivity analysis of the model was performed for specified limitation on the maximum number of duplications in each section. This was considered in order to mimic real life conditions, where different railway sections may or may

not be duplicated due to space restrictions. In this thesis it has been found that the duplication limit greatly affects capacity expansion results. For example, there are capacity bottlenecks that occur, which can only be bypassed by relaxing the duplication limit.

6.1.3 Regulating Competition

In Chapter 5, the competition between train services, train types, railway corridors and alternative capacity metrics were considered. These competing factors were incorporated and captured by formulating various bi- objective mathematical models. This approach was necessary because the mix of trains that use railways is variable, and capacity determination and other planning activities should not be based purely upon a static mix. This bi-objective approach is very applicable to real life and is quite valuable, as it performs a sensitivity analysis of capacity for a wide range of parametric values. In order to solve the bi-objective model, two different solution techniques were applied, namely ECM and WSM. The ECM technique was found to be preferable because the entire Pareto frontier can be identified. In contrast, WSM is only able to identify a small number of points. It is shown that as the number of divisions is increased, the Pareto frontier is more accurately characterised. The number of divisions that are required is an unknown parameter and further analysis is required in future research. It is clear from the numerical investigations there is a certain level of precision that is necessary and any level above this is not beneficial.

It has been shown that the Pareto analysis is not required when comparing different train types. The reason for this is due to the formula used for the calculation of sectional running times. For the other scenarios this is perhaps untrue. The capacity versus expansion bi-objective approach has been found to be equivalent to the iterative application of the capacity expansion model proposed in Chapter 4. Bi-objective optimisation, however, provides a better framework because it explores all possible solutions and automatically finds the best one. It can also perform the same analysis with reduced computational effort. In contrast, the other approach is manual and iterative; thus it requires more effort. Obviously the model in Chapter 4 is necessary because it allows specific capacity expansion scenarios to be solved.

In this thesis, various bi-objective models have been formulated and solved. It is quite clear, however, that there are more than two types of competition in practice, for example there are more than two types of trains, networks consist of more than two corridors, etc. Hence, multi-objective models are required. Fortunately the formulated bi-objective models can be easily extended to three or more objectives by defining extra parameter and decision variables. More specifically, additional objective functions of the same type as shown in Chapter 5 just need to be added. The solution strategies, namely ECM and WSM however need to be recoded to handle the

additional dimensions in the criterion space. Visualisation of the Pareto frontier is not possible for a higher number of dimensions.

6.2 SKILLS OBTAINED

In this Master's degree, this researcher has improved significantly in mathematical modelling skills. To solve the developed models, the CPLEX optimisation software was required. This required learning the OPL Studio programming language, and learning how to use the scripting functionality to solve the models iteratively. The scripting code has been provided in the Appendices.

This Master's degree has also given this researcher the opportunity to improve academic writing skills and increase knowledge of scientific decision making techniques and theories.

6.3 FUTURE RESEARCH

This thesis has focused upon improving the capacity of existing railway networks and specifically focused on developing analytical models on capacity determination. There are many avenues that still need to be investigated in future studies. The most significant areas for further research are as below:

- An investigation of other mechanisms (other than track duplication) for capacity expansion should be performed.
- New capacity expansion approaches should be compared if not merged into a single unified approach in order to identify the best and the most cost effective way of performing capacity expansion in railway networks.
- Multi-objective railway capacity analysis and expansion models are also possible and are necessary to regulate a greater number of competitions that may simultaneously occur.
- Improved versions of ECM and WSM should be developed in order to generate the Pareto frontier in a more intelligent way. For example adaptive versions that have reduced computational requirements are possible.
- The result of this thesis can be integrated into a set of guidelines and policies for railway capacity determination and expansion. A manual should be created for practitioners and railway operators to use the proposed model in practice.

This research topic will be investigated in more depth during my PhD candidature. For example it is my intention to consider multi-modal transportation systems (MMTS) that consist of railway networks and many competing objectives. The expansion of MMTS is also a very important and contemporary topic.

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Appendix 1 : CPLEX Code (Base Model Code)

```
//=========================================================
// Constants
//=========================================================

int TIME = ...;             // The time period duration for the analysis
int nbT = ...;              // Number of train types
int nbS = ...;              // Number of sections
int nbL = ...;              // Number of location
int nbC = ...;              // Number of corridors
int nbIO = ...;            // Number of Input Output points
int maxiter = ...;

//=========================================================
// Ranges (i.e. Indexes)
//=========================================================

range T = 1 .. nbT;      // Range of train types
range S = 1 .. nbS;      // Range of sections
range C = 1 .. nbC;      // Range of corridors

//=========================================================
// Sets
//=========================================================

{int} SECT = {s | s in S};
{int} TRAIN = {t | t in T};

//=========================================================
// Parameters
//=========================================================

float SPD[T] = ...;
float LOS[S] = ...;                 // Length of sections
{int} CORR_SECT[C] = ...;           // Sections present in each corrdior
float SRT[S][T];                    // Sectional running time
float SOT[S][T];                    // Section occupation time
float DD[C][T] = ...;               // Directional distribution corridors
float PD[C][T] = ...;               // Proportional distribution corridors
int MAX_TRK[S] = ...;               // Max number of tracks that can be added in section s
int TRK[S] = ...;                   // the number of tracks currently present in section s

//=========================================================
// Decision Variable
//=========================================================

dvar float ABS_CAP;                 // The total flow of trains
dvar float xc_f[C][T];              // The number of trains utilising each corridor - forward
dvar float xc_r[C][T];              // The number of trains utilising each corridor - backward
dvar float xs_f[S][T];              // The number of trains utilising each section
dvar float xs_r[S][T];              // The number of trains utilising each section
dvar float xc[C];
dvar float used[S][T];              // Time used by train types on each section
dvar float usedTot[S];              // Time used (i.e. occupancy) on each section

//=========================================================
// Pre_processing
//=========================================================

execute INITIALISE
{
        var s,i;
        for(s in SECT)
                for(i in T)
                {
                        SRT[s][i] = 60.0 *LOS[s] / SPD[i];
                        SOT[s][i] = SRT[s][i];
                }
```

```
}

//=========================================================
// Mathematical Formulation
//=========================================================

// Objective function
maximize ABS_CAP;
subject to
{
	ABS_CAP == sum(c in C,i in T) (xc_f[c][i] + xc_r[c][i]);

	// Constraint 0: relationship between xc and xs (xc encapsulates xs)
	forall(s in S, i in TRAIN)
	{
		xs_f[s][i] == sum(c in C: s in CORR_SECT[c]) xc_f[c][i];
		xs_r[s][i] == sum(c in C: s in CORR_SECT[c]) xc_r[c][i];
	};

	forall(c in C)
		xc[c] == sum(i in TRAIN) (xc_f[c][i] + xc_r[c][i]);

	// Constraint 1: proportional distribution
	forall(c in C, i in TRAIN)
		(xc_f[c][i] + xc_r[c][i]) == (PD[c][i]*xc[c]);

	// Constraint 2: directional distribution
	forall(c in C, i in TRAIN)
		xc_f[c][i] == DD[c][i] * (xc_f[c][i] + xc_r[c][i]);

	// Constraint 3: Section saturation and usage constraints
	forall(s in S)
		sum(i in TRAIN) used[s][i] <= TIME*(TRK[s]);// + DUP_TRK[s]);

	forall(s in S)
		usedTot[s] == sum(i in TRAIN) used[s][i];

	forall(s in S,i in TRAIN)
		used[s][i] == SOT[s][i]*xs_f[s][i] + SOT[s][i]*xs_r[s][i];

	// Constraint 5: Positivity Constraints:
	forall(i in TRAIN)
	{
		forall(s in S) xs_f[s][i] >= 0;
		forall(s in S) xs_r[s][i] >= 0;
		forall(c in C) xc_f[c][i] >= 0;
		forall(c in C) xc_r[c][i] >= 0;
	};

}
```

Appendix 2 : CPLEX Code (Static Expansion Model Code)

```
//=====================================================
// Constants
//=====================================================

int TIME = ...;             // The time period duration for the analysis
int nbT = ...;              // Number of train types
int nbS = ...;              // Number of sections
int nbL = ...;              // Number of location
int nbC = ...;              // Number of corridors
int nbIO = ...;            // Number of Input Output points
float BUDGET = ...;      // Budget for infrastructure expansion

//=====================================================
// Ranges (i.e. Indexes)
// NOTE: must have spaces before and after ".."
//=====================================================

range T = 1 .. nbT;      // Range of train types
range S = 1 .. nbS;      // Range of sections
range C = 1 .. nbC;      // Range of corridors

//=====================================================
// Sets
//=====================================================

{int} SECT = {s | s in S};
{int} TRAIN = {t | t in T};

//=====================================================
// Parameters
//=====================================================

float COST_DUP[S] = ...;            // Cost to duplicate section s
float SPD[T] = ...;
float LOS[S] = ...;                 // Length of sections
{int} CORR_SECT[C] = ...;           // Sections present in each corrdior
float SRT[S][T];                    // Sectional running time
float SOT[S][T];                    // Section occupation time
float DD[C][T] = ...;               // Directional distribution corridors
float PD[C][T] = ...;               // Proportional distribution corridors
//float PF[C] = ...;                // Percentage flow on each corridor, i..e between IOPT
int MAX_TRK[S] = ...;               // Max number of tracks that can be added in section s
int TRK[S] = ...;                   // the number of tracks currently present in section s

//=====================================================
// Decision Variable
//=====================================================

dvar float ABS_CAP;   // The total flow of trains
dvar float COST;      // The total spending on infrastructure expansion
dvar float xc_f[C][T];  // The number of trains utilising each corridor - forward
dvar float xc_r[C][T];  // The number of trains utilising each corridor - backward
dvar float xs_f[S][T]; // The number of trains utilising each section
dvar float xs_r[S][T]; // The number of trains utilising each section
dvar float xc[C];
dvar int DUP_TRK[S];  // Number of duplicated track in section s
dvar float used[S][T];  // Time used by train types on each section
dvar float usedTot[S];// Time used (i.e. occupancy) on each section

//=====================================================
// Pre_processing //=====================================================

execute INITIALISE
{
        var s,i;
        for(s in SECT)
                for(i in T)
                {
```

```
                        SRT[s][i] = 60.0 *LOS[s] / SPD[i];
                        SOT[s][i] = SRT[s][i];
                }
}

//=========================================================
// Mathematical Formulation
//=========================================================

// Objective function

//minimize COST;
maximize ABS_CAP;
subject to
{
        ABS_CAP == sum(c in C,i in T) (xc_f[c][i] + xc_r[c][i]);
        //sum(c in C,i in T) (xc_f[c][i] + xc_r[c][i]) >= ABS_CAP;
        COST == sum(s in S) COST_DUP[s] * DUP_TRK[s];

        // Constraint 0: relationship between xc and xs (xc encapsulates xs)
        forall(s in S, i in TRAIN)
        {
                xs_f[s][i] == sum(c in C: s in CORR_SECT[c]) xc_f[c][i];
                xs_r[s][i] == sum(c in C: s in CORR_SECT[c]) xc_r[c][i];
        };

        forall(c in C)
                xc[c] == sum(i in TRAIN) (xc_f[c][i] + xc_r[c][i]);

        // Constraint 1: proportional distribution
        forall(c in C, i in TRAIN)
                (xc_f[c][i] + xc_r[c][i]) == (PD[c][i]*xc[c]);

        // Constraint 2: directional distribution
        forall(c in C, i in TRAIN)
                xc_f[c][i] == DD[c][i] * (xc_f[c][i] + xc_r[c][i]);

        // Constraint 3: Section saturation and usage constraints
        forall(s in S)
                sum(i in TRAIN) used[s][i] <= TIME*(TRK[s] + DUP_TRK[s]);

        forall(s in S)
                usedTot[s] == sum(i in TRAIN) used[s][i];

        forall(s in S,i in TRAIN)
                used[s][i] == SOT[s][i]*xs_f[s][i] + SOT[s][i]*xs_r[s][i];

        // Constraint 5: Positivity Constraints:
        forall(i in TRAIN)
        {
                forall(s in S) xs_f[s][i] >= 0;
                forall(s in S) xs_r[s][i] >= 0;
                forall(c in C) xc_f[c][i] >= 0;
                forall(c in C) xc_r[c][i] >= 0;
        };

        //Restriction on the infrastructure spending
        COST <= BUDGET;
        forall(s in S)
        {
                DUP_TRK[s] >= 0;
                DUP_TRK[s] <= MAX_TRK[s];
        };
}
//=========================================================
// Script to do the incremental analysis of budget
//=========================================================

main
{
        var j = 0.0, def, data, c;
```

```
var status = 0;
var mod = thisOplModel;
var i, sum_i;
while(j <= 200)
{
	def = mod.modelDefinition; // Extract current definitions
	data = mod.dataElements;   // Extract current data
	data.BUDGET = 33*j;

	mod = new IloOplModel(def, cplex); // Define a new model
	mod.addDataSource(data); // Add new data to model
	mod.generate(); // Generate the new model for solving
	if ( cplex.solve() )
	{
		for(c in mod.C)
	{
		sum_i = 0.0;
		for(i in mod.T)
		{
			sum_i = sum_i + mod.xc_f[c][i] + mod.xc_r[c][i];
		}

		}
		writeln("j,",j,",ABS_CAP,", mod.ABS_CAP);
	}
	j = j + 1;
}
}
```

Appendix 3 : CPLEX Code (Dynamic Expansion Model Code)

```
// ==================================================================
// Constants
// ==================================================================

float BIGM = 1e10;
int TIME = ...;             // The time period duration for the analysis
int nbT = ...;              // Number of train types
int nbS = ...;              // Number of sections
int nbL = ...;              // Number of location
int nbC = ...;              // Number of corridors
int nbIO = ...;            // Number of Input Output points
int nbP = ...;             //Number of Period for upgrading railway network
float BUDGET = ...;     // Budget for infrastructure expansion
float ABS_CAP_F = ...; // Given Absolut Capacity
float ABS_CAP_I = ...;  //Initial ABS capacity

// ==================================================================
// Ranges (i.e. Indexes)
// ==================================================================

range T = 1 .. nbT;      // Range of train types
range S = 1 .. nbS;      // Range of sections
range C = 1 .. nbC;      // Range of corridors
range P = 1 .. nbP;      // Range of Periods
range Binary = 0 .. 1;

// ==================================================================
// Sets
// ==================================================================

{int} SECT = {s | s in S};
{int} TRAIN = {t | t in T};
{int} Period = {p| p in P};

// ==================================================================
// Parameters
// ==================================================================

float COST_DUP[S][P] = ...;          // Cost to duplicate section s
float BUD[P] = ...;                  // Budget for infrastructure expansion in Period P
float QBUD[P] = ...;                 //cumulative budget in period p
float SPD[T] = ...;
float LOS[S] = ...;                  // Length of sections
{int} CORR_SECT[C] = ...;            // Sections present in each corrdior
float SRT[S][T];                     // Sectional running time
float SOT[S][T];                     // Section occupation time
float DD[C][T] = ...;                // Directional distribution corridors
float PD[C][T] = ...;                // Proportional distribution corridors
int MAX_TRK[S] = ...;                 // Max number of tracks that can be added in section s
int TRK[S] = ...;                    // the number of tracks currently present in section s
float INTER_ABS_CAP[P] = ...;

// ==================================================================
// Decision Variable
// ==================================================================

dvar int OP;                         // The total period of expansion activities
dvar int Q[P] in Binary;             // Binary variable for restricting capacity
dvar int E[P];                       // Spending in each period
dvar int QTRK[S][P];
dvar float COST;                     // The total spending on infrastructure expansion
dvar float xc_f[C][T][P];            // The number of trains utilising each corridor - forward
dvar float xc_r[C][T][P];            // The number of trains utilising each corridor - backward
dvar float xs_f[S][T][P];            // The number of trains utilising each section
dvar float xs_r[S][T][P];             // The number of trains utilising each section
dvar float xc[C][P];
dvar int DUP_TRK[S][P];              // Number of duplicated track in section s
dvar float used[S][T][P];            // Time used by train types on each section
dvar float usedTot[S][P];            // Time used (i.e. occupancy) on each section
```

```
dvar float A[P];                        // Absolute capacity in Period P
dvar int ntrk[S];
// ==================================================================
// Pre_processinger
// ==================================================================

execute INITIALISE
{
        var s,i;
        for(s in SECT)
        {
                for(i in T)
                {
                        SRT[s][i] = 60.0 *LOS[s] / SPD[i];
                        SOT[s][i] = SRT[s][i];
                }
        }
}

// ==================================================================
// Mathematical Formulation
// ==================================================================

// Objective function

minimize OP;                            //Overall time required
subject to
{
        // Bound on period
         OP >= 1;
         OP <= nbP;

        COST == sum(p in P) E[p];

        // expenditure in period p
        forall(p in P)
        {
                E[p] == sum(s in S) (COST_DUP[s][p] * DUP_TRK[s][p]);
                E[p] <= BUD[p];
        };


        forall (s in S, p in P: p == 1)
                QTRK[s][p] == TRK[s];

        forall (s in S, p in P: p > 1)
                QTRK[s][p] == QTRK[s][p-1] + DUP_TRK[s][p];


        forall (p in P)
        {
                p <= OP-1 + (Q[p] * BIGM);
                p >= OP + ((Q[p]-1) * BIGM);
        };

        forall(s in S, p in P)
                DUP_TRK[s][p] >= 0;

        forall(s in S)
        {
                ntrk[s] == sum(p in P) DUP_TRK[s][p];
                ntrk[s] <= 10;
        }

        // constraint
        forall(p in P)
        {
                A[p] == sum(c in C, i in T) (xc_f[c][i][p] + xc_r[c][i][p]);
                A[p] >= Q[p] * ABS_CAP_F;
                A[p] >= INTER_ABS_CAP[p];
        };
```

```
forall(p in P)
        sum(p_ in P: p_ <= p) E[p_] <= QBUD[p];


forall(p in P: p < nbP)
        E[p+1] <= (1 - Q[p]) * BIGM;


forall (p in P: p > 1 )
        A[p] >= A[p-1];


 forall (p in P: p == 1 )
        A[p] >= ABS_CAP_I;


// ==================================================================
// Static Base Formulation
// ==================================================================

// Constraint 0: relationship between xc and xs (xc encapsulates xs)
forall(s in S, i in TRAIN, p in P)
{
        xs_f[s][i][p] == sum(c in C: s in CORR_SECT[c]) xc_f[c][i][p];
        xs_r[s][i][p] == sum(c in C: s in CORR_SECT[c]) xc_r[c][i][p];
};

forall(c in C, p in P)
        xc[c][p] == sum(i in TRAIN) (xc_f[c][i][p] + xc_r[c][i][p]);

// Constraint 1: proportional distribution
forall(c in C, i in TRAIN, p in P)
        (xc_f[c][i][p] + xc_r[c][i][p]) == (PD[c][i]*xc[c][p]);

// Constraint 2: directional distribution
forall(c in C, i in TRAIN, p in P)
        xc_f[c][i][p] == DD[c][i] * (xc_f[c][i][p] + xc_r[c][i][p]);

// Constraint 3: Section saturation and usage constraints
forall(s in S, p in P)
        sum(i in TRAIN) used[s][i][p] <= TIME * QTRK[s][p];

forall(s in S, p in P)
        usedTot[s][p] == sum(i in TRAIN) used[s][i][p];

forall(s in S,i in TRAIN, p in P)
        used[s][i][p] == SOT[s][i] * (xs_f[s][i][p] + xs_r[s][i][p]);

// Constraint 5: Positivity Constraints:
forall(i in TRAIN, p in P)
{
        forall(s in S) xs_f[s][i][p] >= 0;
        forall(s in S) xs_r[s][i][p] >= 0;
        forall(c in C) xc_f[c][i][p] >= 0;
        forall(c in C) xc_r[c][i][p] >= 0;
};

}
```

Appendix 4 : CPLEX Code (Multi-Objective Scripts Code)

```
// ==================================================================
// Constants
// ==================================================================

int TIME = ...;             // The time period duration for the analysis
int nbT = ...;              // Number of train types
int nbS = ...;              // Number of sections
int nbC = ...;              // Number of corridors
int maxiter = ...;
float w1 = ...;
float w2 = ...;
float wght = ...;

// ==================================================================
// Ranges (i.e. Indexes)
// NOTE: must have spaces before and after ".."
// ==================================================================

range T = 1 .. nbT;      // Range of train types
range S = 1 .. nbS;      // Range of sections
range C = 1 .. nbC;      // Range of corridors

// ==================================================================
// Sets
// ==================================================================

{int} SECT = {s | s in S};
{int} TRAIN = {t | t in T}; // Set of all trains
{int} FRE = ...;            // Set of freight trains
{int} PAS = ...;            // Set of passenger trains

// ==================================================================
// Parameters
// ==================================================================

float COST_DUP[S] = ...;             // Cost to duplicate section s
float SPD[T] = ...;
float LOS[S] = ...;                  // Length of sections
{int} CORR_SECT[C] = ...;            // Sections present in each corrdior
float SRT[S][T];                     // Sectional running time
float SOT[S][T];                     // Section occupation time
float DD[C][T] = ...;                // Directional distribution corridors
float PD_PAS[C][PAS] = ...;          // Proportional distribution corridors
float PD_FRE[C][FRE] = ...;          // Proportional distribution corridors
int TRK[S] = ...;                    // The number of tracks currently present in section s
float M[C][FRE] = ...;               // Freight measure / payoff
float LIM_FRE = ...;                 // Current limit on freight capacity
int flag = ...;
int opt = ...;
float scale_PAS = ...;
float scale_FRE = ...;

// ==================================================================
// Decision Variable
// ==================================================================

dvar float ABS_PAS;
dvar float ABS_FRE;
dvar float N_FRE;
dvar float Z;
dvar float Z1;
dvar float Z2;
dvar float xc_f[C][T];               // The number of trains utilising each corridor - forward
dvar float xc_r[C][T];               // The number of trains utilising each corridor - backward
dvar float xs_f[S][T];               // The number of trains utilising each section
dvar float xs_r[S][T];   // The number of trains utilising each section
dvar float xc[C];
dvar float xc_PAS[C];
dvar float xc_FRE[C];
```

```
dvar float used[S][T];  // Time used by train types on each section
dvar float usedTot[S]; // Time used (i.e. occupancy) on each section

// ==================================================================
// Pre-processing
// ==================================================================

execute INITIALISE
{
	var s,i;
	for(s in SECT)
		for(i in T)
		{
			SRT[s][i] = 60.0 * LOS[s] / SPD[i];
			SOT[s][i] = SRT[s][i];
		}
}

// ==========================================================
// Mathematical Formulation - Multi-objective
// ==========================================================

maximize Z;
subject to
{
	Z1 == ABS_PAS*flag + ABS_FRE*(1 - flag); // Set as first or second objective
	Z2 == wght*(ABS_PAS/scale_PAS) + (1- wght)*(ABS_FRE/scale_FRE);
	Z == opt*Z1 + (1 - opt)*Z2;

	// Compute bi-objectives
	ABS_PAS == sum(c in C, i in PAS) (xc_f[c][i] + xc_r[c][i]);
	ABS_FRE == sum(c in C, i in FRE) (M[c][i]*xc_f[c][i] + M[c][i]*xc_r[c][i]);
	N_FRE == sum(c in C, i in FRE) (xc_f[c][i] + xc_r[c][i]);

	// Imposed restriction on second objective
	 ABS_FRE >= LIM_FRE;

	// Constraint 0: Relationship between xc and xs (xc encapsulates xs)
	forall(s in S, i in TRAIN)
	{
		xs_f[s][i] == sum(c in C: s in CORR_SECT[c]) xc_f[c][i];
		xs_r[s][i] == sum(c in C: s in CORR_SECT[c]) xc_r[c][i];
	};

	forall(c in C)
	{
		xc[c] == sum(i in TRAIN) (xc_f[c][i] + xc_r[c][i]);
		xc_PAS[c] == sum(i in PAS) (xc_f[c][i] + xc_r[c][i]);
		xc_FRE[c] == sum(i in FRE) (xc_f[c][i] + xc_r[c][i]);
	};

	// Constraint 2: Directional distribution
	forall(c in C, i in TRAIN)
		xc_f[c][i] == DD[c][i] * (xc_f[c][i] + xc_r[c][i]);

	// Constraint 3: Section saturation and usage constraints
	forall(s in S)
	{
		sum(i in TRAIN)		used[s][i] <= TIME*TRK[s];

		forall(i in TRAIN) used[s][i] == SOT[s][i]*xs_f[s][i] + SOT[s][i]*xs_r[s][i];
		usedTot[s] == sum(i in TRAIN) used[s][i];
	};

	// Constraint 5: Positivity Constraints:
	forall(i in TRAIN)
	{
		forall(s in S) xs_f[s][i] >= 0;
		forall(s in S) xs_r[s][i] >= 0;
		forall(c in C) xc_f[c][i] >= 0;
		forall(c in C) xc_r[c][i] >= 0;
```

```
        };
};
// ==========================================================
// SCRIPT - E CONSTRAINT METHOD
// ==========================================================

main
{
        var def, data, data2;
        var status = 0;
        var mod = thisOplModel;
        var UB_PAS, UB_FRE;

        // ----------------------------
        // Step 1: Solve for ABS_PAS
        // ----------------------------

        def = mod.modelDefinition;              // Extract current definitions
        data = mod.dataElements;                // Extract current data
        data.flag = 1;                          // Set objective 1
        data.opt = 1;
        data.LIM_FRE = 0.0;                     // Change RHS to another value
        mod = new IloOplModel(def, cplex);      // Define a new model
        mod.addDataSource(data);                // Add new data to model
        mod.generate();                         // Generate the new model for solving
        if ( cplex.solve() )
        {
                writeln("UB_ABS_PAS,", mod.ABS_PAS, ",ABS_FRE,", mod.ABS_FRE,",N_FRE,",mod.N_FRE);
                UB_PAS = mod.ABS_PAS / 1.0; // Division by 1.0 converts result to a float
        }

        // ----------------------------
        // Step 2: Solve for ABS_FRE
        // ----------------------------

        def = mod.modelDefinition;              // Extract current definitions
        data = mod.dataElements;                // Extract current data
        data.flag = 0;                          // Set objective 2
        data.opt = 1;
        data.LIM_FRE = 0.0;                     // Change RHS to another value
        mod = new IloOplModel(def, cplex);      // Define a new model
        mod.addDataSource(data);                // Add new data to model
        mod.generate();                         // Generate the new model for solving
        if ( cplex.solve() )
        {
                writeln("UB_ABS_FRE,", mod.ABS_FRE, ",ABS_PAS,", mod.ABS_PAS,",N_FRE,",mod.N_FRE);
                UB_FRE = mod.ABS_FRE / 1.0;     // Division by 1.0 converts result to a float
        }

        var delta = UB_FRE / mod.maxiter;
        writeln("Delta = ", delta);

        // ----------------------------
        // Step 3
        // ----------------------------

        var dist, norm_1, norm_2, j = 0.0;
        var min_dist = Infinity;
        var min_at, obj_1, obj_2;

        while(j <= mod.maxiter)
        {
                def = mod.modelDefinition;              // Extract current definitions
                data = mod.dataElements;                // Extract current data
                data.flag = 1;                          // Set objective 1
                data.opt = 1;
                data.LIM_FRE = UB_FRE - delta*j;        // Change RHS to another value
                mod = new IloOplModel(def, cplex);      // Define a new model
                mod.addDataSource(data);                // Add new data to model
                mod.generate();                         // Generate the new model for solving
                if ( cplex.solve() )
```

```
{
norm_1 = (mod.ABS_PAS - 0.0) / (UB_PAS - 0.0);
norm_2 = (mod.ABS_FRE - 0.0) / (UB_FRE - 0.0);
dist = Math.sqrt( mod.w1*(1-norm_1)*(1-norm_1) + mod.w2*(1-norm_2)*(1-norm_2));
if (dist < min_dist) { min_dist = dist; min_at = j; obj_1 = mod.ABS_PAS/1.0; obj_2 =
mod.ABS_FRE/1.0; }

writeln("j,",j,",ABS_PAS,",mod.ABS_PAS,",ABS_FRE,",mod.ABS_FRE,",N_FRE,",mod.N_FRE,",norm_1,",norm_1,",nor
m_2,",norm_2,",dist,",dist);
}
j = j + 1;
}
writeln("Best at = ", min_at, " Min Dist = ", min_dist, " Solution = (",obj_1,",",obj_2,")");
}

// ==========================================================
// SCRIPT - WEIGHTED SUM METHOD
// ==========================================================

main
{
var def, data, data2;
var status = 0;
var mod = thisOplModel;

// ----------------------------
// Step 1: Solve for ABS_PAS
// ----------------------------

def = mod.modelDefinition; // Extract current definitions
data = mod.dataElements;  // Extract current data
data.opt = 1; // Use Z1
data.flag = 1; // Set objective ABS_PAS
data.LIM_FRE = 0.0; // Change RHS to another value
mod = new IloOplModel(def, cplex); // Define a new model
mod.addDataSource(data); // Add new data to model
mod.generate(); // Generate the new model for solving
if ( cplex.solve() )
{
writeln("UB_ABS_PAS,", mod.ABS_PAS, ",ABS_FRE,", mod.ABS_FRE,",N_FRE,",mod.N_FRE);
UB_PAS = mod.ABS_PAS / 1.0; // Division by 1.0 converts result to a float
}

//----------------------------
// Step 2: Solve for ABS_FRE
//----------------------------

def = mod.modelDefinition; // Extract current definitions
data = mod.dataElements;  // Extract current data
data.opt = 1; // Use Z1
data.flag = 0; // Set objective ABS_FRE
data.LIM_FRE = 0.0; // Change RHS to another value
mod = new IloOplModel(def, cplex); // Define a new model
mod.addDataSource(data); // Add new data to model
mod.generate(); // Generate the new model for solving
if ( cplex.solve() )
{
writeln("UB_ABS_FRE,", mod.ABS_FRE, ",ABS_PAS,", mod.ABS_PAS,",N_FRE,",mod.N_FRE);
UB_FRE = mod.ABS_FRE / 1.0; // Division by 1.0 converts result to a float
}

//---------------------------------
// Step 3
//---------------------------------

var dist, norm_1, norm_2, j = 0;
var min_dist = Infinity;
var min_at, obj_1, obj_2;
//var c1, c2;

delta = 1.0 / mod.maxiter;
```

```
        writeln("Delta = ", delta);

        while(j <= mod.maxiter)
        {
                def = mod.modelDefinition; // Extract current definitions
                data = mod.dataElements;  // Extract current data
                data.wght = 0.0 + j*delta;
                data.opt = 0; // Use Z2 instead of Z1
                data.scale_PAS = UB_PAS / 1.0;
                data.scale_FRE = UB_FRE / 1.0;

                mod = new IloOplModel(def, cplex); // Define a new model
                mod.addDataSource(data); // Add new data to model
                mod.generate(); // Generate the new model for solving
                if ( cplex.solve() )
                {
                        norm_1 = (mod.ABS_PAS ) / (UB_PAS );
                        norm_2 = (mod.ABS_FRE ) / (UB_FRE );
                        dist = Math.sqrt( mod.w1*(1-norm_1)*(1-norm_1) + mod.w2*(1-norm_2)*(1-norm_2));
                        if (dist < min_dist) { min_dist = dist; min_at = j; obj_1 = mod.ABS_PAS/1.0; obj_2 =
mod.ABS_FRE/1.0; }
                        writeln("j,",j,",wght,",data.wght,",Z,",
mod.Z,",ABS_PAS,",mod.ABS_PAS,",ABS_FRE,",mod.ABS_FRE,",N_FRE,",mod.N_FRE,",norm_1,",norm_1,",norm_2,",norm_2,",
dist,",dist);
                }
                j = j + 1;
        }
        writeln("Best at = ", min_at, " Min Dist = ", min_dist, " Solution = (",obj_1,",",obj_2,")");
}
```